INCH-POUND

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w/CHANGE 6
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USED IN LIEU OF
MIL-STD-1553B
w/CHANGE 4
15 January 1996

DEPARTMENT OF DEFENSE
INTERFACE STANDARD

DIGITAL TIME DIVISION COMMAND/RESPONSE MULTIPLEX DATA BUS

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AMSC: N/A AREA MCCR

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FOREWORD

This standard contains requirements for an Extended Waveform digital time division command/response multiplex data bus for use in systems integration that is capable of coexisting with the 1553B multiplex data bus. Even with the use of this standard, differences may exist between multiplex data buses in different system applications due to particular application requirements and the designer options allowed in this standard. The system designer must recognize this fact and design the multiplex bus controller hardware and software to accommodate such differences.

It should be noted that verification of requirements provides an important part of this standard.

These designer selected options must exist to allow the necessary flexibility in the design of specific multiplex systems in order to provide for the control mechanism, architectural redundancy, degradation concept and traffic patterns peculiar to the specific application requirements.

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1. SCOPE

1.1 Scope.

This change notice establishes the performance, interface, and verification requirements for an extended waveform multiplexing capability (extended 1553) that retains backwards compatibility with the 21 September 1978 release of MIL-STD 1553B and notices 1 through 4.

The overarching requirement is to enable common operability between vendors so as to provide extended waveform capability at the line replaceable unit (LRU) or weapon replaceable assembly (WRA) level by providing sufficient information to implement a compliant interface.

1.2 Application.

When invoked in a specification or statement of work, these requirements will apply to the extended waveform multiplexing capability (extended 1553) and associated equipment as installed in the LRU/WRA developed either alone or as a portion of a weapon system or subsystem. The contractor is responsible for invoking all the requirements of this notice on any and all subcontractors he may employ unless otherwise required by the procuring agency.

Extended 1553 waveform consists of:

a. A broadband multi-carrier waveform (MCW) to support real-time applications such as audio, video and non-real time traffic e.g., file transfers.

b. A baseband filtered waveform (FW) to support communications currently established by devices compliant to MIL-STD-1553B to notice 4 and to reduce generated interference.

c. A baseband trapezoidal waveform (TW) that is identical to the MIL-STD-1553B to notice 4 (see section 5.5).

Communication on the MCW network is similar to that of TW in that messages are controlled by a designated bus controller (BC) with all other nodes being designated as remote terminals (RT)s. Transmission on both A and B buses with reception on either is supported.

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2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461 Electromagnetic Interference Characteristics, Requirements for Equipment

MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems

Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

2.3 Order of precedence.

In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Multi-carrier waveform (MCW) and applicable open systems interconnection (OSI) definitions.

3.1.1 Acknowledgment.

Notification sent from one terminal to another to acknowledge the occurrence of some event (for example, the receipt of a message).

3.1.2 Additive white Gaussian noise (AWGN).

A type of noise with the following properties:

a. Is additive with respect to the information carrying signal.

b. The statistics follow a Gaussian distribution with zero mean.

c. The samples are uncorrelated (resulting in flat spectral characteristics).

3.1.3 Arbitrary Test Frame

A physical layer (PHY) frame used to send vendor-specific arbitrary test signals between terminals.

3.1.4 Basic rate (BR).

Lowest rate of data communications used for broadcast and sending control information.

3.1.5 Binary phase shift keying (BPSK).

A type of binary digital modulation in which the phase of the transmitted signal may assume one of two values which are separated by 180°. The phase of the transmitted signal is determined only by the current data bit.

3.1.6 Bit error rate (BER).

The ratio of the number of bits incorrectly received to the total number of bits transmitted.

3.1.7 Bus command frame (BCF).

A physical layer (PHY) frame broadcasted from the bus controller (BC) that is used to convey MCW commands to all terminals on the bus.

3.1.8 Bus controller (BC).

The terminal assigned the task of initiating information on the data bus. The bus controller acts as an RT when scheduled into a compound transaction.

3.1.9 Clock jitter.

The short term (sample interval) variance of clock phase.

3.1.10 Conformance

Fulfillment of an implementation of all requirements specified herein.

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3.1.11 Compound transaction.

A multi-carrier waveform transaction that consists of a set of commands, terminal responses, and terminal acknowledgements.

3.1.12 Cyclic redundancy check (CRC).

A method of detecting errors in the serial transmission of data. A CRC for a block of data is calculated before transmission and then included with the transmitted data. If the new CRC, calculated on the received data, does not match the CRC originally transmitted with the data, then an error has occurred.

3.1.13 Data link layer (DLL).

Layer 2 of the OSI reference model that provides data transfer across a physical link. At this layer, data packets are encoded and decoded into bits to be transmitted and received respectively across the physical layer (PHY). This layer is further divided into the media access control (MAC), packet adaptation layer (PAL) and logical link control (LLC) sublayers.

3.1.14 Differential binary phase shift keying (DBPSK).

A type of binary digital modulation in which the phase of a transmitted signal is varied by 0°or 180° to convey 1 bit worth of information. The phase of the transmitted signal is determined by the current data bit and the phase of the signal transmitted during the previous data bit.

3.1.15 Discrete Fourier transform (DFT).

An invertible linear transformation that converts a sampled time domain signal into the frequency domain.

3.1.16 Egress.

The direction of data flow from the terminal towards the bus wiring.

3.1.17 Encapsulation.

The process of appending protocol information, in the form of a header and, if applicable, a footer to a service data unit (SDU).

3.1.18 Filtered waveform (FW).

A spectrally friendly variant of the trapezoidal waveform (TW) with controlled out-of-band spectral content.

3.1.19 Frame control (FC).

A section of the PHY frame used to carry information about the transmitted frame.

3.1.20 Frame error rate (FER).

The ratio of the number of PHY frames incorrectly received to the total number of PHY frames transmitted.

3.1.21 Frame payload (PL).

A section of the PHY frame used to carry the information bits transmitted between terminals.

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3.1.22 Header.

Protocol information placed before the data when encapsulating that data in preparation for transmission to a lower layer or over the PHY medium.

3.1.23 Ingress.

The direction of data flow from the bus wiring towards the terminal.

3.1.24 In-band signaling.

Transmission of protocol management information (e.g. connection establishment handshaking) within a frequency range normally used for data transmission.

3.1.25 Interoperability

The condition achieved among communications-electronics systems or items of communications-electronics equipment when information or services can be exchanged directly and satisfactorily between them and/or their users as defined herein (reference section 5.4)

3.1.26 Inverse discrete fourier transform (IDFT).

The inverse of the DFT, thereby converting a sampled frequency domain signal into the time domain.

3.1.27 Logical link control (LLC).

Upper functional part of the DLL that supports initiation of control signal interchange, framing and MAC-sublayer addressing, interpretation of received commands and generation of appropriate responses and actions taken regarding the error/flow control and recovery in the communication links.

3.1.28 MAC management frame (MMF).

A PHY frame used for transmitting MAC management data between terminals.

3.1.29 Media access control (MAC).

Lower functional part of the DLL that supports frame delineation and recognition, addressing and error checking, handling access to shared media, scheduling and acquiring the right to use the underlying physical medium.

3.1.30 Multi-carrier waveform (MCW).

A bandwidth configurable orthogonal frequency division multiplex (OFDM) based communication method to enable the transmission of high data rate signals over existing MIL-STD-1553B cables and couplers.

3.1.31 Network layer (NL).

Layer 3 of the OSI reference model. This layer provides connectivity, routing and path selection as well as logical paths (virtual connections) for transmitting data between two end systems.

3.1.32 Network management.

Generic term used to describe systems or actions that help maintain, characterize, or troubleshoot a network.

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3.1.33 Orthogonal frequency division multiplex (OFDM) symbol.

A unit of transmission that contains one (D)BPSK or quadrature amplitude modulation (QAM) modulated constellation point for each transmitted tone.

3.1.34 Open system interconnection (OSI).

The international standardization program created by International Standards Organization (ISO) and International Telecommunication Union-Telecommunications Standards Sector (ITU-T) to develop data networking standards that facilitate multi vendor equipment interoperability. OSI forms the basic communications model for MCW. Each layer is a logical subsystem; each subsystem is a server for the layer above it and a client of the layer below it; each subsystem has a set of methods and attributes; adjacent subsystems communicate through their common boundary. Peer-to-peer communication is always indirect through connections provided by the server sub-layer below.

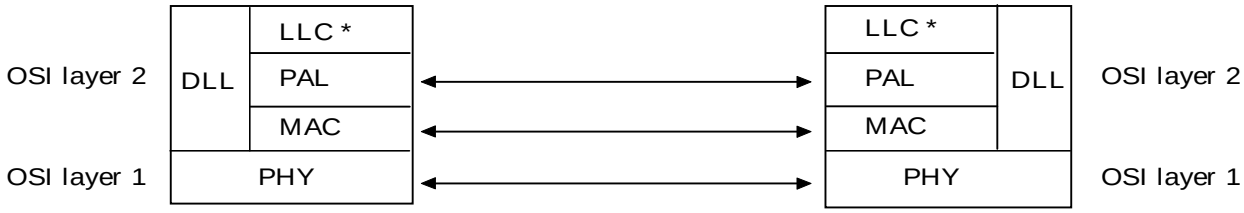
3.1.35 OSI peer-to-peer interfaces.

MCW peer-to-peer OSI interfaces are listed below and illustrated in FIGURE 1.

a. PHY-PHY.

b. MAC-MAC.

c. PAL-PAL.

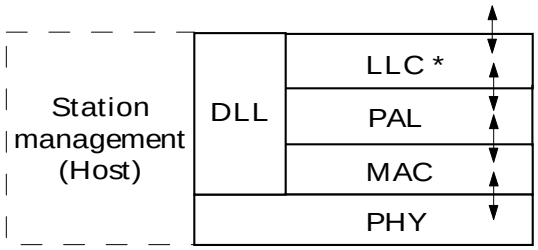


* The logical link control (LLC) sublayer is not defined in this standard.

FIGURE 1. OSI peer-to-peer interfaces.

3.1.36 OSI stacked level interfaces.

SDU interfaces between MCW stacked layers/sublayers are shown in FIGURE 2 as two way arrows, with the OSI layers included for reference. Station management, which is part of the host, is shown in dotted lines.



* The logical link control (LLC) sublayer is not defined in this standard.

FIGURE 2. OSI stacked layered interfaces.

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3.1.37 Orthogonal frequency division multiplexing (OFDM).

A type of digital modulation in which a signal is split into several carriers at different frequencies in such a manner that these carriers do not interfere with each other.

3.1.38 Output noise.

The noise introduced by a terminal to its bus interface when it is not transmitting.

3.1.39 Packet.

A synonym for the LLC protocol data unit (PDU).

3.1.40 Packet adaptation layer (PAL).

A sublayer above the MAC that allows transmission of arbitrary length packets.

3.1.41 Payload.

A synonym for service data unit (SDU).

3.1.42 Peer protocol.

The sequence of message exchanges between two entities in the same layer that utilize the services of the underlying layers to effect the successful transfer of data and/or control information from one location to another.

3.1.43 Peer-to-peer.

A communications model in which each party has the same capabilities and either can initiate a communication session.

3.1.44 Performance measures.

Metrics that provide quantitative or qualitative measures of the performance of systems or subsystems of interest.

3.1.45 PHY block.

A group of 8 consecutive OFDM symbols. If carrying data, the PHY block is forward error correction (FEC) encoded to form a convolutional code word.

3.1.46 PHY frame.

A Multi-carrier waveform protocol data unit at the PHY layer (a MCW PHY PDU).

3.1.47 Physical layer (PHY).

Layer 1 of the OSI reference model. The physical layer defines the electrical, mechanical, procedural, and functional specifications for activating, maintaining, and deactivating the physical link between terminals.

3.1.48 Preamble.

Section of the PHY frame used for PHY level synchronization between terminals.

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3.1.49 Protocol.

A set of rules and formats (semantic and syntactic) which determines the communication behavior of entities within a system to allow interoperability between vendors.

3.1.50 Protocol data unit (PDU).

The combination of data passed from the adjacent higher layer protocol (the service data unit (SDU) or payload) and protocol information for this layer.

3.1.51 Puncturing.

Elimination or omission of some encoded bits in a transmitter.

3.1.52 Quadrature amplitude modulation (QAM).

A type of modulation by which two amplitude modulated signals with the same carrier frequency are combined in quadrature (90° out of phase) into a single signal, thereby doubling the effective data carrying capacity.

3.1.53 Queue ID.

A number used to identify a stream of MAC-SDUs at the source RT. The queue ID is unique on a given RT and is analogous to a sub-address.

3.1.54 Reference terminal.

A terminal defined by the custodian agency that conforms to the requirements of this standard and which can be used as a tool to prove that a vendor's extended 1553 capability is interoperable with other extended 1553 terminals. The reference terminal may contain additional functionality not normally present in a terminal.

3.1.55 Remote terminal (RT).

Any terminal communicating on the bus.

3.1.56 Response frame.

A PHY frame transmitted as a result of a frame exchange command. A response frame may be a terminal response frame (TRF), MAC management frame (MMF), training frame (TF) or arbitrary test frame (ATF).

3.1.57 Service data unit (SDU).

Unit of information from an upper-layer protocol that defines a service request to the adjacent lower-layer protocol. Following the service request, the SDU is encapsulated by protocol information at the lower layer; thus the SDU becomes the payload for the lower-layer protocol.

3.1.58 Signal constellation.

A representation of a digital modulation scheme in the complex plane, obtained by assigning a set of complex numbers to the sequence of transmitted bits.

3.1.59 Signal to interference ratio (SIR).

The ratio of the power of the desired signal, to the power of the total interference signal.

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3.1.60 Signal to noise ratio (SNR).

The ratio of the power of the desired signal to the power of the total noise power.

3.1.61 Signal to noise and distortion ratio (SINAD).

On a per-tone basis for the MCW signal, the ratio of the root mean square (RMS) value of the full-amplitude tone to the RMS sum of all noise and distortion within a frequency band equal to the tone spacing and centered on the frequency of the tone for which the SINAD is being measured.

3.1.62 Sub-band.

A contiguous set of a predefined number of tones of the OFDM signal. Sub-bands start and end at prescribed frequencies.

3.1.63 Subsystem.

The device or functional unit receiving data transfer service from the data bus.

3.1.64 Sync bus command frame (SBCF).

A command frame transmitted periodically by the bus controller to allow RTs to communicate data timing with the bus controller.

3.1.65 Sync symbols.

Synchronization symbols contained in the preamble portion of the PHY frame.

3.1.66 Terminal acknowledgement frame (TAF).

A PHY frame used to indicate whether a response frame was received successfully.

3.1.67 Terminal.

The electronic module necessary to interface the data bus with the subsystem and the subsystem with the data bus. Terminals may exist as separate LRUs or WRAs contained within the elements of the subsystem.

3.1.68 Terminal response frame (TRF).

A PHY frame used for transmitting MAC payload data between terminals.

3.1.69 Throughput.

The PHY information rate, before FEC encoding, for blocks within the payload section of a PHY frame, as depicted in FIGURE 3.

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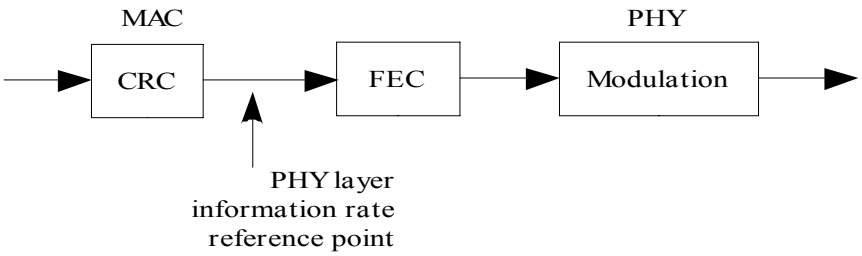


FIGURE 3: PHY layer information rate.

3.1.70 Time division multiple access (TDMA).

A means for multiple terminals to share access to a common physical medium, whereby only one terminal transmits on the medium at a time.

3.1.71 Tone.

A single carrier within the OFDM signal.

3.1.72 Tone map.

A record comprising the punctured convolutional code rate and the number of bits per each tone that can be transmitted under the current channel conditions.

3.1.73 Training frame (TF).

A PHY frame used to determine the data encoding parameters between two MCW terminals.

3.1.74 Trapezoidal waveform (TW).

A waveform identical to that specified in MIL-STD 1553B up to and including notice 4.

3.1.75 Unique Item Identifier (UII).

A message response used for asset management purposes throughout the life cycle of a terminal.

3.1.76 Unit under test (UUT).

A specific system or device that, for testing purposes, is considered to be the terminal being subjected to a test process or procedure.

3.1.77 Variable bit rate (VBR).

Communication using a higher data rate than basic rate by means of a negotiated optimum set of modulation parameters adapted to current channel conditions.

3.2 Trapezoidal waveform (TW) and filtered waveform (FW) definitions.

The definitions pertaining to the TW are contained in 5.5.3. Definitions pertaining to FW are identical to those for TW.

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3.3 Acronyms.

A	analysis
ADC	analog to digital converter
AFE	analog front-end
ATF	arbitrary test frame
AWGN	additive white gaussian noise
BC	bus controller
BCF	bus command frame
BER	bit error rate
BN	bus notification
BPSK	binary phase shift keying
BR	basic rate
CP	cyclic prefix
CS	cyclic suffix
CRC	cyclic redundancy check
D	demonstration
DAC	digital to analog converter
DBPSK	differential binary phase shift keying
DC	direct current
DFT	discrete fourier transform
DLL	data link layer
FC	frame control
FE	frame exchange
FEC	forward error correction
FER	frame error rate
FW	filtered waveform
IDFT	inverse discrete fourier transform
IEC	international electrotechnical commission
IEEE	institute of Electrical and Electronics Engineers
ISO	international standards organization
ITU-T	international telecommunication union-telecommunications standards sector
LGS	long guard slot
LLC	logical link control
LRU	line replaceable unit
LSB	least significant bit
MAC	media access control
Mbps	megabits per second

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MCW	multi-carrier waveform
MGS	medium guard slot
MHz	megahertz
MM	MAC management
MMF	MAC management frame
MSB	most significant bit
NL	network layer
OAM	operation/administration/management
OSI	open systems interconnection
PAL	packet adaptation layer
PDU	protocol data unit
PHY	physical layer
PL	frame payload
POC	proof of compliance
PPM	parts per million.
PSD	power spectral density
QAM	quadrature amplitude modulation
RMS	root mean square
RS	reed-solomon
RT	remote terminal
S	simulation
SBCF	sync bus command frame
SDU	service data unit
SGS	short guard slot
SINAD	signal to noise and distortion ratio
SIR	signal to interference ratio
SNR	signal to noise ratio
T	test
TAF	terminal acknowledgement frame
TDP	time domain processor
TF	training frame
TRF	terminal response frame
TW	trapezoidal waveform
UII	Unique Item Identifier
UUT	unit under test
VCRM	verification compliance reference matrix
VBR	variable bit rate

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VC	virtual connection
VCE	virtual connection end point
VGS	variable guard slot
WRA	weapon replaceable assembly
XML	eXtensible markup language

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4. GENERAL REQUIREMENTS

4.1 MIL-STD-001553B(USAF) Change 6 certification.

In order to prove and measure both conformance with the requirements of this standard and interoperability between terminals from different designs and vendors, individual vendor terminal certification to the extended 1553 capability contained in this standard is contingent on approval by the custodian agency of the following:

a. Verification of electrical and performance requirements in accordance with sections 4.1.1and 5.3; and

b. Performance of interoperability testing in addition to the verification tests above at the discretion of the custodian.

4.1.1 Electrical and performance requirements verification.

Electrical and performance requirements verification is contingent on:

a. Completion and submittal of analysis and simulation data prior to physical test and as defined in the verification compliance reference matrix contained as TABLE I.

b. Completion and submittal of demonstration and test data after analysis/simulation and as defined in the verification compliance reference matrix contained as TABLE I.

c. Specific vendor proof of compliance reports for a. and b. above and as referenced in 5.3.1.1 for all methodologies contained in each line of the VCRM.

TABLE I. Electrical and performance VCRM / POC template.

Section reference	Performance Heading	Verification Method	POC Report Reference	
			Transformer Coupled	Direct Coupled
5.2.1	General.	N/A: Header	N/A	N/A
5.2.2	MCW requirements.	N/A: Header	N/A	N/A
5.2.2.4	Output characteristics.	N/A: Header	N/A	N/A
5.2.2.4.1	Power spectral density.	A,T	vendor supplied	vendor supplied
5.2.2.4.2	Transmit spectral flatness.	S, T	vendor supplied	vendor supplied
5.2.2.4.3	Output signal fidelity.	S, T	vendor supplied	vendor supplied
5.2.2.4.4	Output noise.	A, T	vendor supplied	vendor supplied
5.2.2.4.5	Output transients.	S, T	vendor supplied	vendor supplied
5.2.2.4.6	BC clock accuracy.	A, T	vendor supplied	
5.2.2.4.7	RT clock accuracy.	A, T	vendor supplied	
5.2.2.4.8	Output impedance.	T	vendor supplied	vendor supplied

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TABLE I. Electrical and performance VCRM / POC template – Continued.

Section reference	Performance Heading	Verification Method	POC Report Reference	
			Transformer Coupled	Direct Coupled
5.2.2.4.9	Output timing accuracy.	A,T	vendor supplied	vendor supplied
5.2.2.5	Input characteristics.	N/A: Header	N/A	N/A
5.2.2.5.1	Input signal compatibility.	S, T	vendor supplied	vendor supplied
5.2.2.5.2	Common mode rejection.	T	vendor supplied	vendor supplied
5.2.2.5.3	Input impedance.	T	vendor supplied	vendor supplied
5.2.2.5.4	Susceptibility to noise and interference.	N/A: Header	N/A	N/A
5.2.2.5.4.1	Susceptibility to additive white gaussian noise (AWGN).	A, T	vendor supplied	vendor supplied
5.2.2.5.4.2	Susceptibility to a single narrow-band interference source	S, T	vendor supplied	vendor supplied
5.2.2.5.4.3	Susceptibility to TW interference	S, T	vendor supplied	vendor supplied
5.2.2.5.4.4	Susceptibility to FW interference	S, T	vendor supplied	vendor supplied
5.2.2.6	Test networks.	N/A: Header	N/A	N/A
5.2.2.6.1	Test network #1 (20 foot bus).	S, T	vendor supplied	vendor supplied
5.2.2.6.2	Test network #2 (328 foot bus).	S,T	vendor supplied	vendor supplied
5.2.2.6.3	Test network #3.	S, T	vendor supplied	vendor supplied
5.2.2.6.4	Test network #4 (mixed).	S, T	vendor supplied	vendor supplied
5.2.2.7	Terminal diagnostics.	N/A: Header	N/A	N/A
5.2.2.7.1	Digital loop-back.	A, T	vendor supplied	
5.2.2.7.2	Analog loop-back.	A, T	vendor supplied	
5.2.2.7.3	Data logging support.	D	vendor supplied	
5.2.2.8	Redundant data bus.	N/A: Header	N/A	N/A
5.2.2.8.1	Electrical isolation.	T	vendor supplied	vendor supplied
5.2.2.8.2	Dual redundant data bus.	T	vendor supplied	vendor supplied

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TABLE I. Electrical and performance VCRM / POC template – Continued.

Section reference	Performance Heading	Verification Method	POC Report Reference	
			Transformer Coupled	Direct Coupled
5.2.2.9	Multi-drop support.	A	vendor supplied	vendor supplied
5.2.2.10	Multi-drop communication.	A,T	vendor supplied	vendor supplied
5.2.2.11	MCW interference into TW.	S,T	vendor supplied	vendor supplied
5.2.3	Filtered waveform requirements.	N/A: Header	N/A	N/A
5.2.3.1	Power spectral density.	A, T	vendor supplied	vendor supplied
5.2.3.2	MCW interference into FW.	S, T	vendor supplied	vendor supplied

4.1.2 Interoperability testing.

In order to prove interoperability between terminals, it is in the Government's interest that a vendor's extended 1553 capability may have to be proven interoperable with equipment known to meet the requirements of this standard i.e. a reference terminal.

To this end, the custodian agency reserves the right to maintain a set of interoperability tests in addition to the electrical and performance verification tests contained herein. Vendor test data would need to be submitted to the custodian agency in order to prove interoperability by conformance with these tests. Furthermore, the custodian agency may require samples of the vendor's terminal to be submitted in order to confirm the interoperability with a reference terminal.

4.2 MIL-STD-1553B up to and including notice 4.

The extended capabilities of this standard are cognizant of and backwards compatible with the requirements of MIL-STD-1553B up to and including notice 4. Trapezoidal waveform (TW) contained in 5.5 is identical to MIL-STD-1553B up to and including notice 4.

For reference, all subsections in 5.5 correspond to the equivalent section in MIL-STD-1553B up to and including notice 4 and all figures, tables, references and formats are unchanged as explained in 5.5.

4.3 Extended 1553 referenced to OSI layers.

Trapezoidal waveform requirements contained in 5.5 can be hypothetically mapped to the corresponding physical layer and media access control sublayer of the OSI model. Similarly, MCW protocols are also implemented based primarily on the PHY, MAC and PAL layers of the OSI model.

Standardized communication based on the OSI basic reference model as contained in the document is to be adopted as the basis of extended 1553 communications between vendor implementations.

The following Open System Interconnection specifications are provided for background and information:

4.3.1 OSI Specifications.

ISO/IEC 10022:1996 Open Systems Interconnection Physical Service Definition.

ISO/IEC 2382-26:1996 Information technology — Vocabulary — Part 26: Open Systems Interconnection.

ISO/IEC 8348:2002 Information technology — Open Systems Interconnection — Network service definition.

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ISO/IEC 10731:1994 Information technology — Open Systems Interconnection — Basic Reference Model — Conventions for the definition of OSI services.

ISO/IEC 8886: 1996 Information technology — Open Systems Interconnection — Data Link Service Definition.

ISO/IEC 10028:1993 Information technology — Telecommunications and information exchange between systems — Definition of the relaying functions of a Network layer intermediate system.

Copies of these documents are available online at: International Organization for Standardization: http://www.iso.org/iso/en/prods-services/ISOstore/store.html or from
International Organization for Standardization (ISO)
1, rue de Varembé, Case postale 56
CH-1211 Geneva 20, Switzerland.

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5. DETAILED REQUIREMENTS

5.1 Extended 1533 terminal communication.

An extended 1553 terminal shall be capable of communication using:

a. Stand alone multi-carrier waveform (MCW).

b. Stand alone filtered waveform (FW)/ trapezoidal waveform (TW).

c. MCW and FW/TW communications simultaneously.

For FW/TW communications, the terminal shall:

a. Transmit FW but not TW.

b. Be capable of receiving FW and TW.

FW and TW shall be separate and independent from MCW, including media access functions that pertain to MCW. MCW functions shall not be initialized, set up or controlled through FW or TW.

5.2 Electrical and performance requirements.

The electrical and performance requirements in this section shall be subject to verification as described in 5.3. Each requirement in this section corresponds to its associated test referenced in 5.3.

5.2.1 General.

The performance requirements stated below shall be satisfied over the full range of environmental conditions that the system is required to operate as required by the procuring agency.

5.2.2 MCW requirements.

Unless otherwise indicated, the requirements stated below apply to terminals with both transformer coupled and direct coupled stubs. The load impedance into which output signal characteristics are specified is 75 Ω for transformer coupled stubs and 35 Ω for direct coupled stubs.

5.2.2.1 Reserved.

5.2.2.2 Reserved.

5.2.2.3 Reserved.

5.2.2.4 Output characteristics.

5.2.2.4.1 Power spectral density.

The maximum output PSD at the terminal bus interface within the frequency band configured for communication shall be -77 to -71 dBm/Hz for transformer coupled stubs, and -85 to -79 dBm/Hz for direct coupled stubs. This requirement will not preclude a vendor from configuring a higher in-band transmit level for a particular application.

The PSD of the spurious emissions transmitted outside the band should be no greater than as referenced in TABLE II.

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TABLE II. MCW out-of-band emissions.

Frequency Band	Transformer Coupled Stub	Direct Coupled Stub
0 to 2MHz, and ≥ 45MHz	-134 dBm/Hz	-142 dBm/Hz
40 to 43 MHz	-114 dBm/Hz	-122 dBm/Hz

The transmitted PSDs for transformer and direct coupled stubs are illustrated in FIGURE 4 and FIGURE 5 respectively (assuming the whole 5 to 35 MHz frequency band configured for communication).

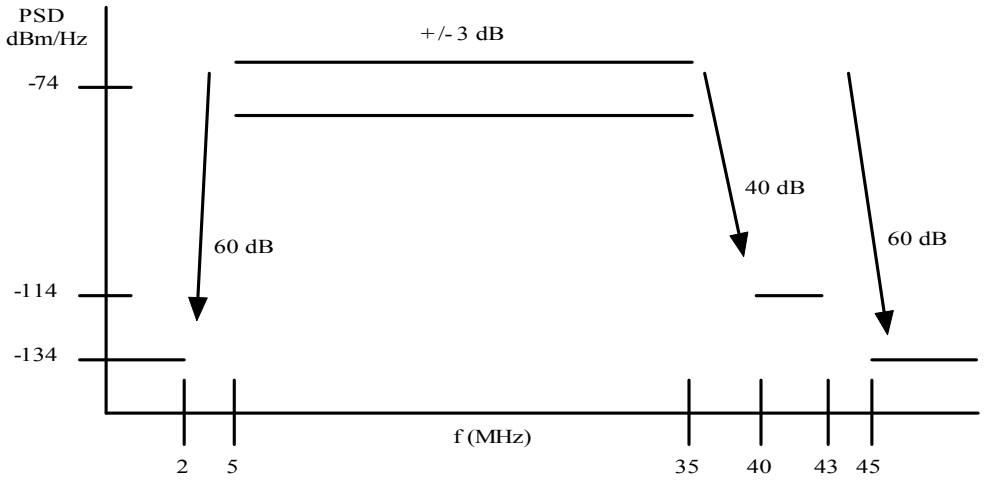


FIGURE 4. MCW transmitted PSD (transformer coupled stubs).

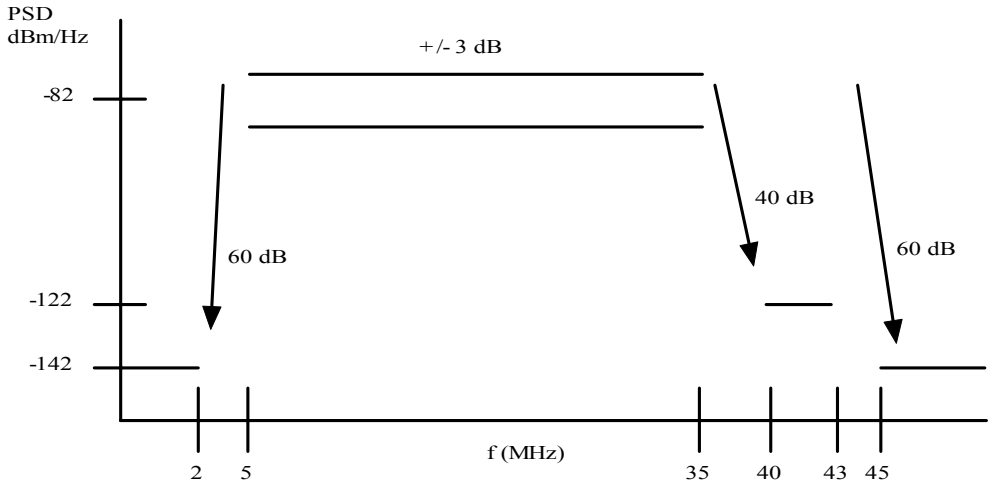


FIGURE 5. MCW transmitted PSD (direct coupled stubs).

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.1.

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5.2.2.4.2 Transmit spectral flatness.

The difference between the maximum and minimum transmit PSD levels within the frequency band configured for communication shall not exceed 4 dB for basic rate and 6 dB for variable bit rate.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.2.

5.2.2.4.3 Output signal fidelity.

Over the 5 to 35 MHz frequency band, the transmitted signal shall have an average signal to noise and distortion ratio of at least 45 dB. The average is defined as the algebraic mean of the per-tone SINAD values (expressed in dB), as defined by the equation below, where N is the number of tones in the 5 to 35 MHz frequency band:

$$\overline{SINAD}\,[dB] = \frac{1}{N} \cdot \sum_{k=1}^{N} SINAD_k\ [dB]$$

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.3.

5.2.2.4.4 Output noise.

When powered down or when receiving, the output noise at the terminal bus interface shall not exceed:

a. -157 dBm/Hz over the 5 to 40 MHz frequency band.

b. 14.0 mV_{rms} for transformer coupled stubs and 5.0 mV_{rms} for direct coupled stubs over the 0 to 5 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.4.

5.2.2.4.5 Output transients.

From the time beginning 7µs after the transmission of the last sample of a PHY frame, the maximum voltage shall be no greater than ± 500 µV peak.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.5.

5.2.2.4.6 BC clock accuracy.

The clock frequency of the bus controller shall be 80 MHz ± 50 ppm. Clock jitter of the bus controller shall be such that the transmitted signal satisfies the requirement in 5.2.2.4.3. The bus controller shall be the source of the system clock for the network.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.6.

5.2.2.4.7 RT clock accuracy.

All terminals shall synchronize their local clock to the system clock of the network which is derived from the clock of the bus controller. The clock frequency and jitter of the remote terminal shall be such that, over the 5 to 35 MHz frequency band, the transmitted signal has an average signal to noise and distortion ratio of at least 45 dB when measured against the clock of the bus controller. The average SINAD is defined as in 5.2.2.4.3.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.7.

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5.2.2.4.8 Output impedance.

The magnitude of the output impedance of a terminal in the 75 kHz to 1 MHz frequency range shall be a minimum of 1 kΩ for transformer coupled stubs, and 2 kΩ for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.8.

5.2.2.4.9 Output timing accuracy.

Within a compound transaction, the timing of the transmitted signal at the bus interface of the transmitting terminal shall be within ± 500 ns of the nominal transmit time. The nominal transmit time shall be derived based on the bus schedule contained within the bus command frame as referenced in FIGURE 6.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.4.9.

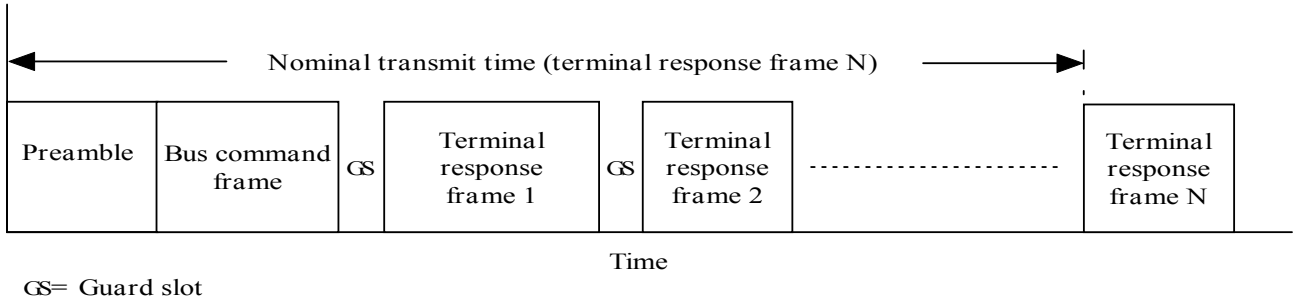


FIGURE 6. Terminal response frame nominal transmit frame

5.2.2.5 Input characteristics.

5.2.2.5.1 Input signal compatibility.

The terminal shall be capable of receiving and operating with the incoming MCW signals specified herein and shall accept valid MCW signals as specified in 5.4. The terminal shall respond to an input signal whose level is within the range of 500 μV_{rms} to 110 mV_{rms} for transformer coupled stubs, and of 500 μV_{rms} to 150 mV_{rms} for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.1.

5.2.2.5.2 Common mode rejection.

The receiver shall have a common mode rejection of at least 30 dB over the 5 to 40 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.2.

5.2.2.5.3 Input impedance.

The magnitude of the input impedance of a terminal in the 75 kHz to 1 MHz frequency range shall be a minimum of 1 kΩ for transformer coupled stubs, and 2 kΩ for direct coupled stubs, in both powered off and receive configurations.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.3.

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5.2.2.5.4 Susceptibility to noise and interference.

The terminal shall exhibit a performance specified as maximum error rate and minimum throughput as stated in the impairment tests described below.

5.2.2.5.4.1 Susceptibility to additive white gaussian noise (AWGN).

The terminal shall exhibit a maximum BER of 10^{-12} at the throughputs specified in TABLE III, with a 15% maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency band when operating in the presence of AWGN at the identified SNR values.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.4.1.

TABLE III. AWGN susceptibility.

SNR [dB]	≥ 50	35	23
Throughput [Mbps]	200	145	65

5.2.2.5.4.2 Susceptibility to a single narrow-band interference source.

The terminal shall exhibit a maximum BER of 10^{-12} and a throughput of 70 Mbps, with a 15 percent maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency band when operating in the presence of a sinusoidal in-band interference source that is 38 dB higher than the per tone power of the received MCW signal.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.4.2.

5.2.2.5.4.3 Susceptibility to TW interference.

The terminal shall exhibit a maximum BER 10^{-12} and a throughput of 200 Mbps, with a 15 percent maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency when operating in the presence of a TW interference. The level of the interference shall be 14 V peak-to-peak for transformer coupled stubs, and 9V peak-to-peak for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.4.3.

5.2.2.5.4.4 Susceptibility to FW interference.

The terminal shall exhibit a maximum BER 10^{-12} and a throughput of 200 Mbps, with a 15 percent maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency band when operating in the presence of a FW interference. The level of the interference shall be 27 V peak-to-peak for transformer coupled stubs, and 9 V peak-to-peak for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.5.4.4.

5.2.2.6 Test networks.

MCW performance shall also be assessed using the test networks described below.

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5.2.2.6.1 Test network #1 (20 foot bus).

This network consists of a 20 foot bus connecting 2 terminals that utilize single stub couplers. The length of each stub is 20 feet for the transformer coupled bus and 2 feet for the direct coupled bus. The performance baseline for this configuration shall be a BER of 10^{-12} and a throughput of 200 Mbps, with a 15 percent maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.6.1.

5.2.2.6.2 Test network #2 (328 foot bus).

This network consists of a 328 foot bus connecting 2 terminals that utilize single stub couplers. The length of each stub is 20 feet for the transformer coupled bus and 2 feet for the direct coupled bus. The performance baseline for this configuration shall be a BER of 10^{-12} and a throughput of 200 Mbps, with a 15 percent maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.6.2.

5.2.2.6.3 Test network #3.

This network consists of a 300 foot bus between End A and End B with 30 resistor terminated stubs comprised of both single-stub and multi-stub couplers. The spacing between the couplers varies from 5 to 70 feet, with the length of each stub being 20 feet for the transformer coupled bus and 2 feet for the direct coupled bus. Two terminals shall be capable of communicating using MCW signaling between the two ends of the network.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.6.3.

5.2.2.6.4 Test network #4 (mixed).

This network is identical to test network #1 except that one of the stubs is transformer coupled (20 foot stub) and the other is direct coupled (2 foot stub). The performance baseline for this configuration shall be a BER of 10^{-12} and a throughput of 200 Mbps, with a 15 percent maximum allowance to account for implementation variations, over the 5 to 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.6.4.

5.2.2.7 Terminal diagnostics.

Terminal diagnostics are applicable to terminals with both transformer and direct coupled stubs. 95% of terminal faults shall be detectable by the self test as initiated by the bus controller. The RT shall neither respond to commands nor communicate while the internal self test is being conducted.

5.2.2.7.1 Digital loop-back.

The terminal shall be capable of initiating a digital loop-back test as installed in the LRU/WRA under operational diagnostic conditions. The digital loop-back test shall demonstrate a minimum throughput of 200 Mbps with a BER of 10^{-12} over the 5 to 35 MHz frequency band.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.7.1.

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5.2.2.7.2 Analog loop-back.

The terminal shall be capable of initiating an analog loop-back test and demonstrate a minimum throughput of 200 Mbps with a BER of 10^{-12} over the 5 to 35 MHz frequency band. The analog loop-back test shall be performed at the level of the terminal's bus interface.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.7.2.

5.2.2.7.3 Data logging support.

In order to supplement performance verification and diagnostics, the terminal shall enable statistical data logging and recording capabilities. As a minimum, it shall be possible for the following metrics to be logged and recorded from power up to power down:

a. Number of synchronization failures (tracking failures) per peer terminal.
b. Number of received PHY frames per peer terminal.
c. Number of frame control CRC failures per peer terminal.
d. Number of payload CRC failures per peer terminal.
e. Number of retransmissions per peer terminal.
f. Current throughput per peer terminal based on the most recent tone map.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.7.3.

5.2.2.8 Redundant data bus.

If redundant data buses are used, the following requirements shall apply to those data buses.

5.2.2.8.1 Electrical isolation.

All terminals shall have a minimum of 45 dB isolation between data buses for all frequencies in the frequency range of 75 kHz to 40 MHz.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.8.1.

5.2.2.8.2 Dual redundant data bus.

The terminal shall transmit the same signal on both buses simultaneously. When receiving, the terminal shall receive on one bus at a time as specified by the bus command. Bus command frames (BCF)s shall be received on either bus; the terminal shall autonomously switch buses for BCF reception purposes only should a fault cause BCF reception failure on the current bus.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.8.2.

5.2.2.9 Multi-drop support.

The terminal shall be capable of supporting communication with 62 other terminals on the bus using MCW signaling and the command-response protocol.

Verification shall be in accordance with the methodology shown in TABLE I.

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5.2.2.10 Multi-drop communication

In a multidrop configuration, the terminal shall be capable of sending and receiving messages in any and all of the 16 possible time slots of a compound transaction without affecting other messages in the same compound transaction.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.10.

5.2.2.11 MCW interference into TW.

A TW terminal shall exhibit a maximum word error rate of 10^{-7}on all words received by the terminal, when operating in the presence of MCW interference of 110 mV_{rms} for transformer coupled stubs, and 150 mV_{rms} for direct coupled stubs. The MCW interference shall occupy the 5 to35 MHz frequency band. The word error rate shall be measured with a 2.1 V peak-to-peak TW input signal for transformer coupled stubs, and a 3 V peak-to-peak TW input signal for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.2.11.

5.2.3 Filtered waveform requirements.

FW requirements shall be identical to those of TW contained in 5.5 with the additional requirements specified in this section.

5.2.3.1 Power spectral density.

The maximum power spectral density up to 1 MHz of the output FW signal shall be -28 dBm/Hz into 75 Ω for transformer coupled stubs, and -34 dBm/Hz into 35 Ω for direct coupled stubs. A terminal shall not generate spurious emissions outside of the 75 kHz to 1 MHz frequency band above the levels specified in TABLE IV. The transmit PSD emission mask is illustrated in FIGURE 7.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.3.1.

TABLE IV. FW out-of-band emissions.

Frequency	0-3 MHz	5 MHz	≥ 15 MHz
PSD for Transformer Coupled Stubs	-28 dBm/Hz	-86 dBm/Hz	-140 dBm/Hz
PSD for Direct Coupled Stubs	-34 dBm/Hz	-92 dBm/Hz	-146 dBm/Hz

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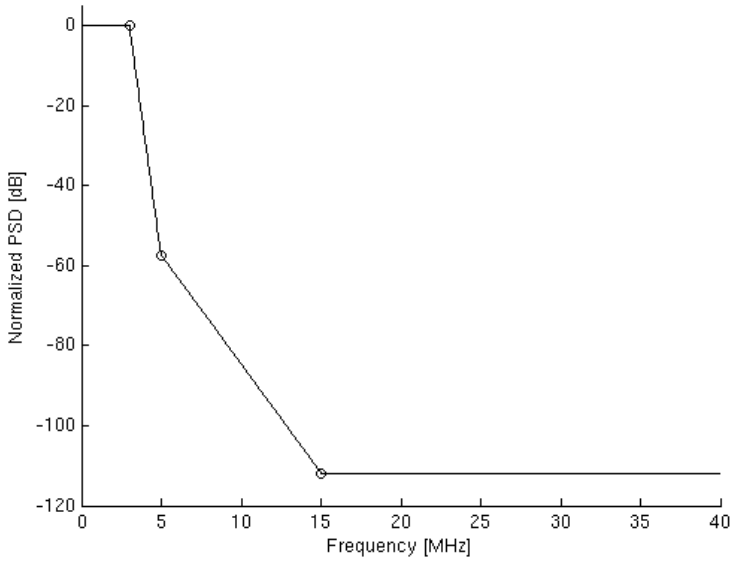


FIGURE 7. FW PSD emission mask (normalized)

5.2.3.2 MCW interference into FW.

The terminal shall exhibit a maximum word error rate of 10^{-7}on all words received by the terminal, when operating in the presence of MCW interference (at a power level of 110 mVrms for transformer coupled stubs and 150 mV rms for direct coupled stubs over the 5 to 35 MHz frequency band). The word error rate shall be measured with a 2.1V peak-to-peak FW input signal for transformer coupled stubs, and a 3 V peak-to-peak FW input signal for direct coupled stubs.

Verification shall be in accordance with the methodologies shown in TABLE I and the test configuration and test procedure contained in 5.3.3.2.

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5.3 Verification of electrical and performance requirements.

5.3.1 General.

5.3.1.1 Proof of compliance (POC) reports.

For each requirement, as indicated in TABLE I, POC reports shall include as a minimum the following:

a. Analysis and simulation data as required.

b. Formal test reports that provide detailed accounts of actual test configurations, specific test equipment and procedures in accordance with the generic configurations and test equipment and procedures contained in 5.3.2.

c. Data presentation as referenced in each test.

The formal test reports referenced in b. shall include an analysis that shows equivalence between the actual test equipment, configurations and procedures used and the generic test configurations and procedures as contained herein. Where more than one UUT is required for a test, they shall each be of the same part number.

5.3.1.2 Verification methods.

Verification methods are defined as follows:

a. Analysis (A): provides a proof that a requirement can be met prior to an actual physical simulation or test.

b. Simulation (S): provides proof that a requirement can be physically met prior to build and test. Simulation may be used in lieu of a test where it is impractical to conduct a physical test. Presentation of simulation results shall include proof that the simulation conditions and the obtained results correspond to what is to be expected had a test using real hardware been conducted.

c. Demonstration (D): provides proof that a requirement can be met through the observation of a suitable event(s) that does not entail parametric measurements.

d. Test (T): involves parametric measurement within stated limits generally involving a test configuration and specific test equipment.

Analysis and simulation as referenced in a. and b. shall provide proofs that include consideration of the wide ranging environmental conditions such as temperature, EMI/EMC and other conditions that the UUT is required to operate in as required by the procuring agency.

Testing shall be conducted under the environmental conditions over which UUTs are required to operate, and shall be performed on UUTs of the same configuration to which simulation and analysis were conducted prior to testing.

5.3.1.3 Verification compliance reference matrix (VCRM).

TABLE I provides the verification methodology for the electrical and performance requirements of 5.2. The table also serves as a template for which vendor POC documentation is referenced for each requirement.

All of the verification methods contained on each line of the table are to be conducted with associated POC documentation for transformer and direct coupled stubs as indicated. Requirements with transformer and direct coupling that are straddled are independent of bus coupling and may be verified for either configuration.

The following examples provide an interpretation to the table.

a. | 5.2.2.4.2 | Transmit spectral flatness. | S, T | means that verification for the requirement shall be via both simulation and test for transformer and direct coupled stubs.

b. | 5.2.2.7.1 | Digital loop-back. | A, T | means that verification for the requirement shall be via both analysis and test (transformer or direct coupled stubs are irrelevant).

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5.3.2 Compliance with MCW requirements.

Unless otherwise indicated, the tests described below apply to terminals with both transformer coupled and direct coupled stubs.

A test environment shall be provided which ensures that interference from external ambient noise sources not included as part of the test are eliminated or minimized.

5.3.2.1 Characteristics of reference couplers and reference cables.

Throughput and BER performance for the susceptibility to noise and interference tests in 5.2.2.5.4 and over the test networks described in 5.2.2.6 is to be based on the following component characteristics:

a. Reference cable: characteristic impedance Z0 of 75 Ω ±5%, maximum attenuation of 2.5 dB / 100 ft at 5 MHz, and maximum attenuation of 7.5 dB / 100 ft at 40 MHz.

b. Reference coupler: isolation resistors of 0.75·Z0 ±2%; wide-band transformer with a turns ratio of 1.41:1, insertion loss of no more than 2 dB anywhere across the 5 to 35 MHz frequency band.

5.3.2.2 Test signals.

Test signals used for performance testing shall be as described below. All tones within the stated frequency range shall be transmitted.

5.3.2.2.1 Test signal #1 (basic rate transmit compound transaction).

This signal consists of an MCW compound transaction at 80 MHz sample rate occupying the 5 to 35 MHz frequency band. The transmission shall contain a bus command frame not requesting acknowledgement followed by a maximum length basic rate (BR) terminal response frame (TRF).

5.3.2.2.2 Test signal #2 (training transmit compound transaction).

This signal consists of an MCW compound transaction at 80 MHz / sample occupying the 5 to 35 MHz frequency band. The transmission shall contain a bus command frame followed by a training frame.

5.3.2.2.3 Test signal #3 (training request compound transaction).

This signal consists of an MCW compound transaction at 80 MHz / sample occupying the 5 to 35 MHz frequency band. The transmission shall contain a bus command frame requesting a training frame to be transmitted by the UUT and sufficient time to accommodate the length of the training frame.

5.3.2.2.4 Test signal #4 (basic rate transmit compound transaction with TAF).

This signal consists of an MCW compound transaction at 80 MHz / sample occupying the 5 to 35 MHz frequency band. The transmission shall contain a bus command frame requesting an acknowledgement and disabling retries, followed by a BR TRF transmitted by the bus controller and sufficient time to accommodate the TAF. The test signal has two variants, bus A and bus B, depending on which bus the RT is to receive on (determined by the bus command).

5.3.2.2.5 Test signal #5.

This signal consists of an MCW compound transaction at 80 MHz / sample occupying the 5 to 35 MHz frequency band. The transmission shall contain a bus command frame followed by enough time to accommodate two TRFs and two TAFs. The bus command shall be configured for the following compound transaction to take place:

a. RT #1 transmits a basic rate TRF to RT #2.

b. RT #2 transmits a basic rate TRF to RT #1.

c. RT #2 returns TAF.

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d. RT #1 returns TAF.

5.3.2.3 Joint throughput and error rate measurement.

Throughput measurements are used to determine the PHY layer information rate over a communication link, while error rate measurements are used to determine the corresponding bit error rate (BER). For verification purposes, BER may be equated to a verifiable frame error rate (FER) with an identifiable number of bits per frame. The FER to be measured shall be derived from the target BER using the equation:

$$FER = 1 - (1 - BER)^{BitsPerFrame}$$

A frame error shall include any fault which causes an error indication to be set in a terminal or, which causes a terminal not to respond to a valid command.

Regardless of whether BER or FER is measured, the test shall be run continuously until the specified performance metrics are achieved or the test fails. The following procedure describes the joint throughput and error rate measurement:

a. Set up the network over which the BER must be measured, connecting a UUT (UUT-1) to one end of the link, and a second UUT (UUT-2) to the other.

b. Configure UUT-1 as bus controller and the UUT-2 as RT.

c. If measuring BER directly, connect a BER counter between the Data In port of the bus controller and the Data Out port of the RT.

d. Configure the bus controller to initiate communication over the bus using a schedule that incorporates the following compound transactions:

   i. A periodically repeated training compound transaction, as specified by test signal #2, transmitted by the bus controller.

   ii. A periodically repeated tone map return compound transaction, containing a bus command frame followed by a tone map message originating from the RT. The training and the tone map return compound transactions should be spaced by an amount of time that is sufficient for the RT to calculate the tone map.

   iii. A repeated sequence of data carrying compound transactions, each containing a bus command frame (requesting acknowledgement) followed by a terminal response frame from bus controller to RT and the terminal acknowledge frame (TAF). Modulation of the transmitted TRFs shall be based on the tone map received by the bus controller during the most recent tone map return compound transaction. The transmitted data bits within the data carrying compound transactions are generated based on a rolling pattern of random bits.

e. If measuring BER directly, read off the bit error rate indication of the counter. If measuring FER, read off the corresponding error count via the modem management software running on the RT and convert the result to BER using the formula above.

f. Retrieve via the modem management software running on the bus controller the PHY information rate reported for the communication link from bus controller to RT.

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5.3.2.4 Output characteristics.

5.3.2.4.1 Power spectral density.

The maximum output PSD shall be measured based on the following test configuration and procedure.

5.3.2.4.1.1 Test objectives.

Refer to the requirements described in 5.2.2.4.1 on page 18.

5.3.2.4.1.2 Test equipment.

The following test equipment shall be used:

a. Spectrum analyzer.

b. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

c. Programmable 50 Ω attenuator.

5.3.2.4.1.3 Test configuration.

Refer to the test configuration in FIGURE 8.

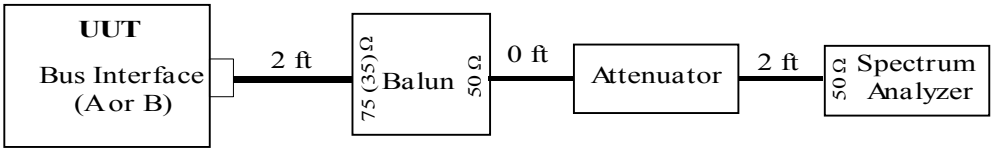


FIGURE 8. Test configuration 1.

5.3.2.4.1.4 Test procedure.

Step 1: Configure the UUT as bus controller (BC).

Step 2: Set the attenuator to prevent overloading the spectrum analyzer front-end when subjected to the full bandwidth MCW signal.

Step 3: Set up the test configuration using the bus A interface.

Step 4: Configure the UUT to transmit compound transactions based on test signal #1, using a rolling data pattern to modulate the terminal response frames.

Step 5: Set the spectrum analyzer to sweep the 0 to 35 MHz frequency band.

Step 6: Set the spectrum analyzer to measure peak power in a 9 kHz resolution bandwidth (dBm/9kHz) and use a linear display mode.

Step 7: Record the frequency at which the highest power is observed.

Step 8: Set the spectrum analyzer to measure quasi-peak power in a 9 kHz resolution bandwidth (dBm/9kHz). Set the center frequency to the one recorded in step 7. Set the span to 200 kHz.

Step 9: Record the maximum quasi-peak power, corrected for the attenuation level of the attenuator and for the insertion loss of the balun.

Step 10: Repeat steps 5 to 9 for the following frequency bands: 0 to 2 MHz, 40 to 43 MHz, 45 to 100 MHz.

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Step 11: Repeat steps 3 to 10 using the bus B interface.

Step 12: Add a 1.05 dB envelope detector correction factor to the recorded maximum quasi-peak powers.

Step 13: Determine the spectrum analyzer's equivalent noise power bandwidth for the 9 kHz filter by measuring the analyzer filter's peak, 3 dB, 6 dB, and 20 dB points and constructing a piecewise linear model of the filter. The equivalent noise power bandwidth is the bandwidth of a rectangular filter whose area is equivalent to that of the piecewise linear model.

Step 14: Calculate and record the maximum power spectral density for the unit under test by taking the values obtained in step 12 and subtracting 10 log 10 (equivalent noise power bandwidth/1Hz).

5.3.2.4.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded maximum quasi-peak power values.

c. The calculated maximum power spectral density levels.

d. Indication of compliance with test objectives.

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5.3.2.4.2 Transmit spectral flatness.

The spectral flatness of the transmitted signal shall be measured based on the following test configuration and test procedure.

5.3.2.4.2.1 Test objectives.

Refer to the requirements described in 5.2.2.4.2 on page 20.

5.3.2.4.2.2 Test equipment.

Refer to the test equipment for the PSD test in 5.3.2.4.1.2 on page 30.

5.3.2.4.2.3 Test configuration.

Refer to the test configuration described in 5.3.2.4.1.3 on page 30.

5.3.2.4.2.4 Test procedure.

Step 1: Carry out steps 1 to 9 of the test procedure for PSD in 5.3.2.4.1.4 on page 30.

Step 2: Set the spectrum analyzer to measure peak power in a 9 kHz resolution bandwidth (dBm/9kHz) and use a linear display mode.

Step 3: Record the frequency at which the lowest power is observed.

Step 4: Set the spectrum analyzer to measure quasi-peak power in a 9 kHz resolution bandwidth (dBm/9kHz). Set the center frequency to the one recorded in step 3. Set the span to 200 kHz.

Step 5: Record the minimum quasi-peak power, corrected for the attenuation level of the attenuator and for the insertion loss of the balun.

Step 6: Repeat steps 1 to 5 using the bus B interface.

Step 7: Repeat steps 1 to 6 using test signal #2 instead of test signal #1.

Step 8: Add a 1.05 dB envelope detector correction factor to the recorded maximum and minimum quasi-peak powers.

Step 9: Determine the spectrum analyzer's equivalent noise power bandwidth for the 9 kHz filter by measuring the analyzer filter's peak, 3 dB, 6 dB, and 20 dB points and constructing a piecewise linear model of the filter. The equivalent noise power bandwidth is the bandwidth of a rectangular filter whose area is equivalent to that of the piecewise linear model.

Step 10: Calculate and record the maximum and minimum power spectral density for the unit under test by taking the values obtained in step 8 and subtracting 10 log 10 (equivalent noise power bandwidth/1Hz).

5.3.2.4.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded maximum and minimum quasi-peak power values.

c. The calculated maximum and minimum power spectral density levels.

d. Indication of compliance with test objectives.

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5.3.2.4.3 Output signal fidelity.

The output signal fidelity shall be measured based on the following test configuration and test procedure.

5.3.2.4.3.1 Test objectives.

Refer to the requirements described in 5.2.2.4.3 on page 20.

5.3.2.4.3.2 Test equipment.

The following test equipment shall be used:

a. Data acquisition system with 50 Ω input impedance and -140 dBm/Hz maximum noise floor.
b. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.
c. Off-line digital signal processing capability.

5.3.2.4.3.3 Test configuration.

Refer to the test configuration in FIGURE 9.

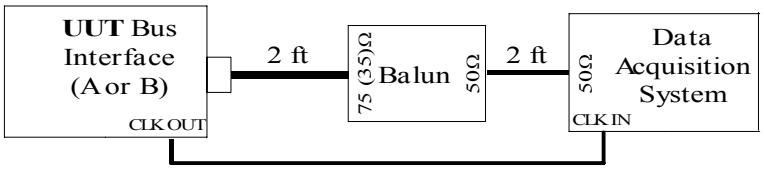


FIGURE 9. Test configuration 2.

5.3.2.4.3.4 Test procedure.

Step 1: Configure the UUT as bus controller.

Step 2: Set up the test configuration using the bus A interface.

Step 3: Configure the UUT to continuously transmit test signal #2.

Step 4: Capture at least one full compound transaction.

Step 5: Determine, using off-line digital signal processing and based on the training frame portion of the captured signal, the SINAD of all transmitted tones. Then, calculate the average according to the relation in 5.2.2.4.3.

Step 6: Repeat steps 2 to 5 using the bus B interface.

5.3.2.4.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. Graphical data showing the determined signal to noise and distortion ratio versus frequency.
c. The calculated average SINAD.
d. Indication of compliance with test objectives.

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5.3.2.4.4 Output noise.

The output noise shall be measured based on the following test configuration and procedure.

5.3.2.4.4.1 Test objectives.

Refer to the requirements described in 5.2.2.4.4 on page 20.

5.3.2.4.4.2 Test equipment.

The following test equipment shall be used:

a. Spectrum analyzer.
b. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.
c. Calibrated low-pass filter/pre-amplifier with a 0 to 5 MHz pass-band.
d. Band-pass filter with a 5 to 35 MHz pass-band.
e. True RMS voltmeter.

5.3.2.4.4.3 Test configuration.

Refer to the test configuration in FIGURE 10.

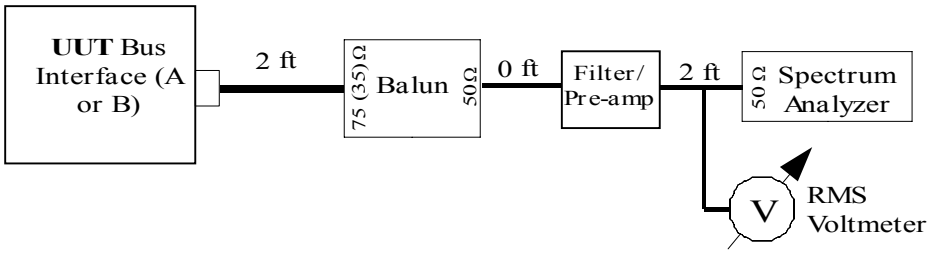


FIGURE 10. Test configuration 3.

5.3.2.4.4.4 Test procedure.

Step 1: Power off the UUT.

Step 2: Set up the test configuration using the bus A interface and the band-pass filter.

Step 3: Measure and record the PSD level in the 5 to 35 MHz frequency band, compensating for the insertion loss of the balun and filter.

Step 4: Repeat steps 1-3 using the bus B interface.

Step 5: Power on the UUT, place it into receive mode and repeat steps 2 to 4.

Step 6: Power off the UUT.

Step 7: Set up the test configuration using the bus A interface and the low-pass filter.

Step 8: Measure and record the RMS voltage level, referring it back to the terminal bus interface (compensating for the voltage ratio of the balun and the loss across the filter).

Step 9: Repeat step 6-8 using the bus B interface.

Step 10: Power on the UUT, place it into receive mode and repeat steps 7 to 9.

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5.3.2.4.4.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing the measured noise power spectrum density levels versus frequency.

c. The measured RMS noise voltage levels.

d. Indication of compliance with test objectives.

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5.3.2.4.5 Output transients.

The output transient signal level shall be measured based on the following test configuration and procedure.

5.3.2.4.5.1 Test objectives.

Refer to the requirements described in 5.2.2.4.5 on page 20.

5.3.2.4.5.2 Test equipment.

The following test equipment shall be used:

a. Oscilloscope (recording) set to 50 Ω input impedance.

b. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

5.3.2.4.5.3 Test configuration.

Refer to the test configuration in FIGURE 11.

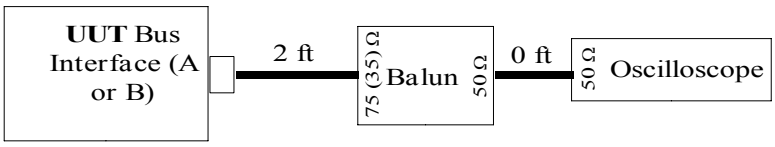


FIGURE 11. Test configuration 4.

5.3.2.4.5.4 Test procedure.

Step 1: Configure the UUT as bus controller.

Step 2: Set up the test configuration using the bus A interface.

Step 3: Configure the UUT to continuously transmit test signal #2.

Step 4: Record the waveform containing the transmission gap between the end of the bus command frame and the start of the training frame.

Step 5: Measure the maximum transient peak voltage level starting at 7 µs after the transmission of the last sample of the bus command frame to the time immediately preceding the start of the subsequent training frame, taking into account the voltage ratio of the balun.

Step 6: Repeat steps 2 to 5 using the bus B interface.

5.3.2.4.5.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded waveforms.

c. The measured maximum transient peak voltage levels.

d. Indication of compliance with test objectives.

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5.3.2.4.6 BC clock accuracy.

The bus controller clock frequency shall be measured based on the following test configuration and test procedure. The clock jitter shall be assessed based on the test procedure in 5.3.2.4.3 on page 33.

5.3.2.4.6.1 Test objectives.

Refer to the requirements described in 5.2.2.4.6 on page 20.

5.3.2.4.6.2 Test equipment.

The following test equipment shall be used:

a. Frequency counter.

b. Terminating resistor RL of 75 Ω ±2% for transformer coupled stubs, and of 35 Ω ±2% for direct coupled stubs.

5.3.2.4.6.3 Test configuration.

Refer to the test configuration in FIGURE 12.

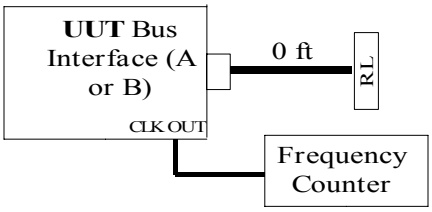


FIGURE 12. Test configuration 5.

5.3.2.4.6.4 Test procedure.

Step 1: Configure the UUT as bus controller.

Step 2: Set up the test configuration by connecting the frequency counter to the clock out test connector of the UUT (or an equivalent test point).

Step 3: Record the clock frequency.

5.3.2.4.6.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The exact measured clock frequency.

c. Indication of compliance with test objectives.

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5.3.2.4.7 RT clock accuracy.

The remote terminal clock accuracy shall be measured based on the following test configuration and test procedure.

5.3.2.4.7.1 Test objectives.

Refer to the requirements described in 5.2.2.4.7 on page 20.

5.3.2.4.7.2 Test equipment.

The following test equipment shall be used:

a. Data acquisition system with 1MΩ input impedance and -140 dBm/Hz maximum noise floor.
b. Programmable test waveform generator.
c. Programmable clock source.
d. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

5.3.2.4.7.3 Test configuration.

Refer to the test configuration in FIGURE 13.

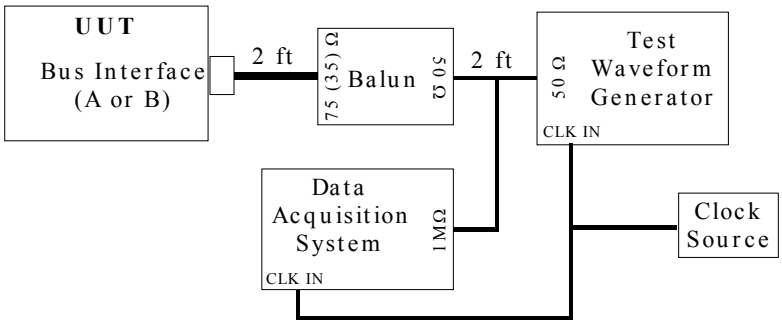


FIGURE 13. Test configuration 6.

5.3.2.4.7.4 Test procedure.

Step 1: Configure the UUT as a RT.

Step 2: Set up the test configuration using either the bus A or bus B interface.

Step 3: Set up the test waveform generator to continuously transmit test signal #3 at a level which does not overload the UUT front-end.

Step 4: Set the programmable clock source to 80 MHz - 50 ppm.

Step 5: Capture at least one full compound transaction.

Step 6: Determine, using off-line digital signal processing and based on the training frame portion of the captured signal, the SINAD of all transmitted tones. Then, calculate the average according to the relation in 5.2.2.4.3 on page 20.

Step 7: Set the programmable clock source to 80 MHz + 50 ppm.

Step 8: Repeat steps 3 to 6.

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5.3.2.4.7.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing the determined signal to noise and distortion ratio versus frequency at both clock frequencies.

c. The calculated average SINAD.

d. Indication of compliance with test objectives.

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5.3.2.4.8 Output impedance.

The output impedance shall be measured based on the following test configuration and test procedure.

5.3.2.4.8.1 Test objectives.

Refer to the requirements described in 5.2.2.4.8 on page 21.

5.3.2.4.8.2 Test equipment.

The following test equipment shall be used:

a. Vector network analyzer.
b. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

5.3.2.4.8.3 Test configuration.

Refer to the test configuration in FIGURE 14.

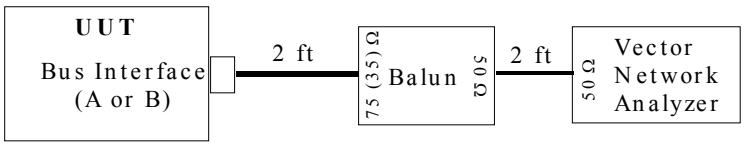


FIGURE 14. Test configuration 7.

5.3.2.4.8.4 Test procedure.

Step 1: Perform a one-port calibration of the network analyzer at the UUT facing port of the balun.

Step 2: Configure the UUT such that the transmit chain is enabled but without transmitting any signal.

Step 3: Set up the test configuration using the bus A interface.

Step 4: Measure the S_{11} reflection coefficient over the 75 kHz to 1 MHz frequency band and record the magnitude of the corresponding output impedance.

Step 5: Repeat steps 3 to 4 using the bus B interface.

5.3.2.4.8.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. Graphical data showing the magnitude of the output impedance versus frequency.
c. Indication of compliance with test objectives.

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5.3.2.4.9 Output timing accuracy.

The timing of the transmitted signal shall be measured based on the following test configuration and test procedure.

5.3.2.4.9.1 Test objectives.

Refer to the requirements described in 5.2.2.4.9 on page 21.

5.3.2.4.9.2 Test equipment.

The following test equipment shall be used:

a. Programmable test waveform generator.
b. Oscilloscope (recording) set to 1MΩ input impedance.
c. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

5.3.2.4.9.3 Test configuration.

Refer to the test configuration in FIGURE 15.

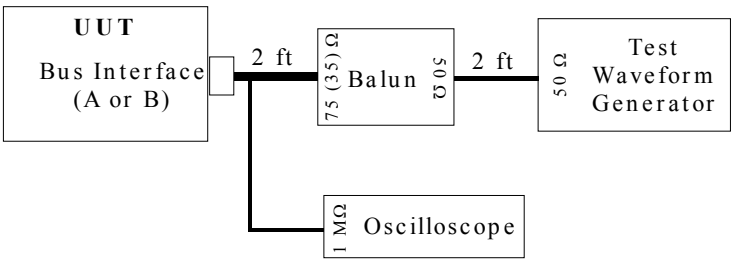


FIGURE 15. Test configuration 8.

5.3.2.4.9.4 Test procedure.

Step 1: Configure the UUT as remote terminal.

Step 2: Set up the test configuration using the bus A interface.

Step 3: Set up the programmable test waveform generator to continuously transmit test signal #3 at a level which does not overload the UUT front-end.

Step 4: Using the oscilloscope, observe and record the signal containing the bus command frame and the start of the terminal response frame.

Step 5: Measure the time from the start of the bus command frame to the start of the terminal response frame.

Step 6: Repeat steps 2 to 5 using the bus B interface.

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5.3.2.4.9.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded waveforms.

c. The nominal transmit time and the measured transmit times.

d. Indication of compliance with test objectives.

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5.3.2.5 Input characteristics.

5.3.2.5.1 Input signal compatibility.

Input signal compatibility shall be verified based on the following test configuration and procedure.

5.3.2.5.1.1 Test objectives.

Refer to the requirements described in 5.2.2.5.1on page 21.

5.3.2.5.1.2 Test equipment.

The following test equipment shall be used:

a. Programmable 50 Ω attenuator.
b. Programmable test waveform generator.
c. Oscilloscope (recording) set to 1 MΩ input impedance.
d. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.
e. True RMS voltmeter.

5.3.2.5.1.3 Test configuration.

Refer to the test configuration in FIGURE 16.

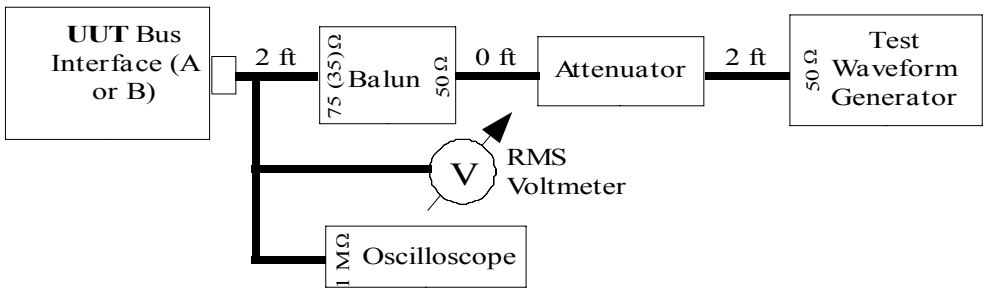


FIGURE 16. Test configuration 9.

5.3.2.5.1.4 Test procedure.

Step 1: Configure the UUT as remote terminal.

Step 2: Set up the test configuration using the bus A interface.

Step 3: Set up the programmable test waveform generator to continuously transmit test signal #1.

Step 4: Adjust the attenuator such that the signal at the UUT is at the highest specified voltage level.

Step 5: Set up the waveform generator to continuously transmit test signal #3.

Step 6: Observe on the oscilloscope the presence of terminal response frames and record the waveform of one compound transaction.

Step 7: Repeat steps 3 to 6 for the lowest specified voltage level.

Step 8: Repeat steps 2 to 7 using the bus B interface.

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5.3.2.5.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded waveforms for the minimum and maximum input signal levels.

c. Indication of compliance with test objectives.

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5.3.2.5.2 Common mode rejection.

The common mode rejection of the receiver shall be measured based on the following test configuration and procedure.

5.3.2.5.2.1 Test objectives.

Refer to the requirements described in 5.2.2.5.2 on page 21.

5.3.2.5.2.2 Test equipment.

The following test equipment shall be used:

a. Programmable test waveform generator.

b. Oscilloscope set to1 MΩ input impedance.

5.3.2.5.2.3 Test configuration.

Refer to the test configuration in FIGURE 17.

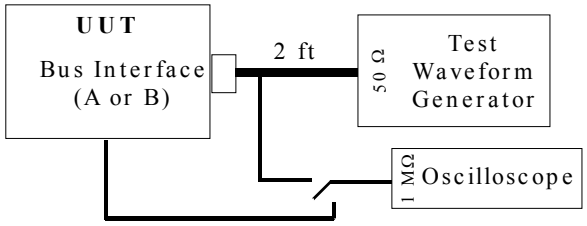


FIGURE 17. Test configuration 10.

5.3.2.5.2.4 Test procedure.

The test shall be performed at the frequencies given in TABLE V.

TABLE V. Common mode rejection test frequencies.

Frequency [MHz]	5	15	25	35	45

Step 1: Configure the UUT for maximum receiver sensitivity.

Step 2: Set up the test configuration using the bus A interface, connecting the signal to the UUT as common mode excitation.

Step 3: Configure the test waveform generator to continuously transmit a sinusoidal signal of frequency chosen from TABLE V. Verify on the oscilloscope the signal injected into the UUT.

Step 4: Connect the oscilloscope to the A/D converter input on the receive path of the UUT.

Step 5: Measure the received differential voltage and calculate the common mode rejection, compensating for the gain of the receive path.

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Step 6: Repeat steps 2 to 5 using the bus B interface.

Step 7: Repeat steps 2 to 6 for the other frequencies listed in TABLE V.

Step 9: Repeat steps 1 to 7 for minimum receiver sensitivity.

5.3.2.5.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing the determined common mode rejections at each test frequency.

c. Indication of compliance with test objectives.

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5.3.2.5.3 Input impedance.

The input impedance shall be measured based on the following test configuration and test procedure.

5.3.2.5.3.1 Test objectives.

Refer to the requirements described in 5.2.2.5.3 on page 21.

5.3.2.5.3.2 Test equipment.

Refer to the test equipment for the output impedance test in 5.3.2.4.8.2 on page 40.

5.3.2.5.3.3 Test configuration.

Refer to the test configuration in FIGURE 14 on page 40.

5.3.2.5.3.4 Test procedure.

Step 1: Perform a one-port calibration of the network analyzer at the UUT facing port of the balun.

Step 2: Configure the UUT to receive only.

Step 3: Set up the test configuration using the bus A interface.

Step 4: Measure the S_{11} reflection coefficient over the 75 kHz to 1 MHz frequency band and record the magnitude of the corresponding input impedance.

Step 5: Repeat steps 3 to 4 using the bus B interface.

Step 6: Power off the UUT and repeat steps 3 to 5.

5.3.2.5.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. Graphical data showing the magnitude of the input impedance versus frequency for both powered off and receive configurations.
c. Indication of compliance with test objectives.

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5.3.2.5.4 Susceptibility to noise and interference.

The noise and interference susceptibility tests shall be run continuously until the specified performance metrics are achieved or the test fails.

5.3.2.5.4.1 Susceptibility to additive white gaussian noise (AWGN).

Throughput and BER performance in the presence of AWGN shall be measured based on the following test configuration and procedure.

5.3.2.5.4.1.1 Test objectives.

Refer to the requirements described in 5.2.2.5.4 on page 22.

5.3.2.5.4.1.2 Test equipment.

The following test equipment shall be used:

a. Reference couplers (2 for transformer and 2 for direct coupled, corresponding to the desired bus coupling).
b. Reference bus segment cables.
c. Terminating resistor RL of 75 Ω ±2%.
d. Calibrated 50 Ω / 75 Ω balun.
e. Programmable 50 Ω attenuator.
f. Programmable test waveform generator.
g. True RMS voltmeter.
h. Oscilloscope set to 1 MΩ input impedance.
i. Low-pass filter with the following characteristics: pass-band stop frequency at 40 MHz, stop-band start frequency at 50 MHz, minimum stop-band attenuation of 60 dB.
i. Error (only if performing direct BER measurement, as described in the joint throughput and error measurement 5.3.2.3).

5.3.2.5.4.1.3 Test configuration.

Refer to the test configuration in FIGURE 18.

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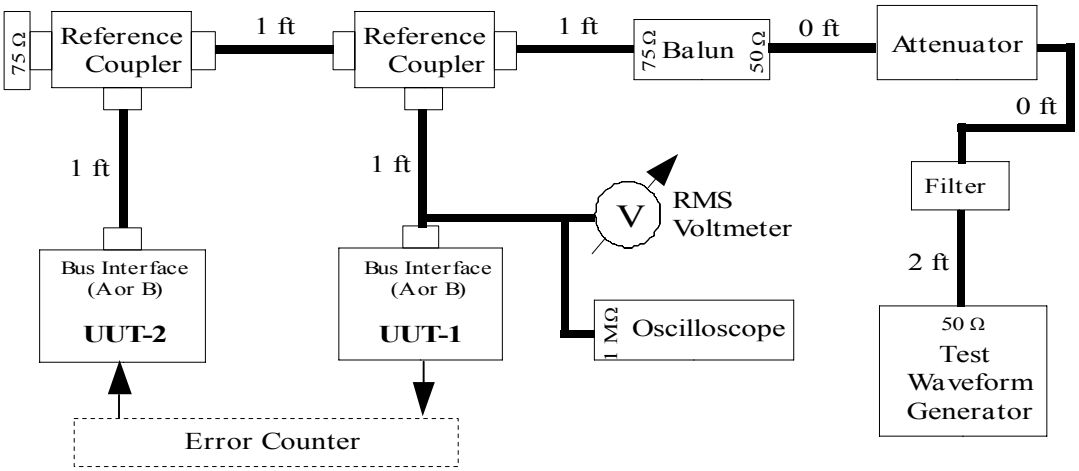


FIGURE 18. Test configuration 11.

5.3.2.5.4.1.4 Test procedure.

Step 1: Configure the UUT-2 as bus controller and UUT-1 as RT.

Step 2: Set up the test configuration using the bus A interface at both terminals.

Step 3: With the test waveform generator and the terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 4: Configure the bus controller to continuously transmit test signal #2 and measure the received RMS signal level with the voltmeter. The bus command shall be such that the UUT is not the intended receiver of the training frame.

Step 5: Reconfigure the bus controller to idle. Set up the test waveform generator to continuously transmit the AWGN.

Step 6: Measure the received RMS noise level with the voltmeter. Adjust the attenuator until the measured RMS value yields an SNR value (with respect to the received signal level measured on step 4) as specified in TABLE III on page 22.

Step 7: Carry out a joint throughput and error rate measurement, as described in 5.3.2.3 on page 29.

Step 8: Record the measured BER and the throughput.

Step 9: Repeat steps 3 to 8 for the other SNR values in TABLE III on page 22.

Step 10: Repeat steps 2 to 9 using the bus B interface.

5.3.2.5.4.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing the measured RMS value of the received MCW signal and of the noise, for each SNR level at which the test was conducted.

c. Tabular data showing the measured throughput and BER value, for each SNR level at which the test was conducted.

d. Indication of compliance with test objectives.

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5.3.2.5.4.2 Susceptibility to a single narrow-band interference source.

Throughput and BER performance in the presence of a single sinusoidal interference source shall be measured based on the following test configuration and procedure.

5.3.2.5.4.2.1 Test objectives.

Refer to the requirements described in 5.2.2.5.4.2 on page 22.

5.3.2.5.4.2.2 Test equipment.

Refer to the test equipment for the AWGN susceptibility test in 5.3.2.5.4.1.2 on page 48.

5.3.2.5.4.2.3 Test configuration.

Refer to the test configuration in FIGURE 18 on page 49.

5.3.2.5.4.2.4 Test procedure.

The test shall be performed at the frequencies given by the following equation

$$f_n = n \cdot 5\,[MHz] + 2.5\,[MHz] + \frac{78.125\,[kHz]}{2}, \quad n = 0 \ldots 7$$

Step 1: Configure the UUT-2 as bus controller and UUT-1 as RT.

Step 2: Set up the test configuration using the bus A interface at both terminals.

Step 3: With the test waveform generator and the terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 4: Configure the bus controller to continuously transmit test signal #2 and measure the received RMS signal level with the voltmeter. The bus command shall be such that the UUT is not the intended receiver of the training frame. Calculate the required interference level as 12 dB higher than the measured RMS level of the received MCW signal. The value of 12 dB is obtained by noting that the per tone power of the received MCW signal is 26 dB below the total MCW power (which is distributed over the 384 tones spanning the 5 to 35 MHz frequency band) and then adding the required 38 dB relative offset.

Step 5: Reconfigure the bus controller to idle. Set up the test waveform generator to continuously transmit the samples of the interference signal of the lowest frequency given above.

Step 6: Measure the received RMS interference level with the voltmeter. Confirm on the oscilloscope the shape of the interference signal. Adjust the attenuator until the measured RMS voltage is at the calculated interference level.

Step 7: Carry out a joint throughput and error rate measurement, as described in 5.3.2.3 on page 29.

Step 8: Record the measured BER and the throughput.

Step 9: Repeat steps 5 to 8 for the other interference signal frequencies.

Step 10: Repeat steps 2 to 9 using the bus B interface.

5.3.2.5.4.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Tabular data showing the measured RMS value of the received MCW signal and of the interference, for each frequency at which the test was conducted.

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c. Tabular data showing the measured throughput and BER value, for each frequency at which the test was conducted.

d. Indication of compliance with test objectives.

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5.3.2.5.4.3 Susceptibility to TW interference.

Throughput and BER performance in the presence of TW interference source shall be measured based on the following test configuration and procedure.

5.3.2.5.4.3.1 Test objectives.

Refer to the requirements described in 5.2.2.5.4.3 on page 22.

5.3.2.5.4.3.2 Test equipment.

The following test equipment shall be used:

a. Reference couplers (2 for transformer and 2 for direct coupled, corresponding to the desired bus coupling).
b. Reference bus segment cables.
c. Terminating resistor RL of 75 Ω ±2%.
d. Programmable 50 Ω wide-band amplifier.
e. Programmable test waveform generator.
f. Oscilloscope set to 1 MΩ input impedance.
g. Calibrated 75 Ω / 50 Ω balun.
h. Low-pass filter with the following characteristics: pass-band stop frequency at 40 MHz, stop-band start frequency at 50 MHz, minimum stop-band attenuation of 60 dB.
ii. Error counter (only if performing direct BER measurement, as described in the joint throughput and error measurement 5.3.2.3).

5.3.2.5.4.3.3 Test configuration.

Refer to the test configuration in FIGURE 19.

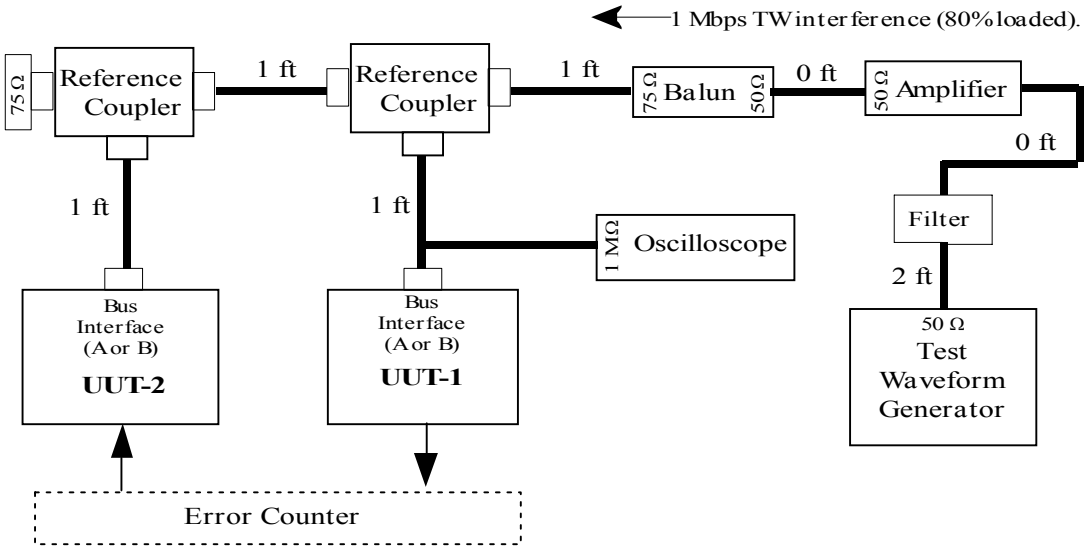


FIGURE 19. Test configuration 12.

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5.3.2.5.4.3.4 Test procedure.

Step 1:Configure the UUT-2 as bus controller and UUT-1 as RT.

Step 2: Set up the test configuration using the bus A interface at both terminals.

Step 3: With the test waveform generator and the terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 4: Set up the test waveform generator to continuously transmit an 80% loaded TW interference signal.

Step 5: Measure the received signal amplitude with the oscilloscope and confirm the shape of the interference signal. Adjust the amplifier until the measured value yields the specified peak-to-peak voltage.

Step 6: Carry out a joint throughput and error rate measurement, as described in 5.3.2.3 on page 29.

Step 7: Record the measured BER and the throughput.

Step 8: Repeat steps 2 to 7 using the bus B interface.

5.3.2.5.4.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The measured throughput and BER value.

c. Indication of compliance with test objectives.

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5.3.2.5.4.4 Susceptibility to FW interference.

Throughput and BER performance in the presence of FW interference source shall be measured based on the following test configuration and procedure.

5.3.2.5.4.4.1 Test objectives.

Refer to the requirements described in 5.2.2.5.4.4 on page 22.

5.3.2.5.4.4.2 Test equipment.

Refer to the test equipment for the TW susceptibility test in 5.3.2.5.4.3.2 on page 52.

5.3.2.5.4.4.3 Test configuration.

Refer to the test configuration in 5.3.2.5.4.3.3 on page 52.

5.3.2.5.4.4.4 Test procedure.

Carry out to the test procedure for the TW susceptibility test in 5.3.2.5.4.3.4 on page 53, replacing the TW interference signal with an FW interference signal.

5.3.2.5.4.4.5 Data presentation.

Refer to the data presentation for the TW susceptibility test in 5.3.2.5.4.3.5 on page 53.

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5.3.2.6 Test networks.

5.3.2.6.1 Test network #1 (20 foot bus).

Throughput and BER performance over test network #1 shall be measured based on the following test configuration and procedure.

5.3.2.6.1.1 Test objectives.

Refer to the requirements described in 5.2.2.6.1on page 23.

5.3.2.6.1.2 Test equipment.

The following test equipment shall be used:

iii. Error counter (only if performing direct BER measurement, as described in the joint throughput and error measurement 5.3.2.3).

a. Reference couplers and reference cable segments for the bus and stubs.

b. 2 bus terminating resistors of 75 Ω ±2%.

5.3.2.6.1.3 Test configuration.

Refer to the test configuration in FIGURE 20 for the transformer coupled bus, and in FIGURE 21 for the direct coupled bus with a UUT connected at either end of the bus.

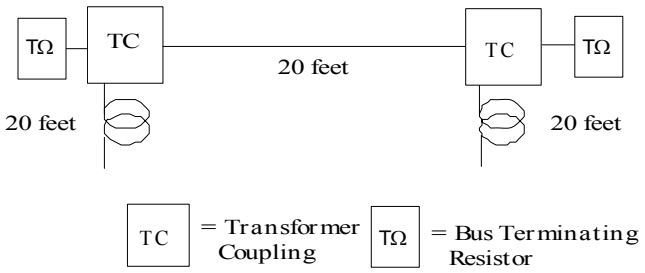


FIGURE 20. Transformer coupled test network #1.

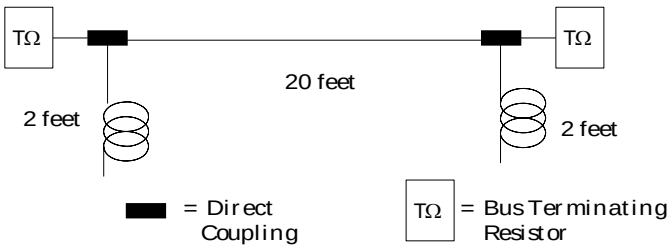


FIGURE 21. Direct coupled test network #1.

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5.3.2.6.1.4 Test procedure.

Step 1: Configure the UUT-2 as bus controller and UUT-1 as RT.

Step 2: Set up the test configuration using the bus A interface at both terminals.

Step 3: Carry out a joint throughput and error rate measurement, as described in 5.3.2.3 on page 29.

Step 4: Record the measured BER and the throughput.

Step 5: Repeat steps 2 to 4 using the bus B interface.

5.3.2.6.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.
b. The measured throughput and BER values.
c. Indication of compliance with test objectives.

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5.3.2.6.2 Test network #2 (328 foot bus).

Throughput and BER performance over test network #2 shall be measured based on the following test configuration and procedure.

5.3.2.6.2.1 Test objectives.

Refer to the requirements described in 5.2.2.6.2 on page 23.

5.3.2.6.2.2 Test equipment.

Refer to the test equipment for test network #1 in 5.3.2.6.1.2 on page 55.

5.3.2.6.2.3 Test configuration.

Refer to the test configuration in FIGURE 22 for the transformer coupled bus, and FIGURE 23 for the direct coupled bus with a UUT connected at either end of the bus. Either the physical test configuration or a network simulator that simulates all physical aspects of the network shall be used.

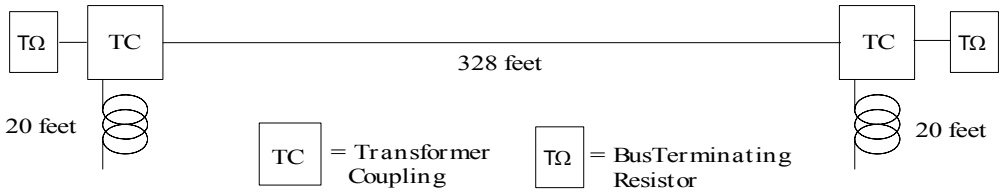


FIGURE 22. Transformer coupled test network #2.

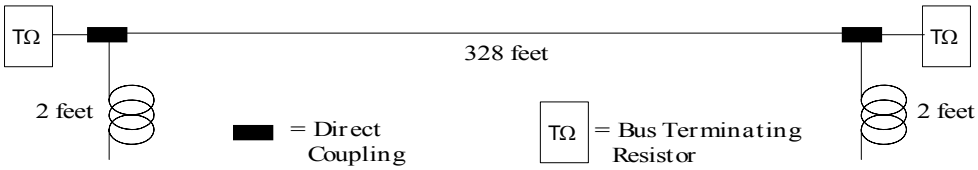


FIGURE 23. Direct coupled test network #2.

5.3.2.6.2.4 Test procedure.

Refer to the test procedure for test network #1 in 5.3.2.6.1.4 on page 56.

5.3.2.6.2.5 Data presentation.

Refer to the data presentation for test network #1 in 5.3.2.6.1.5 on page 56.

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5.3.2.6.3 Test network #3.

Communication over test network #3 shall be verified based on the following test configuration and procedure.

5.3.2.6.3.1 Test objectives.

Refer to the requirements described in 5.2.2.6.3 on page 23.

5.3.2.6.3.2 Test equipment.

The following test equipment shall be used:

a. Reference couplers and reference cable segments for the bus and stubs.

b. 2 bus terminating resistors of 75 Ω ±2%.

c. 30 stub terminating resistors of 1 kΩ ±2% for the transformer coupled bus, and of 2 kΩ ±2% for the direct coupled bus.

5.3.2.6.3.3 Test configuration.

Refer to the test configuration in FIGURE 24 for the transformer coupled bus, and in FIGURE 25 for the direct coupled bus.

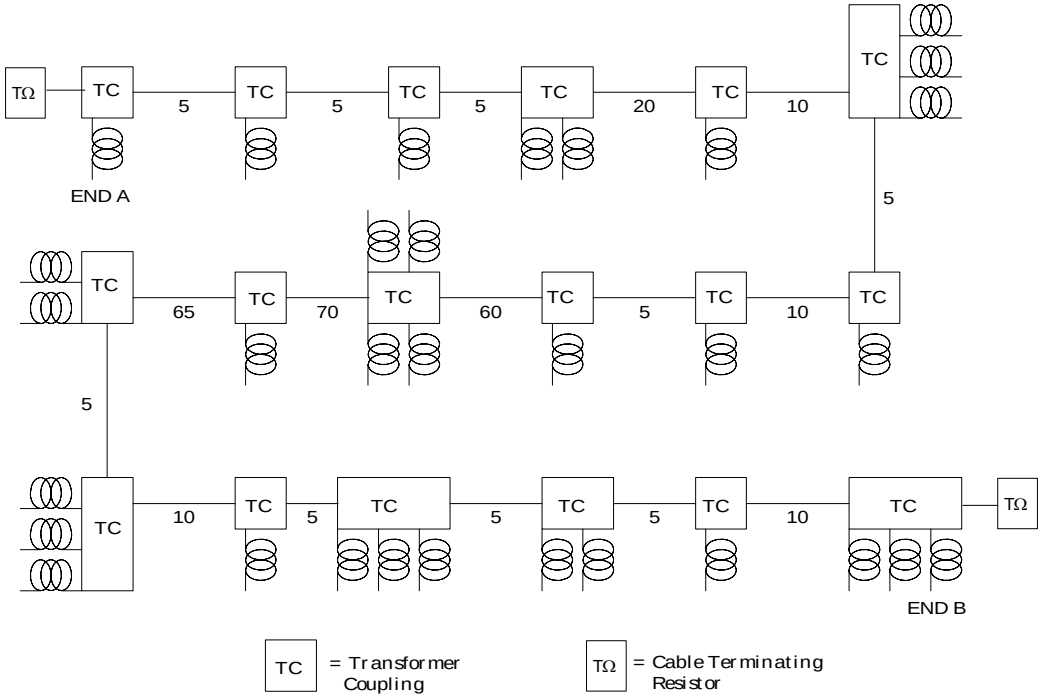


FIGURE 24. Transformer coupled test network #3.

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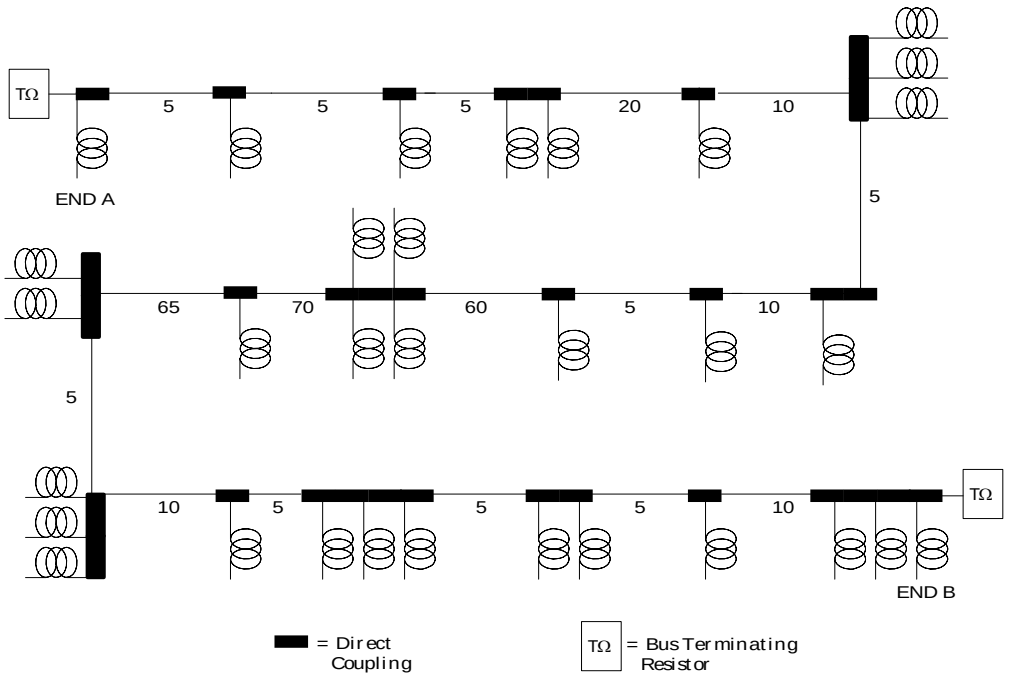


FIGURE 25. Direct coupled test network #3.

5.3.2.6.3.4 Test procedure.

Step 1: Connect the UUT-2 to the stub marked "End A" and the UUT-1 to the stub marked "End B". All other stubs shall be terminated in 1 kΩ for the transformer coupled bus and in 2 kΩ for the direct coupled bus.

Step 2: Set up the test configuration using the bus A interface at both terminals.

Step 3: Configure the UUT-2 as bus controller, and the UUT-1 as remote terminal.

Step 4: Configure the bus controller to continuously transmit test signal #5, where RT1 is the bus controller and RT2 is the UUT. The bus controller shall not transmit TAFs.

Step 5: Confirm via the modem management software running on the bus controller that the acknowledgements originating from both the RT are positive and that the bus controller did not register frame payload CRC failures.

5.3.2.6.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The nature of the observed acknowledgements (positive or negative) and the number of frame payload CRC failures registered at the bus controller.

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5.3.2.6.4 Test network #4 (mixed).

Throughput and BER performance over mixed networks shall be measured based on the following test configuration and procedure.

5.3.2.6.4.1 Test objectives.

Refer to the requirements described in 5.2.2.6.4 on page 23.

5.3.2.6.4.2 Test equipment.

Refer to the test equipment for test network #1 in 5.3.2.6.1.2 on page 55.

5.3.2.6.4.3 Test configuration.

Refer to the test configuration in FIGURE 26. Each UUT shall be connected to either end of the bus coupled opposite to each other.

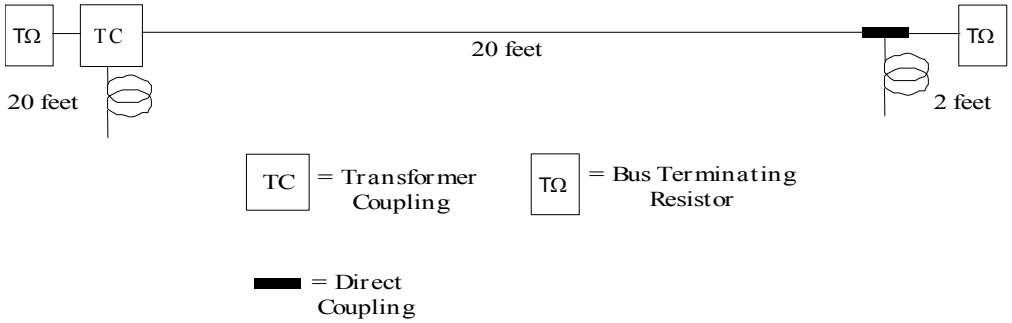


FIGURE 26. Mixed network.

5.3.2.6.4.4 Test procedure.

Step 1: Carry out the test procedure for test network #1 in 5.3.2.6.1.4 on page 56.

5.3.2.6.4.5 Data presentation.

Refer to the data presentation for test network #1 in 5.3.2.6.1.5 on page 56.

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5.3.2.7 Terminal diagnostics.

5.3.2.7.1 Digital loop-back.

Throughput and BER performance in the digital loop-back configuration shall be measured based on the following test configuration and procedure.

5.3.2.7.1.1 Test objectives.

Refer to the requirements described in 5.2.2.7.1 on page 23.

5.3.2.7.1.2 Test equipment.

The test results are obtained from the UUT itself. No other test equipment is required.

5.3.2.7.1.3 Test configuration.

Refer to the test configuration in FIGURE 27.

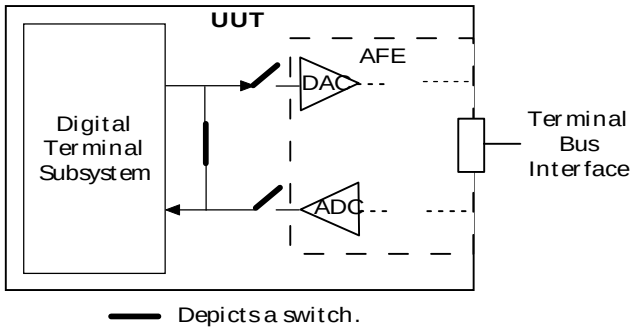


FIGURE 27. Test configuration 13.

5.3.2.7.1.4 Test procedure.

Step 1: Configure the UUT to carry out a digital loop-back self-test. For this purpose, the Analog Front-End (AFE) is completely bypassed by routing the digital signal at the input of the transmitter Digital to Analog Converter (DAC) directly to the output of the receiver Analog to Digital Converter (ADC) while isolating both converters from the “digital terminal.” The self-test shall use signaling that is similar to the one described in 5.3.2.3 on page 29 for the joint throughput and error rate measurement.

Step 2: Record the measured BER and the throughput.

5.3.2.7.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The measured throughput and BER value.

c. Indication of compliance with test objectives.

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5.3.2.7.2 Analog loop-back.

Throughput and BER performance in the analog loop-back configuration shall be measured based on the following test configuration and procedure.

5.3.2.7.2.1 Test objectives.

Refer to the requirements described in 5.2.2.7.2 on page 24.

5.3.2.7.2.2 Test equipment.

The test results are obtained from the UUT itself. No other test equipment is required.

5.3.2.7.2.3 Test configuration.

The test configuration is the stand alone UUT.

5.3.2.7.2.4 Test procedure.

Step 1: Configure the UUT to carry out an analog loop-back self-test using the bus A interface. The self-test shall use signaling that is similar to the one described in 5.3.2.3 on page 29 for the joint throughput and error rate measurement.

Step 2: Record the measured BER and the throughput.

Step 3: Repeat steps 1 to 2 using the bus B interface.

5.3.2.7.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The measured throughput and BER values.

c. Indication of compliance with test objectives.

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5.3.2.7.3 Data logging support.

Conformance with the data logging requirements shall be assessed based on the following test configuration and procedure.

5.3.2.7.3.1 Test objectives.

Refer to the requirements described in 5.2.2.7.3 on page 24.

5.3.2.7.3.2 Test equipment.

None.

5.3.2.7.3.3 Test configuration.

Refer to the test configuration in FIGURE 27 on page 61.

5.3.2.7.3.4 Test procedure.

Step 1: Configure the UUT to carry out a digital loop-back self-test.

Step 2: Demonstrate that the data logger is capable of monitoring and recording the metrics referenced in 5.2.2.7.3 by recording the list of logged parameters and their values.

5.3.2.7.3.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. List of logged metrics and their values.

c. Indication of compliance with test objectives.

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5.3.2.8 Redundant data bus

5.3.2.8.1 Electrical isolation

The electrical isolation between the bus A and bus B interface of the terminal shall be measured based on the following test configuration and procedure.

5.3.2.8.1.1 Test objectives.

Refer to the requirements described in 5.2.2.8 on page 24.

5.3.2.8.1.2 Test equipment.

The following test equipment shall be used:

a. Vector network analyzer.

b. 2 calibrated baluns: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

5.3.2.8.1.3 Test configuration.

Refer to the test configuration in FIGURE 28.

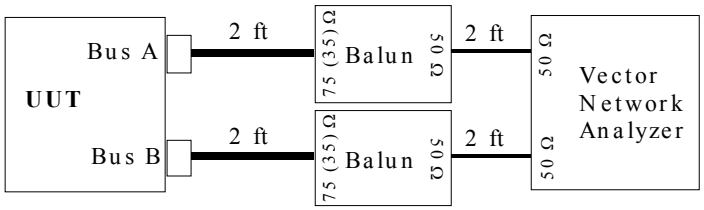


FIGURE 28. Test configuration 14.

5.3.2.8.1.4 Test procedure.

Step 1: Perform a two-port calibration of the network analyzer at the UUT facing port of the baluns.

Step 2: With the UUT powered and idle (not transmitting), measure the S_{21} transmission coefficient over the 75 kHz to 40 MHz frequency band and record the magnitude on a logarithmic scale.

5.3.2.8.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Graphical data showing the level of measured isolation in dB versus frequency.

c. Indication of compliance with test objectives.

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5.3.2.8.2 Dual redundant data bus.

Dual bus transmission and reception on either bus will be based on the following test configuration and procedure.

5.3.2.8.2.1 Test objectives.

Refer to the requirements described in 5.2.2.8.2 on page 24.

5.3.2.8.2.2 Test equipment.

The following test equipment shall be used:

a. Reference couplers (4 for transformer and 4 for direct coupled, corresponding to the desired bus coupling).
b. Reference bus segment cables.
c. 2 bus terminating resistors RL of 75 Ω ±2%.
d. Dual channel oscilloscope (recording) set to 50 Ω input impedance.
iv. 2 calibrated 75 Ω / 50 Ω baluns.

5.3.2.8.2.3 Test configuration.

Refer to the test configuration in FIGURE 29.

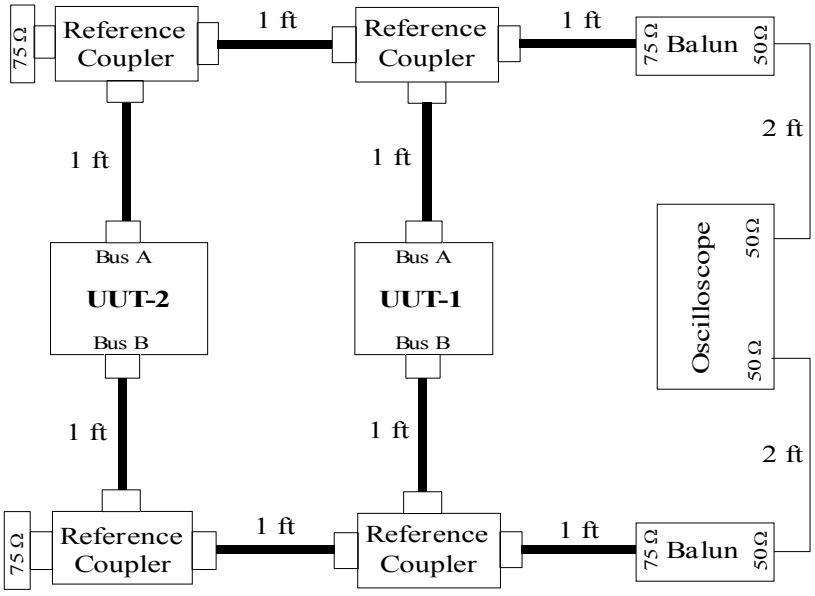


FIGURE 29. Test configuration 15.

5.3.2.8.2.4 Test procedure.

Step 1: Configure UUT-2 as bus controller and UUT-1 as remote terminal.

Step 2: Configure the bus controller to continuously transmit test signal #4 with the bus command specifying reception on bus A.

Step 3: Using the oscilloscope, observe and record simultaneously the waveform of one compound transaction containing the acknowledgement frame on both bus A and bus B.

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Step 4: Disconnect the bus B interface of the bus controller.

Step 5: Using the oscilloscope, observe and record simultaneously the waveform of one compound transaction containing the acknowledgement frame on both bus A and bus B.

Step 6: Confirm via the modem management software running on the bus controller that the acknowledgements originating from UUT-1 are positive.

Step 7: Reconnect the bus B interface and disconnect the bus A interface of the bus controller.

Step 8: Confirm via the modem management software running on the bus controller that the acknowledgements originating from UUT-1 are negative.

Step 9: Repeat steps 2 to 8 while contextually interchanging bus A and bus B in the text.

5.3.2.8.2.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. Recorded compound transaction waveforms and the nature of the observed acknowledgements (positive or negative).

c. Indication of compliance with test objectives.

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5.3.2.9 Multi-drop support

Multi-drop support shall be verified by analysis alone.

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5.3.2.10 Multi-drop communication

Multi-drop communication shall be assessed based on the following test configuration and procedure.

5.3.2.10.1 Test objectives.

Refer to the requirements described in 5.2.2.10 on page 25.

5.3.2.10.2 Test equipment.

The following test equipment shall be used:

a. Reference couplers and reference cable segments for the bus and stubs.

b. 2 bus terminating resistors of 75 Ω ±2%.

5.3.2.10.3 Test configuration.

Refer to the test configuration in FIGURE 30. For a transformer coupled network, the stub lengths shall be L=20 feet, while for a direct coupled network the stub lengths shall be L=2 feet.

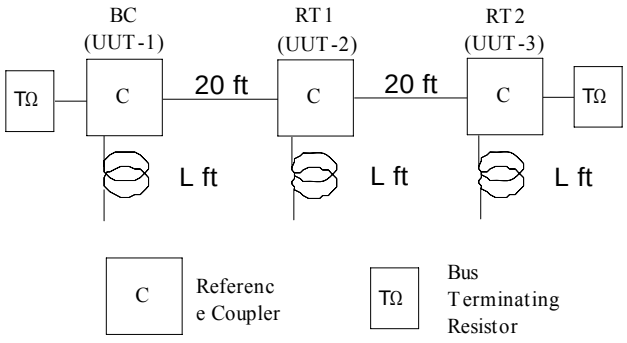


FIGURE 30. Test network #5.

5.3.2.10.4 Test procedure.

Step 1: Configure UUT-1 as bus controller, UUT-2 as RT1 and UUT-3 as RT2.

Step 2: Set up the test configuration using the bus A interface at all terminals by connecting the bus controller to the stub at the end of the bus, the RT2 to the stub at the other end of the bus, and RT1 to the mid-bus stub.

Step 3: Configure the bus controller to establish a schedule with 16 TRFs within a compound transaction. Within the first TRF, RT-2 is to transmit to the bus controller. The remaining TRFs shall be set up for bidirectional communication between bus controller and RT1. All communication (except for RT to bus controller) shall require acknowledgement.

Step 4: Confirm via the modem management software running on the bus controller that all acknowledgements originating from the RTs are positive and that the bus controller did not register frame payload failures..

Step 5: Repeat steps 3 to 4 by scheduling RT-2 transmission within the next TRF up to TRF #16.

Step 6: Repeat steps 3 to 5 by scheduling RT-2 to receive from the bus controller instead of transmitting to the bus controller.

Step 7: Configure the bus controller to establish a schedule with 16 TRFs within a compound transaction and schedule RT-2 to transmit to the bus controller within all TRFs.

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Step 8: Repeat step 4.

Step 9: Repeat steps 7 to 8 by scheduling RT-2 to receive from the bus controller instead of transmitting to the bus controller within all 16 TRFs.

5.3.2.10.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The nature of each observed acknowledgement (positive or negative) and the number of frame payload CRC failures registered at the bus controller.

c. Indication of compliance with test objectives.

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5.3.2.11 MCW interference into TW.

TW word error rate performance in the presence of MCW interference source shall be measured based on the following test configuration and procedure.

5.3.2.11.1 Test objectives.

Refer to the requirements described in 5.2.2.11 on page 25.

5.3.2.11.2 Test equipment.

The following test equipment shall be used:

a. Couplers.
b. Bus segment cables.
c. Terminating resistor RL of 75 Ω ±2%.
d. Calibrated 75 Ω / 50 Ω balun.
e. Programmable 50 Ω amplifier.
f. Test waveform generator.
g. True RMS voltmeter.
h. Oscilloscope set to 1MΩ input impedance.
i. Low-pass filter with the following characteristics: pass-band stop frequency at 40 MHz, stop-band start frequency at 50 MHz, minimum stop-band attenuation of 60 dB.
v. Error Counter (TW words).
j. TW bus tester.

5.3.2.11.3 Test configuration.

Refer to the test configuration in FIGURE 31.

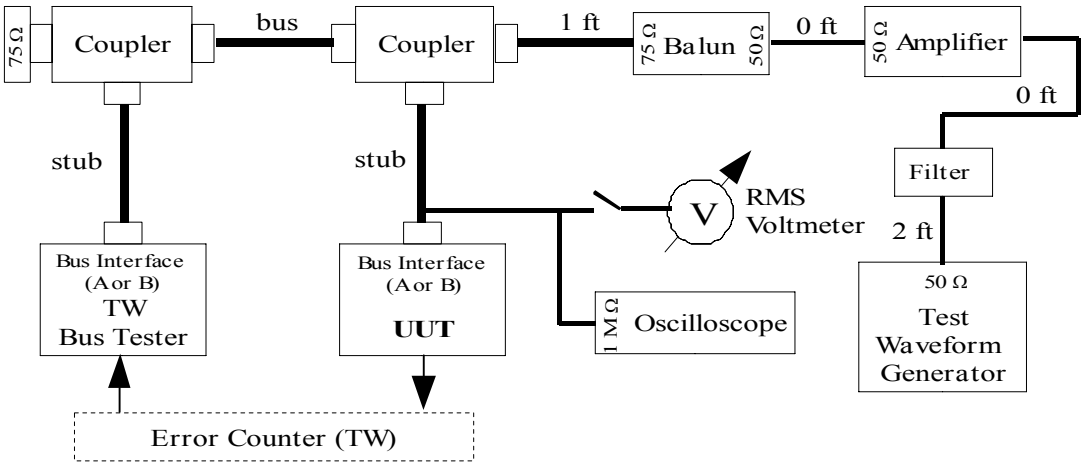


FIGURE 31. Test configuration 16.

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5.3.2.11.4 Test procedure.

Step 1: Set up the test configuration using the bus A interface at both terminals.

Step 2: With the test waveform generator and the terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 3: Configure the TW bus tester to transmit test messages and adjust its output level until the received peak-to-peak voltage at the UUT is at the required level.

Step 4: Reconfigure the TW bus tester to idle. Set up the test waveform generator to continuously transmit test signal #1.

Step 5: Connect the voltmeter. Adjust the amplifier until the measured MCW RMS voltage at the UUT is at the required level.

Step 6: Disconnect the voltmeter and run a word error rate test in accordance with TW certification testing methods.

Step 7: Repeat steps 1 to 6 using the bus B interface.

5.3.2.11.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The measured word error rate.

c. Indication of compliance with test objectives.

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5.3.3 Compliance with FW requirements.

Verification of the FW requirements shall be identical to that required for TW certification with the addition of the verifications contained herein.

A test environment shall be provided which ensures that interference from external ambient noise sources not included as part of the test are eliminated or minimized.

5.3.3.1 Power spectral density.

The PSD of the FW signal shall be measured based on the following test configuration and procedure.

5.3.3.1.1 Test objectives.

Refer to the requirements described in 5.2.3.1 on page 25.

5.3.3.1.2 Test equipment.

The following test equipment shall be used:

a. Spectrum analyzer.

b. Calibrated balun: 75 Ω / 50 Ω for transformer coupled stubs and 35 Ω / 50 Ω for direct coupled stubs.

c. Programmable 50 Ω attenuator.

5.3.3.1.3 Test configuration.

Refer to the test configuration in FIGURE 32.

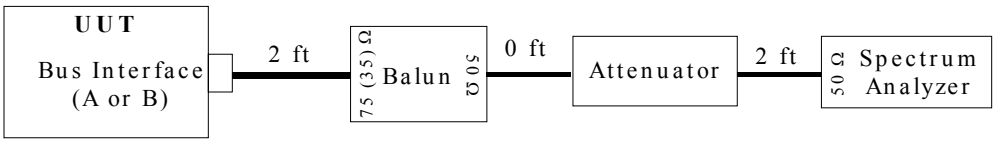


FIGURE 32. Test configuration 17.

5.3.3.1.4 Test procedure.

Step 1: Set the attenuator to prevent overloading the spectrum analyzer front-end when subjected to the full amplitude FW signal.

Step 2: Configure the UUT as bus controller and set it up to continuously transmit messages of at least 16 words with minimum inter-message gap.

Step 3: Set up the test configuration using the bus A interface.

Step 4: Set the spectrum analyzer to sweep the 0 to 1 MHz frequency band.

Step 5: Set the spectrum analyzer to measure peak power in a 9 kHz resolution bandwidth (dBm/9kHz) and use a linear display mode.

Step 6: Record the frequency at which the highest power is observed.

Step 7: Set the spectrum analyzer to measure quasi-peak power in a 9 kHz resolution bandwidth (dBm/9kHz). Set the center frequency to the one recorded in step 6. Set the span to 200 kHz.

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Step 8: Record the maximum quasi-peak power, corrected for the attenuation level of the attenuator and for the insertion loss of the balun.

Step 9: Set the spectrum analyzer to sweep the 1 to 40 MHz frequency band.

Step 10: Measure the quasi-peak power at the frequencies indicated in TABLE IV on page 25 and record the values corrected for the attenuation level of the attenuator and for the insertion loss of the balun.

Step 11: Record the PSD plot for the 1 to 40 MHz frequency band.

Step 12: Repeat steps 3 to 11 using the bus B interface.

Step 13: Add a 1.05 dB envelope detector correction factor to the quasi-peak power values (PSDs) recorded in steps 11 and 12.

Step 14: Determine the spectrum analyzer's equivalent noise power bandwidth for the 9 kHz filter by measuring the analyzer filter's peak, 3 dB, 6 dB, and 20 dB points and constructing a piecewise linear model of the filter. The equivalent noise power bandwidth is the bandwidth of a rectangular filter whose area is equivalent to that of the piecewise linear model.

Step 15: Calculate and record the power spectral density for the unit under test by taking the recorded quasi-peak power levels and subtracting 10 $\log_{10}$ (equivalent noise power bandwidth/1Hz).

5.3.3.1.5 Data presentation.

Data presentation shall be as follows:

a. Detailed test configuration, test equipment and test procedures used for conducting the test.

b. The recorded quasi-peak power values.

c. The calculated power spectral density levels: the maximum level in the 0 to 1 MHz frequency band and the PSD levels at the frequencies indicated in TABLE IV on page 25.

d. The recorded PSD plot with the transmit emissions mask of FIGURE 7 overlayed.

e. Indications of compliance.

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5.3.3.2 MCW interference into FW.

FW word error rate performance in the presence of MCW interference source shall be measured based on the following test configuration and procedure.

5.3.3.2.1 Test objectives.

Refer to the requirements described in 5.2.3.2 on page 26.

5.3.3.2.2 Test equipment.

Refer to the test equipment for the MCW interference into TW test in 5.3.2.11.2 on page 70.

5.3.3.2.3 Test configuration.

Refer to the test configuration in FIGURE 31 on page 70.

5.3.3.2.4 Test procedure.

Step 1: Set up the test configuration using the bus A interface at both terminals.

Step 2: With the test waveform generator and the terminals in the powered on state and not transmitting any signal (idle), confirm on the oscilloscope that there is no other interference present than background noise.

Step 3: Configure the 1553B (TW) bus tester to transmit test messages and adjust its output level until the received peak-to-peak voltage at the UUT is at the required level.

Step 4: Reconfigure the 1553B (TW) bus tester to idle. Set up the test waveform generator to continuously transmit test signal #1.

Step 5: Connect the voltmeter. Adjust the amplifier until the measured MCW RMS voltage at the UUT is 423 mV_{rms} (corresponding to -71 dBm/Hz into 75 Ω over 30 MHz bandwidth) for transformer coupled stubs, and 115 mV_{rms} (corresponding to -79 dBm/Hz into 35 Ω over 30 MHz bandwidth) for direct coupled stubs.

Step 6: Disconnect the voltmeter and run the word error rate test in accordance with TW certification testing methods

Step 7: Repeat steps 1 to 6 using the bus B interface.

5.3.3.2.5 Data presentation.

Refer to the data presentation for the MCW interference into TW test in 5.3.2.11.5 on page 71.

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5.4 MCW interface requirements.

5.4.1 General.

Multi-carrier waveform PHY and MAC interface requirements contained in this section shall provide interoperability with extended 1553 systems.

5.4.2 PHY interface.

5.4.2.1 Overview.

A terminal utilizing MCW is a bandwidth configurable transceiver capable to use 1 to 6 contiguous sub-bands spanning the 5 to 35 MHz frequency band. Each sub-band shall be of 5 MHz bandwidth with 64 tones per sub-band. MCW signaling, in this standard, defines a synchronous time division multiple access (TDMA) communication system.

The block diagram of the MCW transmitter is provided in FIGURE 33.

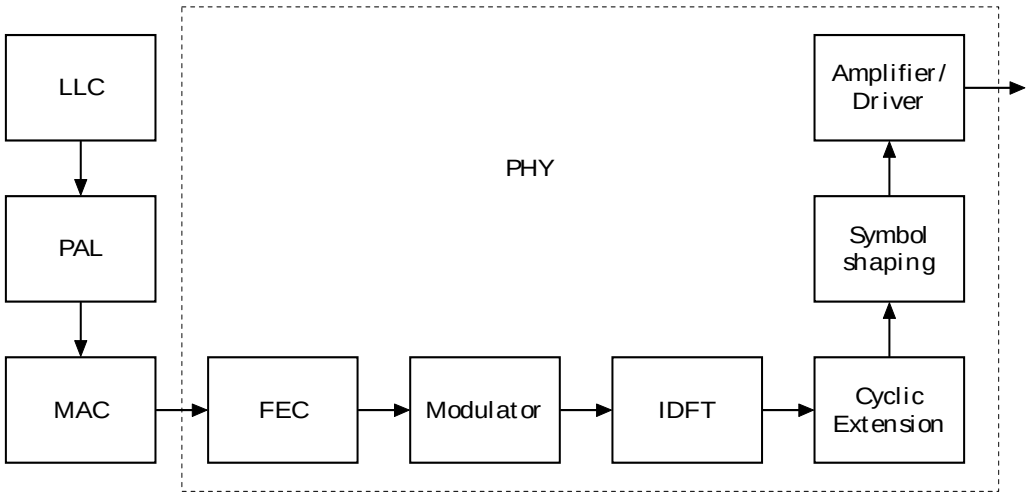


FIGURE 33. MCW PHY transmitter functional block diagram.

The PHY layer shall include forward error correction (FEC) and two modulation types known as basic rate (BR) and variable bit rate (VBR) modulation. Differential binary phase shift keying (DBPSK) shall be used for low rate BR modulation which is employed with frame control information and BR communications (such as bus command frames). Coherent QAM up to 4096 point constellations (12 bits / tone) shall be used for higher rate VBR modulation.

The MCW signal will be processed through a 1024 point inverse discrete Fourier transform (IDFT) and a discrete Fourier transform (DFT).

TABLE VI specifies the sub-band tone number and frequency assignment. The default network shall be configured to use all sub-bands from 1 to 6, i.e. the 5 to 35 MHz frequency band. The only valid configurations are those with contiguous allocations of one or more sub-bands.

TABLE VII lists the basic PHY parameters.

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TABLE VI. Sub-band tone number and frequency assignment.

Sub-band	Tone Number	Frequency [MHz]
1	64 to 127	5 to 9.92
2	128 to 191	10 to 14.92
3	192 to 255	15 to 19.92
4	256 to 319	20 to 24.92
5	320 to 383	25 to 29.92
6	384 to 447	30 to 34.92

TABLE VII. Basic PHY parameters.

Parameter	Value
IDFT / DFT size (N_{dft})	1024
Tone spacing [kHz]	78.125
Sub-band size	64 tones 5 MHz
Max. number of sub-bands	6

5.4.2.2 Frame structure and modulation modes.

The structure of a PHY frame is as shown in FIGURE 34 and shall consist of a synchronization preamble, frame control (FC), and an optional frame payload (PL).

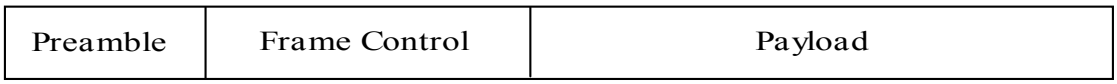


FIGURE 34. PHY frame structure.

TABLE VIII contains the PHY frame parameters. As a minimum, a PHY frame shall contain the preamble and the FC.

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TABLE VIII. PHY frame parameters.

Parameter	Units	Value
preamble size	sync symbols	5
PHY block size	OFDM symbols	8
FC size	PHY blocks	1
PHY frame size (FC +PL)	PHY blocks	1 to 31

5.4.2.2.1 Preamble.

The preamble shall consist of 5 sync symbols used to detect the start of frame, calculate symbol timing and estimate the channel response.

5.4.2.2.1.1 Sync symbols.

Sync symbols shall be generated as described next. If the transmission band spans N_t consecutive tones, then these tones, indexed $k_0 \ldots k_0 + N_t - 1$, are modulated with S_{k_0+k} given below:

$$S_{k_0+k} = \exp\left[j \frac{\pi k^2}{N_t} \right], \; k = 0 \ldots N_t - 1$$

In the above equation k_0 is the index of the first tone within the transmission band.

5.4.2.2.1.2 Time-reversed sync symbols.

In order to distinguish between the bus command frame and all other frames within a MAC compound transaction, the non BCF sync symbols shall be time-reversed as described next. If the bus command frame sync symbol samples are generated by modulating tone k with S_k as given in 5.4.2.2.1.1, then the sync symbols of non bus command frames shall be generated by modulating tone k with the complex conjugates S_k^*.

5.4.2.2.2 Frame control.

The frame control (FC) shall consist of 8 OFDM symbols which are used to carry PHY frame information. FC data shall be protected with a 16-bit CRC code; forward error correction encoded, and transmitted using DBPSK modulation.

5.4.2.2.3 Frame payload.

The frame payload may or may not be present. When present, it shall consist of PHY blocks, with each block containing 8 OFDM symbols. The payload can be data or training.

5.4.2.2.3.1 Data.

The data payload shall be used to carry information using basic rate or variable bit rate modulation. Both BR and VBR data shall be protected with a 32-bit CRC and encoded using FEC.

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5.4.2.2.3.1.1 Basic rate modulation.

Basic rate shall use DBPSK modulation to allow for low rate communications to occur before clock recovery enables higher rate coherent VBR modulation.

5.4.2.2.3.1.2 Variable bit rate modulation.

Variable bit rate modulation is a coherent modulation scheme that uses a negotiated, optimum tone map for data communication over a particular channel. Each tone shall be assigned a bit capacity of 0, 1, 2, 4, 6, 8, 10 or 12 bits being either suppressed, or BPSK or QAM modulated.

5.4.2.2.3.1.3 Training.

The training payload shall be the same as the data payloads except that the OFDM symbols are 4-QAM modulated with pseudo-random data. The training payload shall be used to stimulate the channel with the same statistical signaling as the data. There shall be no CRC protection or FEC encoding for training.

5.4.2.3 FEC.

The FEC encoder for FC and basic rate shall consist of a scrambler, Reed-Solomon and convolutional encoders, interleaver and repetition coding (diversity). The encoder for the VBR payload shall consist of scrambler, Reed-Solomon and convolutional encoder, puncturing, zero padding (as needed) and interleaver. TABLE IX summarizes the usage of the individual encoders and the corresponding code rates.

TABLE IX. Encoders and code rates.

Modulation Type	Convolutional Code	Reed-Solomon Code	Diversity Order
FC	1/2	≤ 239/255	not more than the number of configured sub-bands
BR	1/2	≤ 239/255	
VBR	1/2 or 3/4 or 7/8	≤ 239/255	1

5.4.2.3.1 Scrambler.

The data field shall employ a scrambler of length 127. The input bytes shall be converted into a serial bit stream with the most significant bit (MSB) first and least significant bit (LSB) last. The scrambler shall use the polynomial:

$$S(x) = x^7 + x^4 + 1$$

The scrambler shall be of the form shown in FIGURE 35. The initial value loaded into the shift registers shall be 0x7F for PL and 0x40 for FC. The scrambler's state shall be reinitialized at the beginning of each PHY block.

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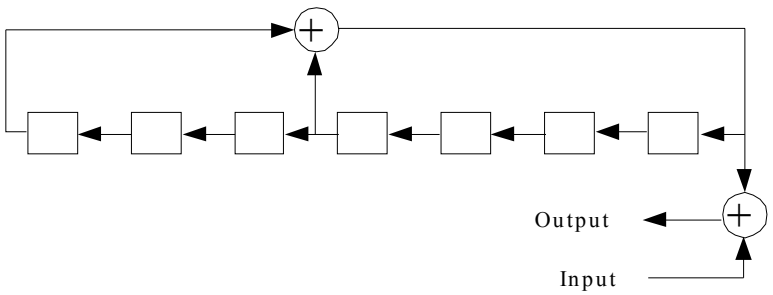


FIGURE 35. Data scrambler.

5.4.2.3.2 Reed-Solomon encoder.

A (255, 239) Reed-Solomon code defined over the GF (256) Galois field (RS symbol length of 8 bits) shall be used in order to provide some level of burst error correction capability. Based on the number of bits carried in each PHY block, it may be necessary to use a shortened version of the RS code, i.e., a rate $(239-s)/(255-s)$ code, where s can be an integer between 0 and 225. The number of redundant RS symbols is 16 and the code can correct errors of up to 8 RS symbols (bytes).

The GF (256) Galois field shall be defined by the primitive element α using the mapping below and the field generator polynomial $f(x) = x^8 + x^4 + x^3 + x^2 + 1$ (435 octal). In forming the RS symbols, the MSB first and LSB last convention shall be used.

$$\begin{aligned} \alpha^0 &= 00000001 \quad \alpha^4 = 00010000 \\ \alpha^1 &= 00000010 \quad \alpha^5 = 00100000 \\ \alpha^2 &= 00000100 \quad \alpha^6 = 01000000 \\ \alpha^3 &= 00001000 \quad \alpha^7 = 10000000 \end{aligned}$$

The code generator polynomial shall have as zeros the first 16 powers (1 through 16) of the primitive element α:

$$g(x) = \prod_{k=1}^{16} (x - \alpha^k)$$

Within the Reed-Solomon codeword, the message bits shall be first (in the order they arrived at the input) followed by the parity bits. When more than one Reed-Solomon code word is contained within one PHY block, all code words shall be of equal length. Denoting by b the number of bytes within one PHY block at the input to the convolutional encoder (calculated based on tone map and puncture rate), the number N_{RS} and length n of the code words shall be found as follows:

$$N_{RS} = \text{floor}\left(\frac{b+239}{255}\right)$$

$$n = \min\left\{254, \text{floor}\left(\frac{b}{N_{RS}}\right)\right\}$$

5.4.2.3.3 Convolutional encoder.

The data field shall be encoded with a convolutional encoder with code rate R = 1/2. The convolutional encoder shall use the generator polynomials $Y = 133_8$ and $X = 171_8$ as shown in FIGURE 36. The bit denoted as "X" shall be output from the encoder before the bit denoted as "Y." Higher data rates shall be derived from the basic code by puncturing.

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The RS bytes shall be converted into a serial bit stream with the MSB first and LSB last with the convolutional encoder operating on a PHY block. At the end of each PHY block there shall be 8 flush bits (8 zeros) inserted at the input in order to return the convolutional encoder to the all zero state.

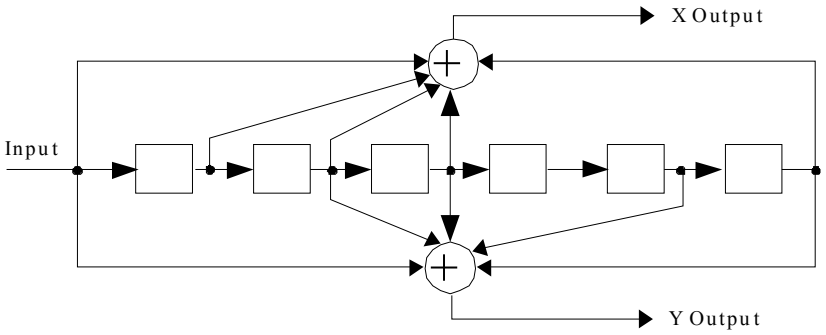


FIGURE 36. Convolutional encoder.

5.4.2.3.4 Puncturing.

In order to increase the maximum possible bit rate with VBR modulation, puncturing may be used to create rate 7/8 and rate 3/4 convolutional codes. The puncturing matrix for the rate 7/8 code is shown in FIGURE 37 with the punctured bits highlighted in gray. The corresponding output bit stream of the puncture block is shown in FIGURE 38. Bits X0 and Y0 are the first ones out of the convolutional encoder.

X0	X1	X2	X3	X4	X5	X6
Y0	Y1	Y2	Y3	Y4	Y5	Y6

FIGURE 37. Puncturing matrix for code rate 7/8.

X0	Y0	Y1	Y2	Y3	X4	Y5	X6

FIGURE 38. Output bit stream for code rate 7/8.

The puncturing matrix for the rate 3/4 code is shown in FIGURE 39 while the output bit stream is shown in FIGURE 40.

X0	X1	X2
Y0	Y1	Y2

FIGURE 39. Puncturing matrix for code rate 3/4.

X0	Y0	Y1	X2

FIGURE 40. Output stream for code rate 3/4.

5.4.2.3.5 Zero padding.

Frequently, the length of the (optionally punctured) convolutional code word is not exactly the size of the interleaver matrix. In this case, zero padding shall be performed before interleaving in order to fill the gap.

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5.4.2.3.6 Interleaver.

The purpose of the interleaver is to spread out time-localized error bursts to the whole PHY block in order to present more random-like errors to the convolutional decoder.

The interleaver matrix shall hold all the bits within one PHY block (corresponding to 8 OFDM symbols). Each OFDM symbol shall contain all the bits carried by all the available tones. The number of bits per tone shall be decided by the modulation of each tone. The interleaver parameters are provided in TABLE X.

TABLE X. Interleaver parameters.

Parameter	Value
Interleaver matrix width (W)	8
Column-to-column shift (S)	6

FIGURE 41 shows the order of writing to the interleaver matrix (8 byte block example). The gray cells represent the first cell that is written to in each row.

0	1	2	3	4	5	6	7
8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23
24	25	26	27	28	29	30	31
32	33	34	35	36	37	38	39
40	41	42	43	44	45	46	47
48	49	50	51	52	53	54	55
56	57	58	59	60	61	62	63

FIGURE 41. Writing to the interleaver matrix.

FIGURE 42 shows the order of reading from the interleaver matrix (8 byte block example). Gray cells represent the first cell which is read from in each column.

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0	10	20	30	32	42	52	62
1	11	21	31	33	43	53	63
2	12	22	24	34	44	54	56
3	13	23	25	35	45	55	57
4	14	16	26	36	46	48	58
5	15	17	27	37	47	49	59
6	8	18	28	38	40	50	60
7	9	19	29	39	41	51	61

FIGURE 42. Reading from the interleaver matrix.

The number of bits per OFDM symbol when N_t tones are used for transmission shall be calculated as:

$$Bits_per_OFDMsymbol = \sum_{k=1}^{N_t} b_k$$

where b_k is the number of bits carried on the k-th of the N_t tones. Then, the number of bits in one PHY block is:

$$Bits_per_PHYblock = 8 \cdot Bits_per_OFDMsymbol$$

In practice, the data is stored as a vector instead of a matrix. In this case, the element at $index_{out}$ in the output vector shall be obtained by reading out the element at $index_{IN}$ in the input vector, as given by the relation:

$$index_{IN} = W \cdot index_{OUT} - (N - 1 - S \cdot W) \cdot \text{floor}\left(\frac{W \cdot index_{OUT}}{N}\right) \quad \text{modulo } N$$

In the equation above $index_{OUT}$ is from 0 to $N-1$, N is equal to $Bits_per_PHYblock$, W is the interleaver width and S is the column-to-column shift.

5.4.2.3.7 Repetition coding (diversity).

Repetition coding (diversity) of basic rate and FC shall be the same. The diversity order is a network-wide programmable parameter with values from 1 to a maximum that is equal to the number of sub-bands configured for communication. When the number of available tones is less than 128 (two sub-bands), the diversity order shall be 1. When the number of available tones N_t is at least 128, the diversity order D shall be chosen to be a divisor of N_t. The bits sent over the lower $M = N_t/D$ available tones shall be repeated on the higher frequency tones. Denoting by b_k, $k = 0 \ldots M-1$, the bits sent over the lower M tones, the repetition shall be according to the relation:

$$b_{M \cdot p+k} = b_k \,, \quad p = 1 \ldots D-1 \,, \quad k = 0 \ldots M-1$$

Given the diversity order, the number of bytes within one PHY block at the input to the convolutional encoder (denoted by b in 5.4.2.3.2 on page 79) shall be calculated according to the relation:

$$b = \frac{M}{2} - 1$$

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The division by 2 accounts for the convolutional code rate and the "-1" for the flush byte. Note: M is guaranteed to be an even number by the diversity rules. For the default network configuration (using the 5 to 35 MHz frequency band), the default diversity order is $D = 4$.

5.4.2.4 Training generator.

This is a pseudo-random number generator used to generate bits which then select points in a 4-QAM constellation during training. The training generator shall use two feed-back shift registers as shown in FIGURE 43. At the start of each training frame, the shift registers shall be initialized to the values shown below:

$$S_1(x) = X^7 + X^4 + 1, \text{ initial value} = 0x7F$$

$$S_2(x) = X^6 + X^3 + 1, \text{ initial value} = 0x3F$$

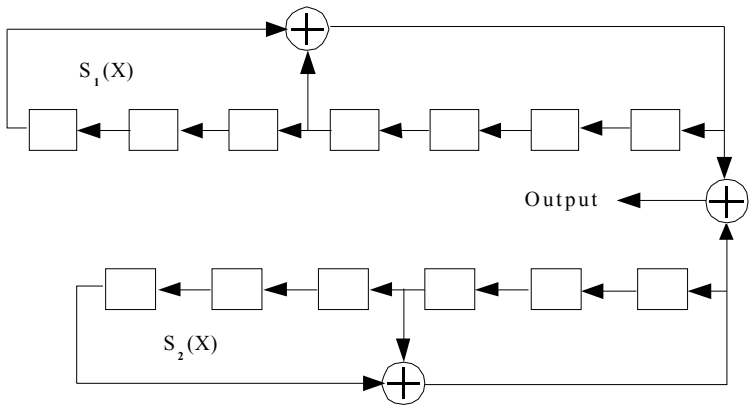


FIGURE 43. Training generator.

5.4.2.5 OFDM symbol generation.

5.4.2.5.1 Data formatter.

The data formatter shall organize the bits from the FEC into a format that is compatible with the QAM modulator. For mapping to a 2^{2M} QAM constellation, the first M bits shall be mapped to the in-phase component while the next M bits shall be mapped to the quadrature component. In both cases, the last bit in shall be placed on the least significant position.

5.4.2.5.2 Modulator.

The modulator shall perform QAM, BPSK and DBPSK modulation as well as sync phase modulation.

5.4.2.5.2.1 QAM modulator.

The modulator shall accept coded bits from the data formatter and map them to complex signal constellation points scaled such that all signal constellations have almost the same average energy. The result shall be a complex number for each tone.

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FIGURE 44 shows the block diagram of the QAM modulator.

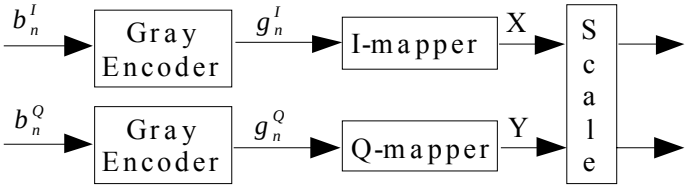


FIGURE 44. QAM modulator.

The Gray encoder blocks shall encode the b_n^I in-phase bits and the b_n^Q quadrature bits according to the relation:

$$g_{M-1} = b_{M-1},$$
$$g_k = g_{k+1} \text{ xor } b_k,\ 0 \le k \le M-2$$

where the value of M depends on the constellation size (e.g. $M = 6$ for 2^{12}-QAM). The mapper blocks shall translate the Gray coded bits to X and Y signal coordinates in the QAM constellation. In particular, taking the decimal value of $\{g_n^I\},\ n \in \{0,\ldots,M-1\}$ the in-phase component shall be mapped with zero corresponding to the most negative and the full scale to the most positive coordinate. For the quadrature component the mapping shall be the exact mirror of the in-phase mapping: zero shall correspond to the most positive and the full scale to the most negative coordinate. This mapping, as illustrated in FIGURE 45, shall be according to the relations:

$$X = -\left(2^M - 1\right) + 2 \cdot g^I$$
$$Y = +\left(2^M - 1\right) - 2 \cdot g^Q$$

where $g^I, g^Q \in \{0,\ldots,2^M - 1\}$

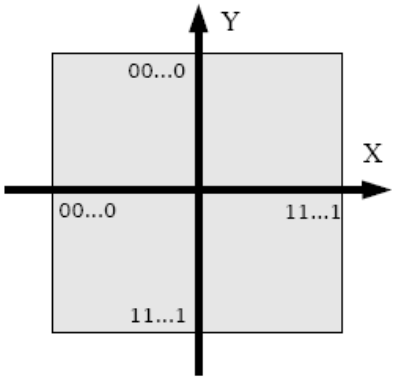


FIGURE 45. I/Q mapping.

The scaler shall be common to both in-phase and quadrature channels and assures the approximately same average constellation energy. The scaling values are given in TABLE XI.

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TABLE XI. Constellation scaling values.

	BPSK	4-QAM	16-QAM	64-QAM	256-QAM	1024-QAM	4096-QAM
Scale Factor	1	1/2	1/4	1/8	1/16	1/32	1/64

5.4.2.5.2.2 BPSK modulator.

The BPSK modulator shall perform the 0→1 and 1→ -1 mapping as shown in TABLE XII, where I refers to the in-phase and Q the quadrature component of the output.

TABLE XII. BPSK modulation.

Input Bit	I-out	Q-out
0	1	0
1	-1	0

5.4.2.5.2.3 DBPSK modulator.

The DBPSK modulator shall change the phase of the output by π when the input bit is 1, as shown in TABLE XIII. The initial phase of the output shall be zero. Modulation of the basic rate payload shall continue that of the FC, without resetting the memory of the DBPSK modulator.

TABLE XIII. DBPSK modulation.

Input Bit	Output Phase Change
0	0
1	π

5.4.2.5.2.4 Sync phase modulation.

Sync phase modulation is a per-tone multiply of the complex constellation point (BPSK, QAM, DBPSK) with the complex sync phasor of the corresponding tone. The complex numbers modulating the tones shall be the ones given in 5.4.2.2.1.1 on page 77. Sync phase modulation shall be applied for the FC as well as the data and training payloads.

5.4.2.5.3 IDFT.

After a constellation point for each tone is chosen, complex conjugates of these values shall be input to the upper frequency half of the IDFT in a manner that ensures the time domain samples are all real. Any residual imaginary component shall be discarded. The IDFT inputs corresponding to tone numbers 0 to 63 and 448 to 511 (covering the 0 to 5 MHz and 35 to 40 MHz frequency bands) shall be set to zero. The IDFT shall be defined as in the equation below:

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$$X_k = \frac{1}{N} \cdot \sum_{n=0}^{N-1} x_n \exp\left[j \cdot 2\pi \frac{nk}{N} \right], \quad k = 0 \ldots N-1$$

5.4.2.5.4 Time domain processing.

FIGURE 46 illustrates the TDP operation for FC and PL while FIGURE 47 illustrates the TDP operation for the preamble. These operations shall consist of cyclic extension, windowing, and overlap with the previous symbol.

To provide transmit spectral shaping and thus reduce spectrum splatter, the OFDM symbol shall be multiplied by a raised-cosine window. The amount of roll-off shall be fixed and tied to the transmit window size. The raised-cosine shaping shall use the asymmetric window given by the equation below:

$$y(n) = \frac{1}{2}\left[1 - \cos\left(\frac{n \cdot \pi}{n_{win}}\right)\right], \quad n = 0 \ldots n_{win}$$

The rising edge of the window shall be defined by samples $n = 0 \ldots n_{win} - 1$ and the falling edge by samples $n = n_{win} \ldots 1$. The last (5th) sync symbol shall borrow n_{borrow} samples from the first OFDM symbol in the FC, which becomes that much shorter.

The default values and the allowed range of the TDP parameters are provided in TABLE XIV.

TABLE XIV. TDP parameters.

Parameter	Valid Range (samples)	Default Value (samples)
tx window size (n_win)	16	n/a
cyclic prefix (CP) + cyclic suffix (CS) size	128 * k + n_win, where 0 ≤ k ≤ 8	272
CP size (n_prefix)	0 … 1024	240
CS size (n_suffix)	n_win + (0 … 1024)	32
1st sync symbol size	n_dft + n_win = 1040	n/a
2nd to 4th sync symbol size	n_dft = 1024	n/a
samples borrowed by 5th sync symbol from 1st FC OFDM symbol (n_borrow)	64	n/a
5th sync symbol size	n_dft + n_borrow = 1088	n/a
1st FC OFDM symbol size	n_dft + n_prefix + n_suffix – n_borrow	1232
2nd to 8th FC OFDM symbol and all PL OFDM symbol size	n_dft + n_prefix + n_suffix	1296
PHY frame overlap size	n_win	n/a

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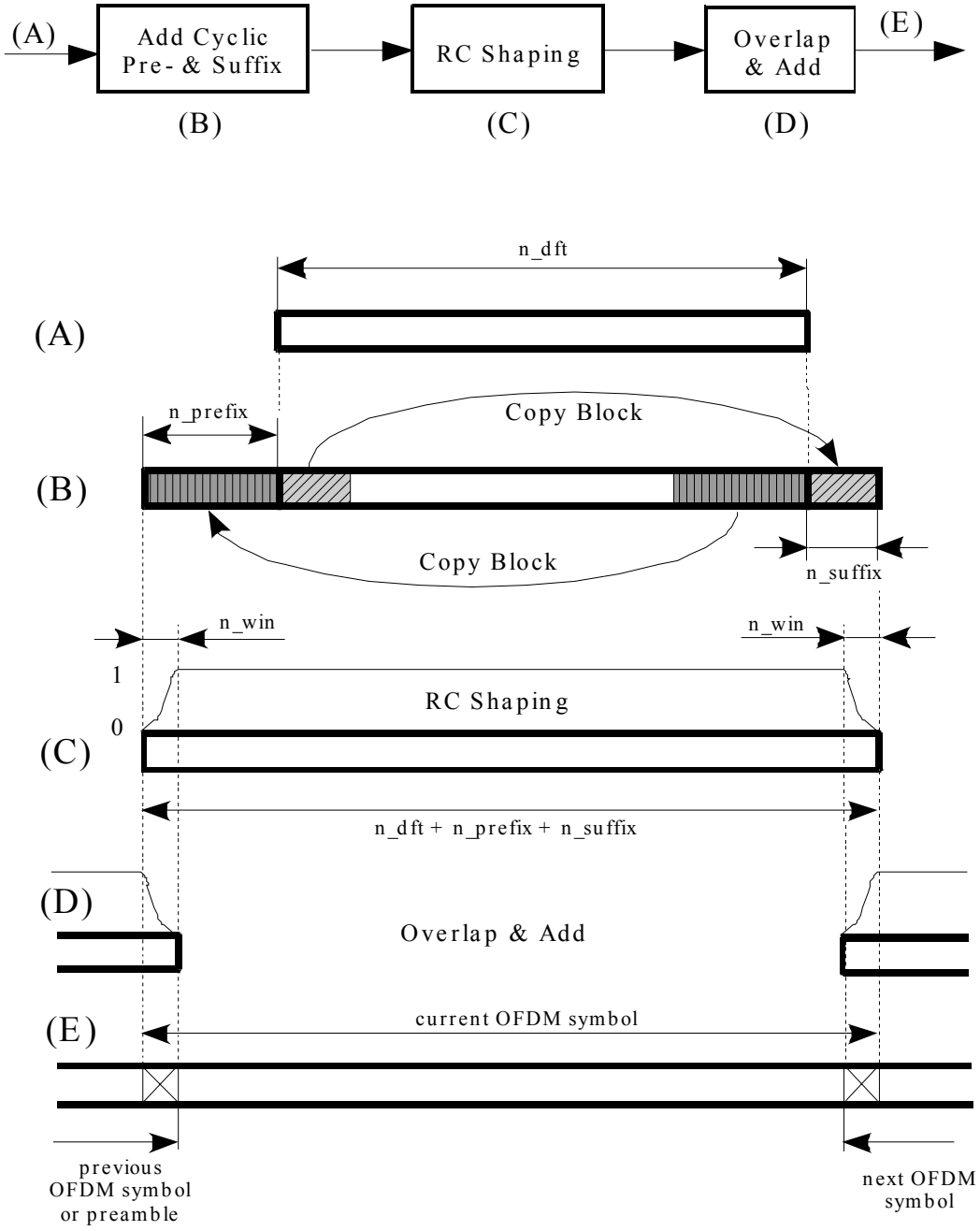
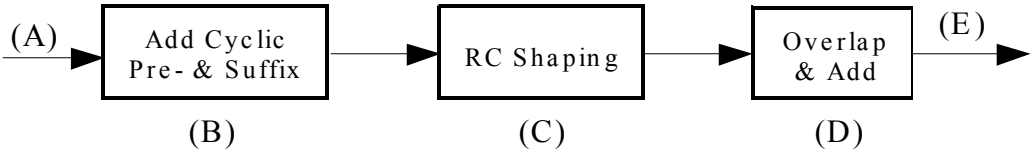


FIGURE 46. Time domain processing of FC and PL.

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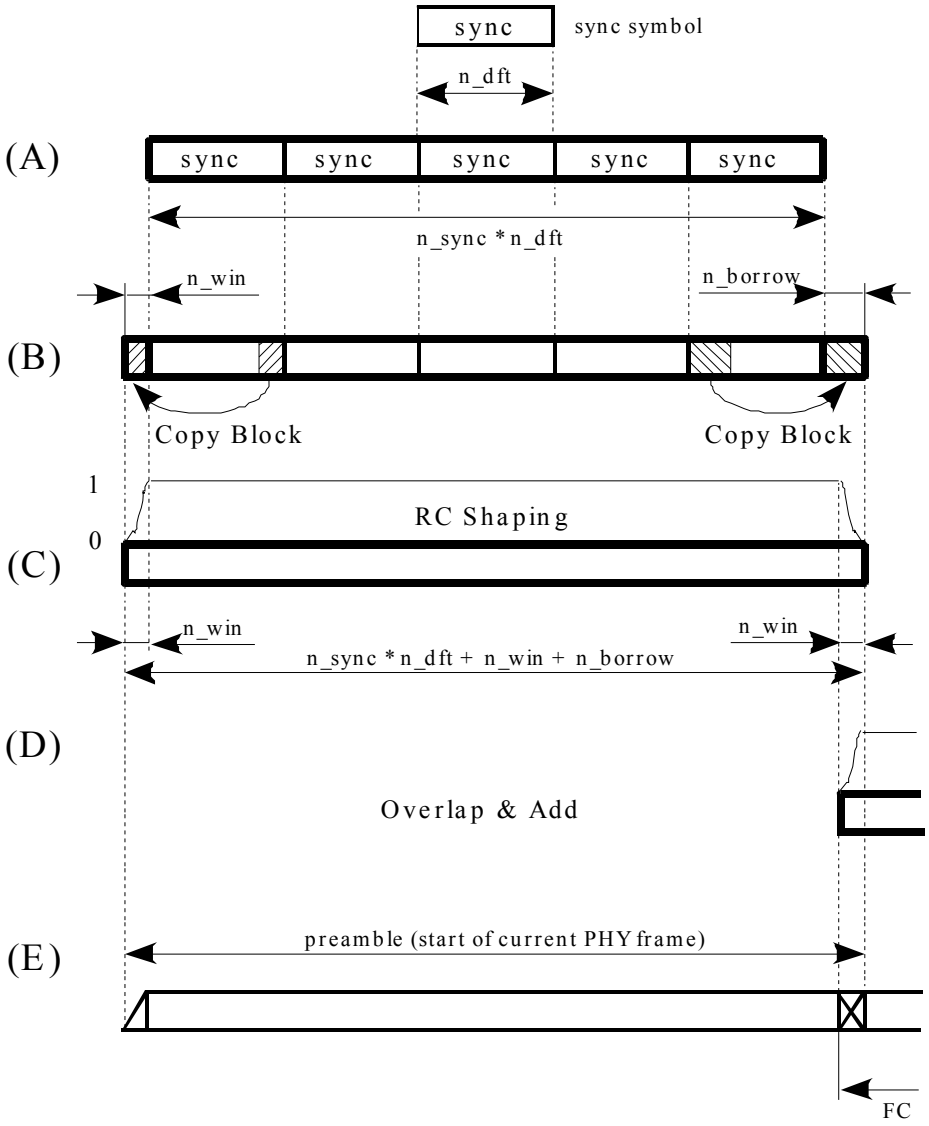


FIGURE 47. Time domain processing of preamble.

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5.4.3 MAC protocol.

5.4.3.1 General.

Each terminal on a bus shall have a unique address, assigned during system design.

All RTs shall be capable of being assigned any unique address from decimal address 0 (000000) through decimal address 62 (111110). The address shall be established through an external connector, which is part of the system wiring and connects to the RT. Changing the unique address of a RT shall not require the physical modification or manipulation of any part of the RT. The RT shall, as a minimum, determine and validate its address during power-up conditions. No single point failure shall cause a terminal to validate a false address. The RT shall not respond to any messages if it has determined its unique address is not valid.

These addresses shall be used at the PHY layer for communications between terminals. The PHY layer shall control the timing and length of MAC layer transmissions as directed from the bus controller, through a configured bus schedule.

5.4.3.2 Command-response protocol.

While the command-response protocol has some similarities to the TW protocol, the two protocols are not the same. In particular, while a TW transaction consists of one command-response message exchange, a MCW transaction may consist of a set of several command-response message exchanges. This section describes the MCW command-response protocol under the following headings:

a. Compound transactions.

b. PHY frames.

c. Guard slots.

5.4.3.2.1 Compound transaction.

MCW shall use compound transactions to control and coordinate communications among RTs and the bus controller. A compound transaction shall consist of a bus command frame containing a set of commands, corresponding response frames, and corresponding terminal acknowledgement frames (TAF)s with guard slots in between. The generic structure of a compound transaction is shown in FIGURE 48.

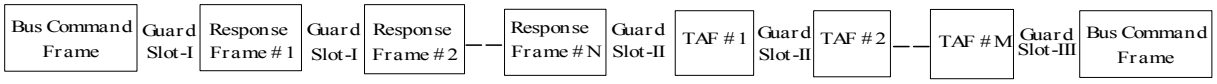


FIGURE 48. Generic compound transaction structure.

The generic compound transaction shall consist of the following components:

a. A bus command frame (BCF).

b. Guard slot-I following the BCF.

c. A response frame for each individual frame exchange (FE) command in the BCF.

d. Guard slot-I between adjacent response frames.

e. Guard slot-II between the last response frame and the first terminal acknowledgement frame (TAF)

f. A TAF for each response frame if requested by the bus controller as indicated in BCF.

g. Guard slot-II between adjacent TAFs

h. Guard slot-III before the next BCF.

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The BCF shall contain a set of commands from the bus controller to indicate what, how much, and in what sequence information should be sent from a source RT to a destination RT.

The commands shall be of two types:

a. Frame exchange (FE) commands; and

b. Bus notification (BN) commands.

FE commands shall cause RTs to subsequently transmit response frames and TAFs. Corresponding to a FE command in the BCF, each individual response frame can consist of:

a. Terminal response frame (TRF) that carries MAC payload data (non management).

b. MAC management frame (MMF) that contains MAC operations/administration/management (OAM) data.

c. Training frame (TF) that is used to derive the tone map between RTs (including the bus controller).

d. Arbitrary test frame (ATF), vendor specific.

The response frames shall be transmitted by the source terminals before any TAFs are sent by the destination terminals, in the same order as the corresponding FE commands in the BCF.

FE commands shall indicate whether a TAF is required for the associated response frame. This being the case, TAFs shall be transmitted by the destination terminals in the same order as the response frames with the first TAF transmitted after guard slot-II. Actual composition and scheduling of compound transactions is application dependent.

Guard slots are described in 5.4.3.2.3; a summary of guard slot configurations is listed in TABLE XV. The exact value setting on short, medium and long guard slots is application dependent.

TABLE XV. Guard slot configurations.

	Guard slot – I	**Guard slot – II**	**Guard slot - III**
No response frame, or TAF	N/A	N/A	Short/Variable
One response frame, one TAF	Short	Medium	Long/Variable
One response frame, no TAF	Short	N/A	Long/Variable
One response frame, no TAF, if the associated FE command is not the first one in the BCF (not contained in the FC)	Medium	N/A	Long/Variable
One response frame, one TAF, if the associated FE command is not the first one in the BCF (not contained in the FC)	Medium	Medium	Long/Variable
More than one response frame, but only one TAF, corresponding to the last response frame	Short	Medium	Long/Variable
All other cases	Short	Short	Long/Variable

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In applications where data synchronization is required at a higher level, sync BCFs (SBCF)s may be generated periodically. The period between SBCFs may be adjusted precisely by lengthening guard slot-III preceding the next SBCF using the variable guard slot (VGS) bus notification (BN) command as described in 5.4.3.2.2.3.2.1. Guard slot-III shall not be shorter than what is indicated in TABLE XV.

For each unsuccessfully received response frame as indicated by the associated TAF, the bus controller may schedule retransmission by including a retry command in a subsequent compound transaction. The exact handling of retransmission is not specified since it is application specific.

5.4.3.2.2 Frames.

At the command-response protocol level (i.e.the MAC sub-layer of the DLL), the basic unit of MCW information exchange between terminals shall be a PHY frame. The different types of PHY frames are discussed under the following headings:

a. General PHY frame structure.
b. Bus command frame (BCF).
c. Terminal response frame (TRF).
d. Terminal acknowledgement frame (TAF).
e. MAC management frame (MMF).
f. Training frame (TF).
g. Arbitrary test frame (ATF).

5.4.3.2.2.1 General PHY frame structure.

The general structure of a PHY frame is shown in FIGURE 34 on page 76.

A PHY frame shall consist of three sections:

a. A preamble section used for synchronization between terminals.
b. A frame control (FC) section, which contains terminal address information, the length of the frame, etc.
c. A frame payload (PL) section, which contains any data being transmitted between the terminals. The PL is optional depending on the frame type.

5.4.3.2.2.1.1 Preamble section.

This section shall be added by the PHY layer (referenced in 5.4.2.2.1 on page 77).

5.4.3.2.2.1.2 Frame control (FC) section.

The size of the frame control (FC) section is dependent on the diversity order and the number of configured sub-bands with a minimum size of 15 bytes. Transmission shall be by DBPSK modulation, protected with a 16 bit CRC.

The first 15 bytes of the FC shall contain information to determine the type and size of the data encoded in the frame payload section, as shown in TABLE XVI. The reserved bytes shall be set to zero. If the FC is larger than 15 bytes, the remaining bytes past the CRC-16 field shall be padded with zeros but shall not be included in the CRC-16 calculation.

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TABLE XVI. Frame control (FC) fields.

Field	Byte	Bit Number	Number of Bits
Frame type	0	7-5	3
Frame size		4-0	5
Variable field [31-24]	1	7-0	32
Variable field [23-16]	2	7-0	
Variable field [15-8]	3	7-0	
Variable field [7-0]	4	7-0	
Reserved	5	7-0	8
…	…	…	…
Reserved	12	7-0	8
CRC-16 [15-8]	13	7-0	16
CRC-16 [7-0]	14	7-0	
Zero padding	15+	…	Variable

5.4.3.2.2.1.2.1 Frame type (FT) field.

The 3-bit Frame type (FT) field shall specifiy the type of data encoded in the frame payload section as shown in TABLE XVII.

TABLE XVII. Frame type (FT) field.

Value	Name
0	Terminal response
1	Bus command
2	Sync bus command
3	MAC management
4	Terminal acknowledgement
5	Training
6	Arbitrary test
7	Reserved

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5.4.3.2.2.1.2.2 Frame size field.

The Frame size field shall specify the size of the frame payload in 8-symbol blocks. Valid values of this field shall be 1- 30 except for TAFs where this field shall be set to 0.

5.4.3.2.2.1.2.3 Variable field.

The use of the Variable field shall depend upon the frame type. See the sections on frame type as referenced in TABLE XVII for specific use.

5.4.3.2.2.1.2.4 CRC-16 field.

The CRC-16 field shall be used to detect errors in the FC section and shall use the generator polynomial: $G(X) = X^{16} + X^{12} + X^5 + 1$. For CRC-16 calculations, each byte shall be transmitted LSB first. Bytes 0-12 shall be included in the calculation in network order (Byte 0 first). The initial value of the calculation shall be all ones (0xFFFF).

5.4.3.2.2.1.3 Frame payload section.

The optional frame payload section referenced in FIGURE 49 shall consist of a sequence of bytes followed by a 4-byte CRC-32 field for error detection. The usage of the data portion of the frame payload shall depend on the frame type and is described in detail in the corresponding sections. The length of the frame payload section shall be in multiples of PHY blocks.

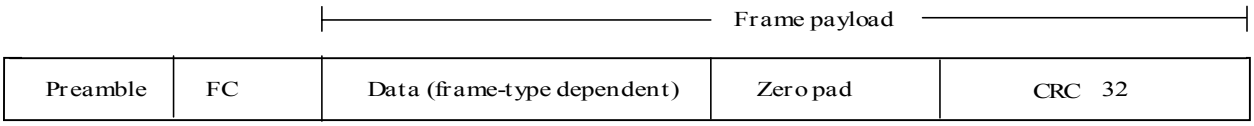


FIGURE 49. Frame payload section.

Unless otherwise specified, all multiple byte fields in the frame payload shall be defined in network byte order (most significant byte transmitted first).

5.4.3.2.2.1.3.1 Zero pad.

If the entire usable space of the frame payload is not used, the unused bytes shall be filled with zeroes.

5.4.3.2.2.1.3.2 CRC-32 field.

The CRC field shall always occupy the last 4 bytes of the frame payload section except for ATF, and shall use the generator polynomial:

$$G(X) = X^{32} + X^{26} + X^{23} + X^{22} + X^{16} + X^{12} + X^{11} + X^{10} + X^8 + X^7 + X^5 + X^4 + X^2 + X + 1.$$

For CRC-32 calculations, each byte shall be transmitted least significant bit LSB first. All bytes from the header and data sections are included in the calculation in network order (Byte 0 first). The initial value of the calculation shall be all ones (0xFFFFFFFF).

5.4.3.2.2.2 Bus command frame (BCF).

The BCF shall be used to convey commands from the bus controller to all terminals on the bus. The BCF may contain from 1 to 16 commands that define the bus schedule of the compound transaction. The BCF shall be transmitted using BR modulation to ensure that it can be received by all terminals.

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The order of the commands shall determine the transmission order of the response frames and corresponding TAFs, during the compound transaction. Each bus command shall contain information regarding:

a. Which terminal is to transmit.

b. How much data is to be transmitted.

c. The destination terminal for the transmission.

d. How long the transmission is to last, and

e. Whether the transmission is to be a retry of an earlier unsuccessful transmission.

Each bus command shall be 4 bytes long. The first command shall be contained in the variable field of the FC section with additional commands, if present, contained in the frame payload. The structure of the BCF is shown in FIGURE 50.

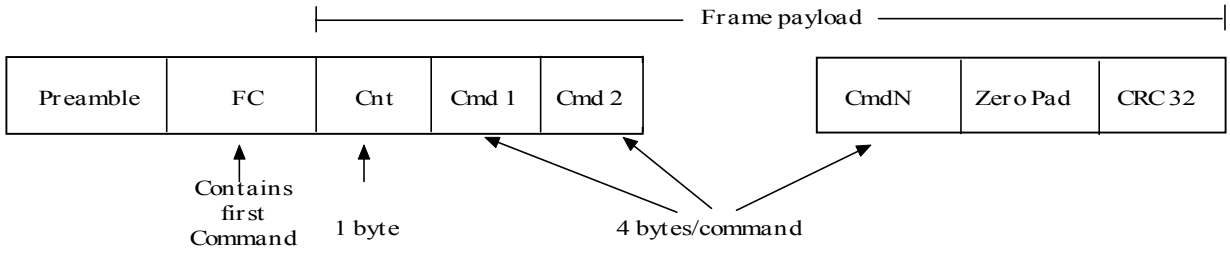


FIGURE 50. Bus command frame (BCF) structure.

The size of the BCF shall be configured at startup and remain fixed during operation. The frame payload size shall be in multiples of basic rate PHY blocks and shall be at least one PHY block. Zero padding shall be added as necessary to position the CRC-32 field in the last 4 bytes of the frame payload.

5.4.3.2.2.2.1 Frame control fields.

For BCFs, the FC frame type field shall be set to either 1 (bus command) or 2 (sync bus command). The variable field shall contain the first 4-byte command as referenced in the following sections.

For conveying bus commands, the two BCF types shall have identical functions. Type 2 BCFs, or sync BCFs (SBCF)s shall provide the additional function of ensuring data synchronization among terminals on a bus by providing more precise timing characteristics.

SBCFs shall be transmitted periodically at a precisely programmed rate by the bus controller. An SBCF shall indicate the start of a group of compound transactions as well as a synchronization point for packet adaptation layer (PAL) operations. Refer to Section 5.4.4 on page 107. A variable number of type 1 or “regular” BCFs shall be transmitted between SBCFs to define the precise composition of compound transactions to fill the gap.

Elsewhere in this document, the term BCF refers to either type of BCF unless otherwise specified.

5.4.3.2.2.2.2 Count field.

The count field shall indicate how many additional commands are contained in the frame payload as shown in TABLE XVIII. Valid values shall be 0 to15. The four MSBs of the field are reserved and shall be set to zero. The count field shall be followed by additional 4-byte commands.

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TABLE XVIII. BCF count field.

Field	Byte	Bit Number	Bits	Definition
Reserved	0	7-4	4	Set to zeroes
Count	0	3-0	4	Number of additional bus commands (0-15)

5.4.3.2.2.2.3 Bus commands.

Each 4-byte command in the BCF shall initiate an action or send a notification on the bus. The command type field in the upper 3 bits of the first byte shall determine the format of the remaining bits and the definition of the action to be executed as follows:

a. Command types 1-5 shall be frame exchange commands and cause a frame exchange to be scheduled.

b. Command type 6 shall be a notification command and shall not cause a frame exchange to be scheduled. It shall be used to convey information contained within the command from the bus controller to the RTs.

Valid command type values are shown in TABLE XIX.

TABLE XIX. Bus command frame (BCF) command categories.

Command Type	Definition	Description
0	Reserved	Reserved
1	Terminal response	Source RT transmits a TRF to destination RT.
2	Training	Source RT transmits a TF to destination RT
3	Tone map	Source RT transmits a tone map MMF to destination RT.
4	MAC management (MM)	Source RT transmits a non-tone map MMF to destination RT
5	Arbitrary test frame (ATF)	Source RT transmits an ATF to destination RT
6	Bus notification	bus controller notification to RTs
7	Reserved	Ignored by RTs.

5.4.3.2.2.2.3.1 Frame exchange (FE) commands.

Command type 1-5 commands shall be FE commands and have the format described in TABLE XX.

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TABLE XX. Frame exchange (FE) command fields.

Field	Byte	Bit Number	Bits	Applies to Command Type
Command type	0	7-5	3	N/A
PHY block count		4-0	5	1,2,3,4,5
Rx bus	1	7	1	1,2,3,4,5
Acknowledgement flag		6	1	1,2,3,4,5
Source RT		5-0	6	1,2,3,4,5
Destination RT	2	7-2	6	1,2,3,4,5
Tone map index		1-0	2	1,3,4
Queue ID	3	7-1	7	1
Retry flag/ clock sync only		0	1	1,2

5.4.3.2.2.2.3.1.1 PHY block count field.

The PHY block count field shall specifiy the number of 8-symbol PHY blocks in the frame payload section of the scheduled PHY frame. Valid values shall be 1-30.

5.4.3.2.2.2.3.1.2 Rx bus field.

The Rx bus field shall specifiy the bus used by the destination RT to receive the frame. A value of 0 shall indicate the A bus, and a value of 1, the B bus.

5.4.3.2.2.2.3.1.3 Acknowledgement flag field.

The acknowledgement flag field shall be set to 1 to command the destination RT to transmit a TAF back to the bus controller, indicating the receive status of a scheduled frame. No acknowledgement shall be transmitted if this field is set to 0. This bit shall be set to 0 for broadcast frames.

5.4.3.2.2.2.3.1.4 Source RT field.

The source RT field shall specify the source terminal of the response frame. Valid values shall be 0-62.

5.4.3.2.2.2.3.1.5 Destination RT field.

The destination RT field shall specify the destination terminal of the response frame. Valid values shall be 0-62 for unicast frames. A value of 63 (0x3f) shall indicate that all RTs shall process the received frame with the basic rate tone map and with the acknowledgement flag set to 0.

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5.4.3.2.2.2.3.1.6 Tone map index field.

The tone map Index field shall indicate the tone map to be used by the source RT to send the response frame to the given destination RT. The source RT shall maintain independent tone maps for each destination RT so the index shall apply only to the specified source-destination RT pair. Valid values are referenced in TABLE XXI.

TABLE XXI. Tone map index field.

Tone Map Index	Definition	Description
0	Basic rate	Use basic rate tone map. This is understood by all RTs and must be used for broadcast frames.
1	VBR 1	Use VBR tone map 1 configured for the source-destination RT pair.
2	VBR 2	Use VBR tone map 2 configured for the source-destination RT pair.
3	VBR 3	Use VBR tone map 3 configured for the source-destination RT pair.

5.4.3.2.2.2.3.1.7 Queue ID field.

The Queue ID field shall indicate which queue to use for sourcing data. Valid values shall be 0-127.

5.4.3.2.2.2.3.1.8 Retry/clock sync only flag.

The retry/clock sync only flag shall serve two purposes depending on the command type.

a. For terminal response frames (TRF)s command type 1, the retry/clock sync only flag shall serve as a retry flag where a value of 1 indicates that the scheduled frame is a retry of a previous transmission.

b. For training frames (TF)s, command type 2, the retry/clock sync only flag shall serve as a clock sync only flag, where a value of 1 indicates that the TF is used for synchronization only and not for tone map generation.

5.4.3.2.2.2.3.2 Bus notification commands.

Command Type 6 commands shall be bus notification commands with the format described in TABLE XXII.

TABLE XXII. Bus notification command fields.

Field	Byte	Bit Number	No. of Bits	Definition
Command type	0	7-5	3	Set to 6
Notification subtype		4-0	5	See TABLE XXIII
Target RT	1	7-2	6	Command target
Reserved		1-0	2	Zeros
Reserved	2	7-0	8	Zeros
Reserved	3	7-6	2	Zeros
Variable guard slot [5-0]		5-0	6	Variable guard slots

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5.4.3.2.2.2.3.2.1 Notification subtype field.

The Notification subtype field shall define the action described in TABLE XXIII.

TABLE XXIII. Notification subtype field.

Notification Subtype	Definition	Description
0	Null event	This notification shall be used when the bus controller has no other command to broadcast. The bus controller shall continuously transmit BCFs to fill the bus and maintain data synchronization with RTs. The RTs shall take no action following reception of this notification subtype.
1	Variable guard slot	This notification shall direct the RTs to override the next scheduled guard slot with a variable guard slot duration specified in the variable guard slot field. The override shall occur once. The duration shall be configurable to 1024k + (1024 – 16) samples, where 0 ≤k≤63. The target RT shall be set to broadcast. This notification shall be transmitted as the last or only bus command in the regular BCF immediately preceding the next SBCF in order to adjust the SBCF transmission period precisely.
2 to 31	Reserved	

5.4.3.2.2.3 Terminal response frame (TRF).

The Terminal response frame (TRF) shall contain the transmitted data that was requested by a single command from the bus controller. Its structure is shown in FIGURE 51.

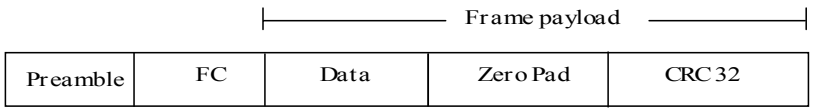


FIGURE 51. TRF structure.

5.4.3.2.2.3.1 Frame control fields.

For TRFs, the FC frame type field shall be set to 0 (terminal response). The variable field is unused and shall be set to 0.

5.4.3.2.2.3.2 Data section.

The data to be transmitted shall be contained in the data section. This data shall be extracted from the source RT Queue ID specified in the most recent BCF by the bus controller. The destination RT shall receive the frame by listening concurrently on the bus.

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5.4.3.2.2.3.3 Unicast frames.

When a terminal response frame (TRF) is destined for a unique destination, the bus controller may command the destination RT to send a TAF during the acknowledgement phase of the compound transaction. This shall be achieved by setting the acknowledgement flag to 1 in the FE command.

The TRF shall contain data that is being sent between two terminals on the network. The optimum encoding between those two terminals shall be determined by the unique network characteristics between those terminals. Thus, the data encoding used in the TRF may be specific to the source-destination pair between which data is being exchanged as defined by the tone map used.

5.4.3.2.2.3.4 Broadcast frames.

When a terminal response frame (TRF) is destined for the broadcast address (63 or 0x3F), the bus controller will suppress the generation of TAFs during the acknowledgement phase of the compound transaction. This will be achieved by setting the acknowledgement flag to 0 in the frame exchange command. The terminal response frame shall use the basic rate tone map to ensure that all RTs can receive the broadcast frame.

5.4.3.2.2.4 Terminal acknowledgement frame (TAF).

The terminal acknowledgement frame (TAF) shall be used to indicate whether a response frame was received successfully. As shown in FIGURE 52, the TAF has no frame payload.

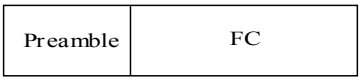


FIGURE 52. TAF structure.

5.4.3.2.2.4.1 Frame control fields.

For terminal acknowledgement frames (TAF)s the FC frame type field shall be set to 4 (terminal acknowledgement). The variable field shall be used to convey the frame reception status to the bus controller. Valid values are described in TABLE XXIV.

TABLE XXIV. TAF variable field status code.

Bit Number	Bits	Definition
31-29	3	See TABLE XXV
28-0	29	Reserved

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TABLE XXV provides the TAF response types.

TABLE XXV. Terminal acknowledge response types.

Response Type	Name	Description
0	ACK	For TRFs and MMFs, a TAF indicates that the frame was received with correct FC and frame payload CRCs. For TFs and ATFs, a TAF indicates that the frame was received with a correct FC CRC.
1	NACK	The TRF or MMF was not received successfully due to a frame payload CRC error.
2	FAIL	The TRF or MMF had correct FC and frame payload CRCs but the destination RT did not currently have the resources to process the frame.
3	NO_RX	The destination RT did not receive the expected frame or the FC CRC is incorrect.
4-7	Reserved	

5.4.3.2.2.5 MAC management frame (MMF).

MAC management frames (MMF)s shall be handled in a similar way to terminal response frames (TRF)s but shall be used for MAC management (MM) communications instead of application communications. The frame payload section shall contain a MM message, which shall be forwarded to the destination MM entity.

MM messages shall be used to exchange management and test messages between terminals. The format of the MM message is shown in FIGURE 53.

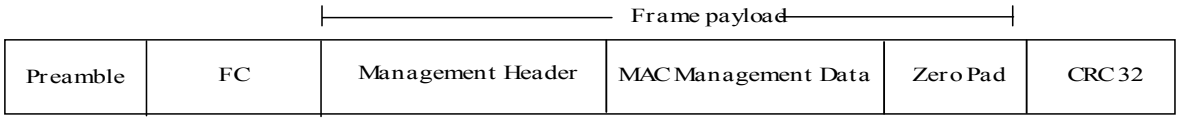


FIGURE 53. MM message format.

MMFs shall be transmitted using the BR tone map to ensure that they can be demodulated by all RTs.

5.4.3.2.2.5.1 Frame control fields.

For MMFs, the FC Frame Type field shall be set to 3 (MAC Management). The Variable Field is unused and shall be set to 0.

5.4.3.2.2.5.2 Management header fields.

The 4-byte message header is described in TABLE XXVI.

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TABLE XXVI. MM header fields.

Field	Byte	Bit Number	Bits	Definition
Reserved	0	7-0	8	Set to zeroes.
Message version	1	7-6	2	Message version
Message type		5-0	6	Message type
Message data length	2	7-0	8	Message data length [15-8]
	3	7-0	8	Message data length [7-0]

5.4.3.2.2.5.2.1 Message version field.

The message version field shall be used to differentiate between this version of the protocol and other past or future versions. It is currently set to 0.

5.4.3.2.2.5.2.2 Message type field.

The message type field shall indicate the contents of the message. Message types are described in TABLE XXVII.

5.4.3.2.2.5.2.3 Message data length.

The message data length field shall indicate the number of valid bytes in the data portion of the message. The length does not include the management header, zero pad and CRC fields.

TABLE XXVII. MM message types.

Message Type	Name
0	Tone map
1	Sync bus command frame count
2	Test
3 to 63	Reserved

5.4.3.2.2.5.3 Message types.

5.4.3.2.2.5.3.1 Tone map message.

A Tone map message shall be a variable length message which specifies the modulation parameters of the reverse path, to be used by the destination RT in a future transmission of frames back to the source RT. The tone map message format is shown in TABLE XXVIII.

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TABLE XXVIII. Tone map message.

Field	Byte	Bit Number	Bits	Definition
Reserved	0	7-4	4	Set to zeroes.
PUNC		3-2	2	Puncturing rate: 0=1/2 1=3/4 2=7/8 3=Reserved
TMI		1-0	2	Tone map index: 0=BR 1=VBR1 2=VBR2 3=VBR3
TONMOD	1	7-6	2	Tone 2 Modulation [1-0]
		5-3	3	Tone 1 Modulation [2-0]
		2-0	3	Tone 0 Modulation [2-0]
TONMOD	2	7	1	Tone 5 Modulation [0]
		6-4	3	Tone 4 Modulation [2-0]
		3-1	3	Tone 3 Modulation [2-0]
		0	1	Tone 2 Modulation [2]
TONMOD	3	7-5	3	Tone 7 Modulation [2-0]
		4-2	3	Tone 6 Modulation [2-0]
		1-0	2	Tone 5 Modulation [2-1]
TONMOD	N	...	...	...
		...	...	...
		...	...	...

5.4.3.2.2.5.3.1.1 Tone modulation field (TONMOD).

Each tone shall be defined by a 3-bit tone modulation field. These fields shall be packed into as many bytes as required, starting from the LSB of byte 1 of the message. Eight tones are thus defined every three bytes. The first tone and the number of consecutive tones used shall be defined at RT configuration time and are the same for all RTs on the same bus. The tone 0 modulation field shall apply to the first tone (lowest frequency) used.

The tone modulation field defines the modulation type as shown in TABLE XXIX.

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TABLE XXIX. Tone modulation types.

TONMOD Value	Bits/Tone	Description
0	0	Tone is unused
1	1	Coherent BPSK
2	2	Coherent 4-QAM
3	4	Coherent 16-QAM
4	6	Coherent 64-QAM
5	8	Coherent 256-QAM
6	10	Coherent 1024-QAM
7	12	Coherent 4096-QAM

5.4.3.2.2.5.3.2 Sync bus command frame (SBCF) count.

This 4-byte message shall contain a count of the number of SBCFs transmitted by the bus controller as referenced in TABLE XXX. This count shall be used by RTs to determine whether it is data synchronized with the bus controller. Only the bus controller shall be permitted to send this message and it shall use the broadcast RT address.

TABLE XXX. Sync bus command frame (SCBF) count field.

Field	Byte	Bit Number	Bits	Definition
Sync bus command frame count	0	7-0	8	Count [31-24]
	1	7-0	8	Count [23-16]
	2	7-0	8	Count [15-8]
	3	7-0	8	Count [7-0]

5.4.3.2.2.5.3.3 Test message.

Test messages shall be variable length messages which provide means of invoking test functions (commands) and retrieving results (responses). When a test function is invoked, the unit under test (UUT) shall respond to indicate that it has started a test but it may take itself off-line afterward and may ignore further communications until the test function is completed.

The test message format is shown in TABLE XXXI.

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TABLE XXXI. Test message.

Field	Byte	Bit Number	Bits	Definition
RESP	0	7	1	Response bit
TESTCMD	0	6-0	7	Test command
TRANSID	1	7-0	8	Transaction ID
P1	2	7-0	8	Parameter 1
P2	3	7-0	8	Parameter 2
REGAUTH	4	7-0	8	Registration authority (optional)
VID[0-2]	5	7-0	24	Unique vendor ID (Optional)
	6	7-0		
	7	7-0		
VDATA	8+	…	…	Vendor-specified data (optional)

5.4.3.2.2.5.3.3.1 Response bit (RESP).

The response bit shall indicate whether the message is an invocation of the test command (RESP=0) or a response to a test command (RESP=1).

5.4.3.2.2.5.3.3.2 Test command field (TESTCMD).

The test command field shall indicate the test function the message pertains to, as described in TABLE XXXII.

TABLE XXXII. Test command (TESTCMD) field.

TESTCMD	Description
0	Retrieve part identification
1	Start test
2	Retrieve test results
3-127	Reserved

5.4.3.2.2.5.3.3.3 Transaction ID field.

The transaction ID (TRANSID) shall be a random number from 0 to 255 that shall be used to correlate a test response to the test command that invoked the function. The UUT shall copy this field from the test command to the test response message.

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5.4.3.2.2.5.3.3.4 Parameter fields.

The Parameter fields (P1, P2) extend the test commands and provide a method of returning results summaries. The meaning of these fields shall be dependent on the test command invoked as described in TABLE XXXIII.

TABLE XXXIII. Parameter field usage.

Test Message	P1	P2
Retrieve part identification	Page number	Page size
Start test command	0 = Reserved	0
	1 = SELF_TEST	0
	2-127 = Reserved	Reserved
	128-255 = Vendor-specific	Vendor-specific
Start test response	0 = Test invoked 1 = Error	0
Retrieve test results command	0	0
Retrieve test results response	Number of tests passed	Number of tests failed

A SELF_TEST command shall cause the UUT to run a comprehensive set of tests able to detect at least 95% of terminal faults. Refer to 5.2.2.7 on page 23.

5.4.3.2.2.5.3.3.5 Registration authority field (REGAUTH).

External standards bodies such as IEEE and PCI-SIG are registration authorities that assign and maintain lists of vendor-unique identifiers. The Registration authority (REGAUTH) field combined with a corresponding vendor-unique identifier is guaranteed to be a unique 32 bit number as a vendor-unique data identifier. The usage of the REGAUTH and VID fields is described in TABLE XXXIV

If REGAUTH is 0 (MIL-STD), the meaning of the VID field is defined as herein. The format of the VDATA field is dependent on the VID field.

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TABLE XXXIV. Registration authority (REGAUTH) and vendor ID (VID) fields.

Description	REGAUTH	VID[0]	VID[1]	VID[2]
MIL-STD	0	0	0x0000 = UII 0x0001 = reserved ... 0xFFFF = reserved	
IEEE Organizationally unique ID (OUI)	1	24 bit OUI		
IEEE Individual address block (IAB)	2	0	0000 + 12 bit IAB	
PCI Vendor ID (VID)	3	0	16 bit VID	
Reserved	4 -255	N/A		

The REGAUTH and VID fields, along with the following VDATA are optional if the parameter fields are sufficient for conveying test information.

5.4.3.2.2.5.3.3.6 Vendor data (VDATA).

The optional vendor data area will contain vendor-specific test information. Its format is defined by the equipment vendor.

5.4.3.2.2.5.3.3.7 Unique item Identification.

The unique item identifier (UII) shall be used for asset management purposes throughout the life cycle of a terminal. It shall be returned as a result of a retrieve part identification command message (TESTCMD = 0). A REGAUTH of 0 (MIL-STD) and a VID of 0 (UII) in the response message shall indicate that the VDATA area contains UII information presented as a null terminated UII string, which is typically in extensible markup language (XML) format. The format and content of the UII string is not defined in this standard.

The entire UII string may be too large to fit into a single test message in which case the page number and page size parameters shall allow retrieval of long strings using multiple pages. The page size shall indicate the size of a page in words (4 bytes) and the page index shall indicate the requested page. The UII string shall be the concatenation of consecutive pages of non-null bytes starting from page 0. The last valid page shall contain the remainder of the UII string, if any, followed by a terminating null byte. A request for a page beyond the valid portion of the UII string shall return all zero data to indicate the end of the string.

5.4.3.2.2.6 Training frame (TF).

Training frames (TF)s shall be handled in a similar way to terminal response frames (TRF)s but used to aid in determining optimum tone maps for a given source-destination RT pair. The frame payload section, including the space reserved for the CRC-32 field, shall contain training generator data that modulates to pre-determined signals.

5.4.3.2.2.7 Arbitrary test frame (ATF).

Arbitrary test frames (ATF)s shall be used to send arbitrary (vendor specific) test data in the frame payload section of the PHY frame. The entire frame payload section, whose length is indicated in the Frame control (FC) section, shall be available for use including the space reserved for the CRC-32 field.

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5.4.3.2.3 Guard slots.

Guard slots are minimum idle periods between frames that are required to accommodate channel and processing delays between terminals.

Three types of guard slots shall be defined at system startup. A fourth type, the variable guard slot (VGS), shall be dynamically defined during the system operation.

a. Short guard slot (SGS)

b. Medium guard slot (MGS)

c. Long guard slot (LGS)

d. Variable guard slot (VGS)

5.4.3.2.3.1 Short guard slot (SGS).

SGSs are required in order to cope with the unequal channel delays from different RTs. The duration shall be configurable to 1024k + (1024 -16) samples, where $0 \leq k \leq 63$. Short guard slots shall be used exclusively except for situations described in 5.4.3.2.3.2 to 5.4.3.2.3.4.

5.4.3.2.3.2 Medium guard slot (MGS).

MGSs are required in order to accommodate receiver processing latencies. The duration shall be configurable to 1024k + (1024 -16) samples, where $0 \leq k \leq 63$. MGS shall be used instead of an SGS under the following conditions:

a. Between a response frame and its associated acknowledgement frame if they are adjacent in the compound transaction.

b. Between a BCF and the first response frame if its associated FE command is contained in the frame payload section of the BCF.

5.4.3.2.3.3 Long guard slot (LGS).

LGSs are required in order to accommodate receiver processing latencies following completion of a compound transaction. The duration shall be configurable to 1024k + (1024 -16) samples, where $0 \leq k \leq 63$. LGS shall be used instead of SGS or MGS between the last non-BCF and the next BCF.

5.4.3.2.3.4 Variable guard slots (VGS).

In order to maintain a precise period between SBCFs, the guard slot between the next SBCF and the preceding MAC frame shall be adjustable. The bus controller shall dynamically transmit a variable guard slot (VGS) notification command in the last BCF sent before the next SBCF to inform the RTs to change the duration of the last guard slot. Refer to section 5.4.3.2.2.2.3.2 for details on notification commands.

5.4.4 Packet adaptation layer (PAL) services.

5.4.4.1 Overview of PAL services.

The Packet Adaptation Layer (PAL) provides peer logical link control (LLC) entities with the ability to exchange PAL service data units (PAL-SDUs) and uses the underlying services offered by the physical layer (PHY). This layer performs segmentation and reassembly between the MAC sublayer and the LLC sublayer.

The PAL receives PAL-SDUs from the LLC and passes MAC-SDUs to the MAC sublayer in the egress direction, and receives MAC-SDUs from the MAC sublayer and passes PAL-SDUs to the LLC in the ingress direction. At this layer, the timing of PAL-SDU transfers is not tied to the bus controller.

FIGURE 54 shows the context of the PAL within the OSI Layer 2.

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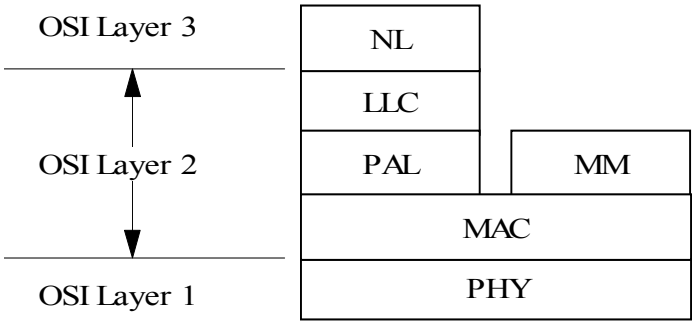


FIGURE 54. Packet adaptation layer context.

This sublayering is necessary because the LLC layer expects to be able to transmit and receive arbitrary length packets of its choosing but the MAC layer imposes a rigid set of constraints over the timing and length of the transmission as dictated by the bus controller.

PAL service data units (PAL-SDUs) are transmitted using the payload section of the terminal response frames (TRF)s.

PAL operation shall be coordinated with sync bus command frame (SBCF) reception. Upon arrival of a SBCF, any incomplete segmentation and reassembly operations shall be discarded and restarted.

Segmentation and reassembly operations shall not be affected by regular BCFs, allowing PAL-SDUs to be transmitted across multiple MAC frames.

The PAL sublayer functions are illustrated in FIGURE 55.

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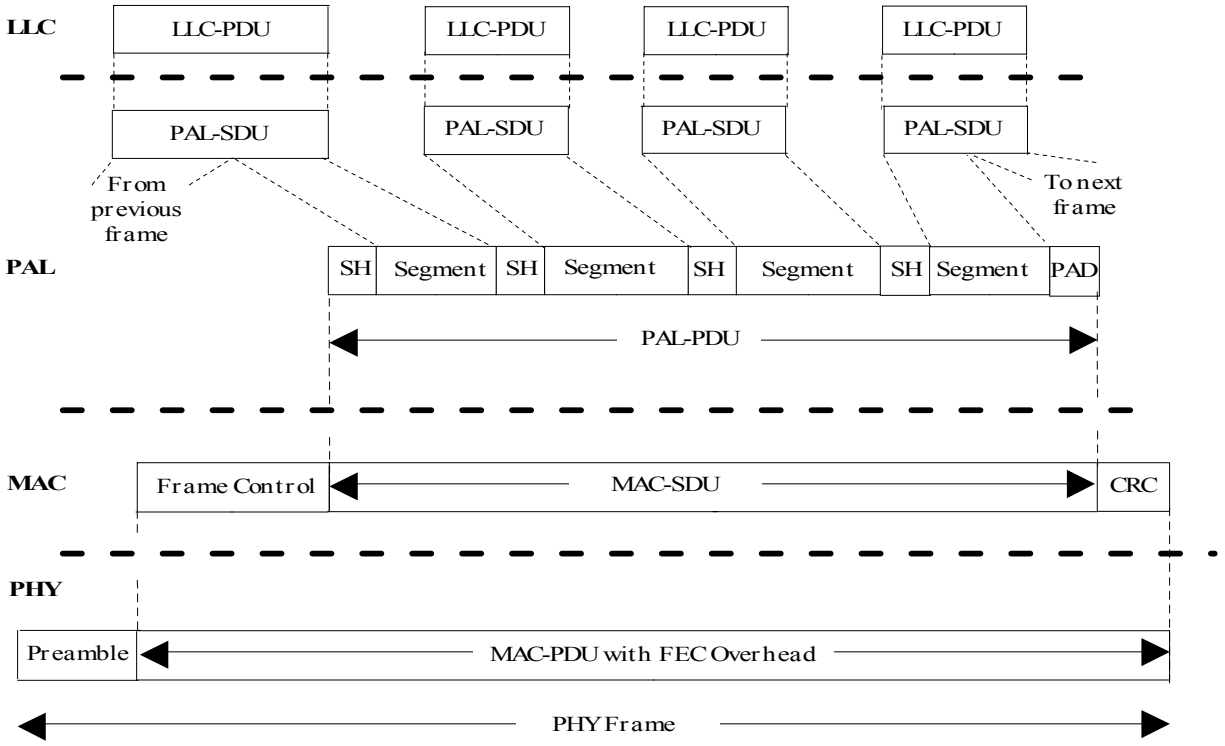


FIGURE 55. PAL sublayer functions.

5.4.4.2 MAC management (MM) entity.

The MM entity shall handle bus controller/RT communications. From a PAL perspective, the MM shall handle data synchronization and retries due to transmission errors. The MM may function as a bus controller or an RT.

An MM entity functioning as a bus controller shall be responsible for sending BCFs and receiving TAFs according to a given bus schedule. It shall be cognizant of the status of all frame exchanges on the bus and may request transmission retries or aborts as required.

An MM entity functioning as an RT receives BCFs and programs its MAC sublayer to generate the appropriate primitives to the PAL sublayer.

5.4.4.3 Egress operation.

The packet adaptation layer (PAL) sublayer shall provide the means to queue PAL service data units (PAL-SDU) data asynchronously to the bus and shall allow the MAC layer to demand data from a specified queue when a scheduled frame exchange arrives. It shall also provide the means to retry and abort PAL-SDUs already queued as required. The PAL sublayer shall aquire PAL-SDUs, segment them if necessary, and fit them into MAC service data units (MAC-SDUs). The PAL sublayer shall optimize bus utilization by attempting to fill the MAC-SDU to maximum capacity. FIGURE 56 shows the MAC-SDU structure.

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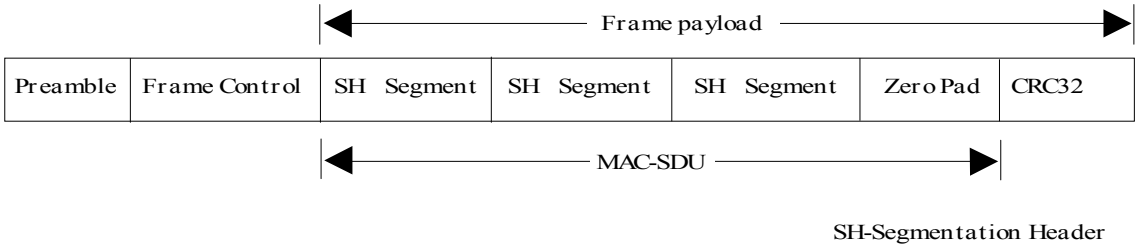


FIGURE 56. MAC-SDU structure.

A MAC-SDU may contain, in order:

a. The end segment of the previous PAL-SDU.

b. Zero or more full PAL-SDUs.

c. The start segment of the next PAL-SDU.

A TRF may also consist solely one middle segment of a PAL-SDU. The open ends of PAL-SDU segments shall not be adjacent to other PAL-SDUs or PAL-SDU segments within a TRF.

The MAC sublayer shall command the PAL sublayer to output MAC-SDUs from a given queue, as determined by the bus schedule, to be passed to the PHY layer as TRFs.

In the event of unsuccessful transmission, the MAC sublayer may request the PAL sublayer to retry a previous transmission at the command of the bus controller.

5.4.4.3.1 Segmentation headers (SH).

Each segment in a MAC-SDU shall be prepended with a segmentation header (SH) on egress to allow reassembly at the destination. The PAL sublayer shall generate the SH at the time of transmission as it needs to consider the size of the current MAC-SDU in the calculation. The SH may have to be regenerated to allow for changes in MAC-SDU size between retries.

The SH is four bytes long containing the fields defined in TABLE XXXV.

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TABLE XXXV. Segmentation header.

Field	Byte	Bit Number	Bits	Definition
Last header flag	0	7	1	Last header in this MAC-SDU
Last segment flag	0	6	1	Last segment of this PAL-SDU
PAL-SDU number	0	5-0	6	Sequence number of this PAL-SDU
Segment number	1	7-4	4	Sequence number of this segment
Segment length	1	3-0	4	Segment length [11-8]
	2	7-2	6	Segment length [7-2]
	2	1-0	2	Set to zero
Reserved	3	7-0	8	Reserved

5.4.4.3.1.1 Last header flag.

The last header flag shall be set if this segment is the last or only one in the MAC-SDU. Any remaining space in the MAC-SDU shall be padded with zeroes.

5.4.4.3.1.2 Last segment flag.

The last segment flag shall be set if this segment is the last or only one in the PAL-SDU. If all previous segments of the PAL-SDU were received successfully, this segment shall complete the PAL-SDU.

5.4.4.3.1.3 PAL-SDU number.

The PAL-SDU number shall be a 6-bit number incremented for each PAL-SDU. This number shall differentiate among PAL-SDUs to prevent inadvertent reassembly of unrelated segments and shall wrap around from 63 to 0.

5.4.4.3.1.4 Segment number.

The segment number is a 4-bit number incremented for each segment of the current PAL-SDU. The number shall restart at zero for the first or only segment of the PAL-SDU and wrap around from 15 to 0.

5.4.4.3.1.5 Segment length.

The segment length indicates the length of the data portion of this segment in bytes. A segment can be up to 2048 bytes long. The length shall be a multiple of 4 bytes.

Segment headers shall always be aligned to 4-byte boundaries relative to the start of the frame payload section. Segments shall be padded to 4-byte boundaries with zeros if necessary to ensure that subsequent segment headers are 4-byte aligned.

Depending on the size of the frame payload section, additional pad bytes past the last segment of the MAC-SDU may be required to fill unused space before the CRC-32 field. It should be noted that the CRC-32 field is not necessarily 4-byte aligned relative to the start of the frame payload section.

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Check the source to verify that this is the current version before use.

5.4.4.3.1.6 Null segmentation header (SH).

If the PAL sublayer has no data to transmit when requested, it transmits a null SH to indicate an empty MAC-SDU. This null SH shall have all fields set to zero except for the last header flag, which is set to one to indicate that there are no segments to follow.

5.4.4.4 Ingress operation.

The MAC sublayer shall receive TRFs from the destination RT and pass the resulting MAC-SDU to the PAL sublayer. The PAL sublayer shall strip and reassemble received MAC-SDUs back into PAL-SDUs. Complete PAL-SDUs shall be passed up to the LLC for processing.

5.4.4.5 PAL limits.

The PAL shall have the limits shown in TABLE XXXVI.

TABLE XXXVI. PAL maximum limits.

Parameter	Value
Maximum MAC-SDU size	32,763 bytes
Maximum MAC- SDUs/PAL-SDU	32
Maximum PAL - SDU length	2048 bytes
Maximum PAL - SDUs/MAC-SDU	32

5.4.4.6 PAL data services.

FIGURE 57 shows the interface between the LLC and the PAL.

The following PAL data service primitives shall be provided to the LLC:

a. PAL-SEND.request.

b. PAL-SEND.indication.

c. PAL-RECEIVE.indication.

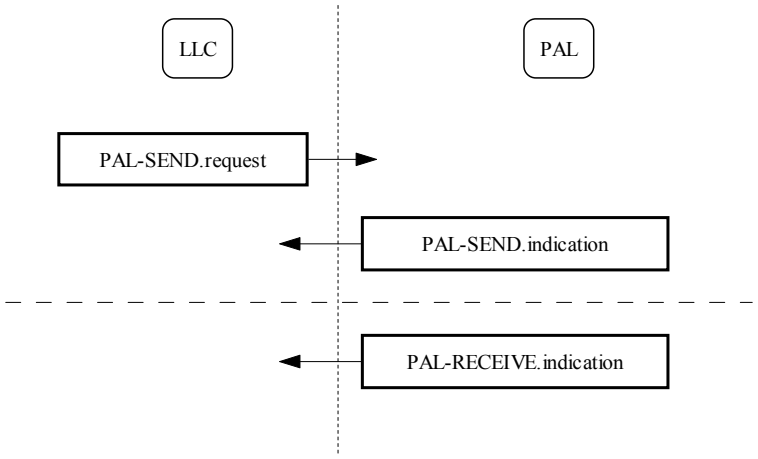


FIGURE 57. PAL data service primitives.

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Check the source to verify that this is the current version before use.

5.4.4.6.1 PAL-SEND.request.

5.4.4.6.1.1 Function.

This primitive shall request the transfer of a PAL-SDU from the LLC layer to the specified destination LLC layer.

5.4.4.6.1.2 Semantics of the service primitive.

PAL-SEND.request (
QueueID,
PAL-SDU Data
)

The QueueID parameter shall indicate the queue on which the PAL-SDU will be sent.

The PAL-SDU Data parameter shall contain the PAL-SDU packet data to be transmitted.

5.4.4.6.1.3 When generated.

This primitive shall be generated by the LLC layer entity whenever a PAL-SDU is to be transferred to a peer LLC layer entity.

5.4.4.6.1.4 Effect of receipt.

The receipt of this primitive shall cause the PAL sublayer to queue PAL-SDU data to the specified queue if there is room for the entire PAL-SDU. A PAL-SEND.indication shall be sent to the LLC.

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Check the source to verify that this is the current version before use.

5.4.4.6.2 PAL-SEND.indication.

5.4.4.6.2.1 Function.

This primitive shall indicate status information from the PAL to the LLC.

5.4.4.6.2.2 Semantics of the service primitive.

PAL-SEND.indication (
QueueID,
Status
)

The QueueID parameter shall indicate the queue that the status indication applies to.

The Status parameter shall contain a status code as referenced in TABLE XXXVII.

TABLE XXXVII. PAL-SEND.indication status codes.

Status Code	Description
ACK	The PAL-SDU has been accepted into the queue.
NACK	The PAL-SDU has not been accepted into the queue because the queue is full or in error.

5.4.4.6.2.3 When generated.

This primitive shall be generated by the PAL entity in response to a PALSEND request.

5.4.4.6.2.4 Effect of receipt.

Action based on the receipt of this primitive is implementation dependent.

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Check the source to verify that this is the current version before use.

5.4.4.6.3 PAL-RECEIVE.indication.

5.4.4.6.3.1 Function.

This primitive shall indicate the arrival of PAL-SDU data from the PAL to the LLC.

5.4.4.6.3.2 Semantics of the service primitive.

PAL-RECEIVE.indication (
PAL-SDU Data
)

The PAL-SDU Data parameter shall contain the PAL-SDU data received.

5.4.4.6.3.3 When generated.

This primitive shall be passed by the PAL sublayer to the LLC entity whenever a complete PAL-SDU has been reassembled successfully.

5.4.4.6.3.4 Effect of receipt.

The LLC shall process the PAL-SDU.

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5.4.5 MAC sublayer.

Operation of the MAC sublayer (MAC) is closely coupled with the bus controller. MAC sublayer services are divided into two types:

MAC data (MD) services – provides the means to transfer application data to and from upper layers.

MAC management (MM) services – provides the means to configure the operation of the bus.

5.4.5.1 MAC data services.

FIGURE 58 shows the interface between the PAL and the MAC.

The following MAC service primitives are provided to the PAL:

a. MD-SEND.command.

b. MD-SEND.response.

c. MD-SEND.indication.

d. MD-RECEIVE.indication.

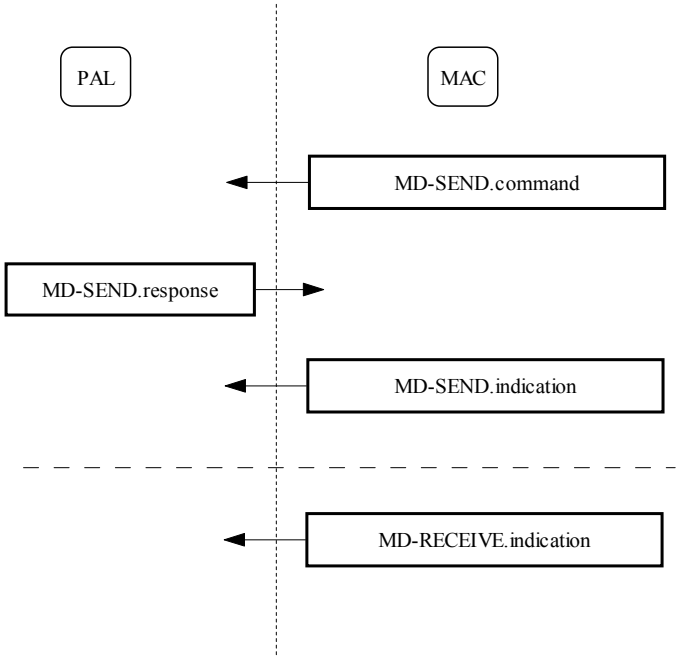


FIGURE 58. MAC data service primitives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.1.1 MD-SEND.command.

5.4.5.1.1.1 Function.

This primitive shall request the transfer of a MAC-SDU from the PAL sublayer to the MAC sublayer.

5.4.5.1.1.2 Semantics of the service primitive.

MD-SEND.command (
QueueID,
Length,
)

The QueueID parameter shall indicate the queue from which MAC-SDU data is requested. The Length parameter shall indicate the length of the scheduled TRF.

5.4.5.1.1.3 When generated.

This primitive shall be generated by the MAC sublayer at the start of a scheduled TRF slot where this RT is the source as configured in the bus schedule.

5.4.5.1.1.4 Effect of receipt.

The receipt of this primitive by the PAL sublayer shall cause it to generate a MD-SEND.response primitive.

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Check the source to verify that this is the current version before use.

5.4.5.1.2 MD-SEND.response.

5.4.5.1.2.1 Function.

This primitive shall pass MAC-SDU data from the PAL sublayer to the MAC sublayer.

5.4.5.1.2.2 Semantics of the service primitive.

MD-SEND.response (
QueueID,
Data
)

The QueueID parameter shall indicate the queue that the response applies to.

The Data parameter shall contain MAC-SDU data. The size of the data shall be equal to the Length parameter of the previously sent MD-SEND.command primitive.

5.4.5.1.2.3 When generated.

This primitive shall be generated by the PAL sublayer in response to a MD-SEND.command primitive. The PAL shall insert SHs as required to adapt PAL-SDU data into PAL-PDUs. Unused space after the last segment shall be padded with zeroes.

5.4.5.1.2.4 Effect of receipt.

The reception of this primitive by the MAC sublayer shall cause it to pass the resulting MAC-PDU to the PHY.

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Check the source to verify that this is the current version before use.

5.4.5.1.3 MD-SEND.indication.

5.4.5.1.3.1 Function.

This primitive shall pass status information from the MAC sublayer to the PAL sublayer regarding the last transmission.

5.4.5.1.3.2 Semantics of the service primitive.

MD-SEND.indication (
QueueID,
Status
)

The QueueID parameter shall indicate the queue that the response applies to.

The Status parameter shall indicate the status of the last transmission as shown in TABLE XXXVIII.

TABLE XXXVIII. MD-SEND.indication status codes.

Status Code	Description
ACK	The last PAL-PDU was received successfully by the destination RT and the PAL may now prepare to send the next PAL-PDU. If the last PAL-PDU contained the last segment of a PAL-SDU and/or contained entire PAL-SDUs, those PAL-SDUs may be marked as sent. An ACK may also be encountered if too many retries have been attempted and the bus controller has decided to move on to the next PAL-PDU. Any PAL-SDUs otherwise considered to have been marked as sent are in fact dropped.
RETRY	The last PAL-PDU was not received successfully by the destination RT. The PAL shall re-send PAL-SDU data starting from the same point in the egress queue as in the previous PAL-PDU.
SYNC	An SBCF was encountered since the last PAL-PDU was sent and the PAL-PDU was received successfully by the destination RT. If the last PAL-PDU contained the last segment of a PAL-SDU and/or contained entire PAL-SDUs, those PAL-SDUs may be marked as sent. The PAL shall re-send PAL-SDU data starting from at least the start of the PAL-SDU not marked as sent. The PAL may choose an earlier PAL-SDU boundary than specified depending on the implementation.
SYNC-RETRY	An SBCF was encountered since the last PAL-PDU was sent and the PAL-PDU was not received successfully by the destination RT. The PAL shall resend PAL-SDU data starting from at least the start of the earliest PAL-SDU not marked as sent, even if segments of that PAL-SDU were sent successfully as part of previous PAL-PDUs. The PAL may choose an earlier PAL-SDU boundary than specified depending on the implementation.

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Check the source to verify that this is the current version before use.

5.4.5.1.3.3 When generated.

When the MM receives a BCF containing terminal reponse FE commands, it shall instruct the MAC sublayer to send this primitive to the PAL sublayer for each applicable FE command. The status code will depend on the state of the FE retry flag and whether a SBCF was received since the previous PAL-PDU transmissions.

5.4.5.1.3.4 Effect of receipt.

The reception of this primitive by the PAL sublayer shall cause it to adjust its queues as described in TABLE XXXVIII.

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Check the source to verify that this is the current version before use.

5.4.5.1.4 MD-RECEIVE.indication.

5.4.5.1.4.1 Function.

This primitive shall pass MAC-SDU data from the MAC sublayer to the PAL sublayer.

5.4.5.1.4.2 Semantics of the service primitive.

MD-RECEIVE.indication (
Data,
)

The Data parameter shall contain the MAC-SDU data received.

5.4.5.1.4.3 When generated.

This primitive shall be generated by the MAC sublayer to the PAL sublayer when a TRF is received from the PHY.

5.4.5.1.4.4 Effect of receipt.

The reception of this primitive by the PAL sublayer shall cause it to queue the MAC-SDU data for reassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.2 MAC management (MM) services.

FIGURE 59 shows the interface between the MM and the MAC.

MAC management services listed below are called to configure operation of the bus:

a. MM-BCMD.command.
b. MM-BCMD.indication.
c. MM-ACK.indication.
d. MM-SEND.command.
e. MM-SEND.response.
f. MM-RECEIVE.indication.
g. MM-TRAIN.command.
h. MM-TRAIN.response.
i. MM-TRAIN.indication.
j. MM-TEST.command.
k. MM-TEST.response.
l. MM-TEST.indication.

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Check the source to verify that this is the current version before use.

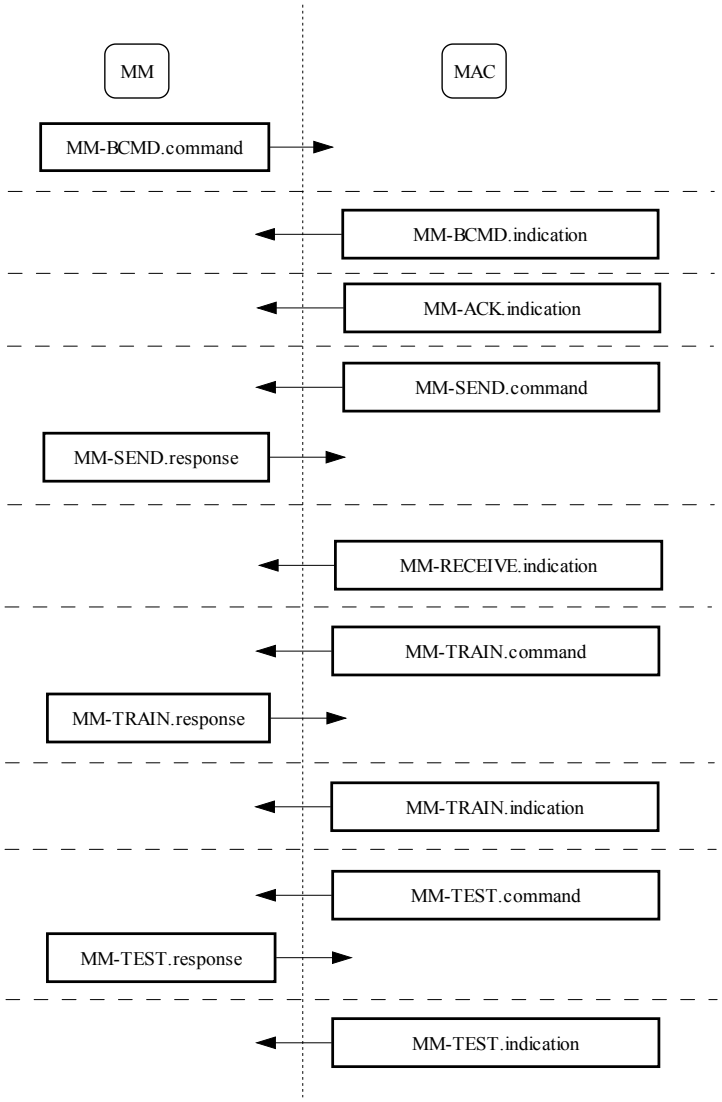


FIGURE 59. MAC management (MM) service primitives.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.2.1 MM-BCMD.command.

5.4.5.2.1.1 Function.

This primitive shall command a terminal configured as a bus controller to send a BCF.

5.4.5.2.1.2 Semantics of the service primitive.

MM-BCMD.command (
Type,
Count,
Command1,
Command2,
...
CommandN
)

The Type parameter shall indicate whether a regular (Type 1) or sync (Type 2) bus command is sent.

The Count parameter shall indicate the number of commands to send. The range shall be from 1 to 16.

The Command parameters shall be a sequence of 4-byte bus commands.

5.4.5.2.1.3 When generated.

This primitive shall be generated by the MM entity at the start of a compound transaction if this terminal is the bus controller.

5.4.5.2.1.4 Effect of receipt.

The reception of this primitive by the MAC sublayer will cause it to send a BCF.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.2.2 MM-BCMD.indication.

5.4.5.2.2.1 Function.

This primitive shall indicate the reception of a BCF from the PHY.

5.4.5.2.2.2 Semantics of the service primitive.

MM-BCMD.indication(
Type,
Count,
Command1,
Command2,
...
CommandN
)

The Type parameter shall indicate whether a regular (Type 1) or Sync (Type 2) BCF was received.

The Count parameter shall indicates the number of commands received. The range shall be from 1 to 16.

The Command parameters shall be a sequence of 4-byte bus commands.

5.4.5.2.2.3 When generated.

This primitive shall be generated by MAC sublayer when a BCF is received from the PHY if this terminal is an RT.

5.4.5.2.2.4 Effect of receipt.

Reception of this primitive by the MM entity, shall cause the structure and actions for the forthcoming compound transaction to be determined. These actions shall pertain to each command contained in the BCF. For each FE command, in which this RT is the source, the MAC sublayer shall generate a corresponding MD-SEND.command, MM-SEND.command, MM-TRAIN.command or MM-TEST.command.

MAC protocol rules shall determine the time within the compound transaction for which the command is generated. Each FE command, for which this RT is the destination, shall indicate the time during which the RT shall listen on the bus. All other FE commands shall serve as placeholders for transmissions not involving this RT, and are necessary for determining compound transaction timing. The MM entity shall also process bus notification (BN) commands.

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Check the source to verify that this is the current version before use.

5.4.5.2.3 MM-ACK.indication.

5.4.5.2.3.1 Function.

This primitive shall pass TAF status from the MAC sublayer to the MM entity.

5.4.5.2.3.2 Semantics of the service primitive.

MM-ACK.indication (
Status
)

The Status parameter shall contain the status code from the TAF as summarized in TABLE XXXIX. For details refer to TABLE XXV on page 100.

TABLE XXXIX. MM-ACK.indication status codes.

Status code	Description
ACK	The frame was received correctly.
NACK	The frame reception encountered an error.
FAIL	
NO_RX	

5.4.5.2.3.3 When generated.

This primitive shall be generated by the MAC sublayer to the MM entity when a TAF is received from the PHY.

5.4.5.2.3.4 Effect of receipt.

The recept of this primitive by the MM entity shall allow it to determine whether frames need to be retried. This primitive is only meaningful if the terminal is a bus controller.

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Check the source to verify that this is the current version before use.

5.4.5.2.4 MM-SEND.command.

5.4.5.2.4.1 Function.

This primitive shall request the transfer of a MMF from the MM entity to the MAC sublayer.

5.4.5.2.4.2 Semantics of the service primitive.

MM-SEND.command (
Length
)

The Length parameter shall indicate the length of the scheduled MMF.

5.4.5.2.4.3 When generated.

This primitive shall be generated by the MAC sublayer at the start of a scheduled MMF slot, where this RT is the source as configured in the bus schedule.

5.4.5.2.4.4 Effect of receipt.

The receipt of this primitive by the MM entity shall cause it to generate a **MM-SEND.response** primitive, in response to both tone map and non-tone map MAC management bus commands. The MM entity shall determine the message type to send according to the order of commands in the last BCF.

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Check the source to verify that this is the current version before use.

5.4.5.2.5 MM-SEND.response.

5.4.5.2.5.1 Function.

This primitive shall pass MMF payload data from the MM entity to the MAC sublayer.

5.4.5.2.5.2 Semantics of the service primitive.

MM-SEND.response (
Data
)

The Data parameter shall contain MMF payload data. The size of the payload shall be equal to the Length parameter of the previously sent **MM-SEND.command** primitive.

5.4.5.2.5.3 When generated.

This primitive shall be generated by the MM entity in response to a **MM-SEND.command** primitive.

5.4.5.2.5.4 Effect of receipt.

The reception of this primitive by the MAC sublayer shall cause it to pass the MMF Frame to the PHY.

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Check the source to verify that this is the current version before use.

5.4.5.2.6 MM-RECEIVE.indication.

5.4.5.2.6.1 Function

This primitive shall pass MMF payload data from the MAC sublayer to the MM entitiy.

5.4.5.2.6.2 Semantics of the service primitive.

MM-RECEIVE.indication (
Data
)

The Data parameter shall contain the payload data received.

5.4.5.2.6.3 When generated.

This primitive shall be generated by the MAC sublayer to the MM entity when a MM frame is received from the PHY.

5.4.5.2.6.4 Effect of receipt.

The reception of this primitive by the MM entity shall cause it to process the message.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.2.7 MM-TRAIN.command.

5.4.5.2.7.1 Function.

This primitive shall request the transfer of a TF from the MM entity to the MAC sublayer.

5.4.5.2.7.2 Semantics of the service primitive.

MM-TRAIN.command (
Length
)

The Length parameter shall indicate the length of the scheduled TF.

5.4.5.2.7.3 When generated.

This primitive shall be generated by the MAC sublayer at the start of a scheduled TF slot where this RT is the source as configured in the bus schedule.

5.4.5.2.7.4 Effect of receipt.

The receipt of this primitive by the MM entity shall cause it to generate a **MM-TRAIN.response** primitive.

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Check the source to verify that this is the current version before use.

5.4.5.2.8 MM-TRAIN.response.

5.4.5.2.8.1 Function.

This primitive shall pass TF data from the MM entity to the MAC sublayer.

5.4.5.2.8.2 Semantics of the service primitive.

MM-TRAIN.response (
Data
)

The Data parameter shall contain the pseudorandom training generator sequence specified in 5.4.2.4 on page 83. This data does not go through the FEC blocks; it shall be inserted directly into the data formatter for modulation.

5.4.5.2.8.3 When generated.

This primitive shall be generated by the MM to the MAC sublayer in response to a MM-TRAIN.command primitive.

5.4.5.2.8.4 Effect of receipt.

The reception of this primitive by the MAC sublayer shall cause it to pass the TF to the PHY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.2.9 MM-TRAIN.indication.

5.4.5.2.9.1 Function.

This primitive shall indicate the reception of a TF from the PHY.

5.4.5.2.9.2 Semantics of the service primitive.

MM-TRAIN.indication (

data

)

The Data parameter shall contain the raw training data as received by the PHY after deformatting but without FEC processing.

5.4.5.2.9.3 When generated.

This primitive shall be generated by the MAC sublayer to the MM entity when a TF is received from the PHY.

5.4.5.2.9.4 Effect of receipt.

The reception of this primitive by the MM entity shall cause it to analyze the received data and calculate the optimum tone map that may be used by the source of the TF to send to this RT. Tone map information shall be transmitted later to the source of the TF using tone map MMFs and put into effect under the control of the bus controller.

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Check the source to verify that this is the current version before use.

5.4.5.2.10 MM-TEST.command

5.4.5.2.10.1 Function.

This primitive shall request the transfer of an ATF from the MM entity to the MAC sublayer.

5.4.5.2.10.2 Semantics of the service primitive.

MM-TEST.command (
Length
)

The Length parameter shall indicate the length in PHY blocks of the scheduled ATF.

5.4.5.2.10.3 When generated.

This primitive shall be generated by the MAC sublayer at the start of a scheduled ATF slot where this terminal is the source as configured in the bus schedule.

5.4.5.2.10.4 Effect of receipt.

The receipt of this primitive by the MM entity shall cause it to generate a **MM-TEST.response** primitive.

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Check the source to verify that this is the current version before use.

5.4.5.2.11 MM-TEST.response.

5.4.5.2.11.1 Function.

This primitive shall pass ATF payload data from the MM entity to the MAC sublayer.

5.4.5.2.11.2 Semantics of the service primitive.

MM-TEST.response (
Data
)

The Data parameter shall contain vendor-specific ATF data. The size of the data shall be such that the duration of the transmission on the bus does not exceed the length parameter of the associated MM-TEST.command primitive

5.4.5.2.11.3 When generated.

This primitive shall be generated by the MM entity in response to a **MM-TEST.command** primitive.

5.4.5.2.11.4 Effect of receipt.

The reception of this primitive by the MAC sublayer shall cause it to pass the ATF to the PHY.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.4.5.2.12 MM-TEST.indication.

5.4.5.2.12.1 Function.

This primitive shall indicate the reception of an ATF from the PHY

5.4.5.2.12.2 Semantics of the service primitive.

MM-TEST.indication (

Data

)

The Data parameter shall contain vendor-specific ATF data.

5.4.5.2.12.3 When generated.

This primitive shall be generated by the MAC sublayer to the MM entity when an ATF is received from the PHY.

5.4.5.2.12.4 Effect of receipt.

The response of the MM entity, upon receipt of an ATF, is vendor specific.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:48Z
Check the source to verify that this is the current version before use.

5.5 Trapezoidal Waveform (TW).

This standard is cognizant of the requirements of MIL-STD-1553B (to Notice 4) in that extended capabilities are added to its existing capabilities. This section therefore contains the requirements and characteristics of the TW and is representative of MIL-STD-1553B (to notice 4).

Note: For reference, all sections in 5.5 including the foreword below, correspond to MIL-STD-1553B (to notice 4). Figures, tables, references and formats contained therein are revised to conform to MIL-STD-001553B Change 6.

E.g. Section 4.3.3.5 in MIL-STD-1553B (to notice 4) corresponds to 5.5.4.3.3.5 under section 5.5 in this document, FIGURE 60 in this section corresponds to FIGURE 1 in MIL-STD-1553B (to notice 4).

The following differences also exist between the MIL-STD-1553B (to notice 4) and the TW specification herein.

a. MIL-STD-1553B (to notice 4) table of contents is not included.

b. MIL-E-6051 Electromagnetic Compatibility Requirements, Systems has been superseded by MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems (reference 5.5.2.1)

c. Unallocated paragraphs in MIL-STD-1553B (to Notice 4) are termed as “reserved”.

FOREWORD

This standard contains requirements for a digital time division command/response multiplex data bus for use in systems integration. Even with the use of this standard, differences may exist between multiplex data buses in different system applications due to particular application requirements and the designer options allowed in this standard. The system designer must recognize this fact and design the multiplex bus controller hardware and software to accommodate such differences. These designer selected options must exist to allow the necessary flexibility in the design of specific multiplex systems in order to provide for the control mechanism, architectural redundancy, degradation concept and traffic patterns peculiar to the specific application requirements. Appendix, Section 5.5.10 selects those options which shall be required and further restricts certain portions of the standard for the use in all dual standby redundant applications for the Army, Navy, and Air Force.

5.5.1 SCOPE

5.5.1.1 Scope.

This standard establishes requirements for digital, command/response, time division multiplexing (Data bus) techniques. It encompasses the data bus line and its interface electronics illustrated on FIGURE 60, and also defines the concept of operation and information flow on the multiplex data bus and the electrical and functional formats to be employed.

5.5.1.2 Application.

When invoked in a specification or statement of work, these requirements shall apply to the multiplex data bus and associated equipment which is developed either alone or as a portion of a weapon system or subsystem development. The contractor is responsible for invoking all the applicable requirements of this Military Standard on any and all subcontractors he may employ.

5.5.2 REFERENCE DOCUMENTS

5.5.2.1 Issue of document.

The following document, of the issue in effect on date of invitation for bid or request for proposal, forms a part of the standard to the extent specified herein.

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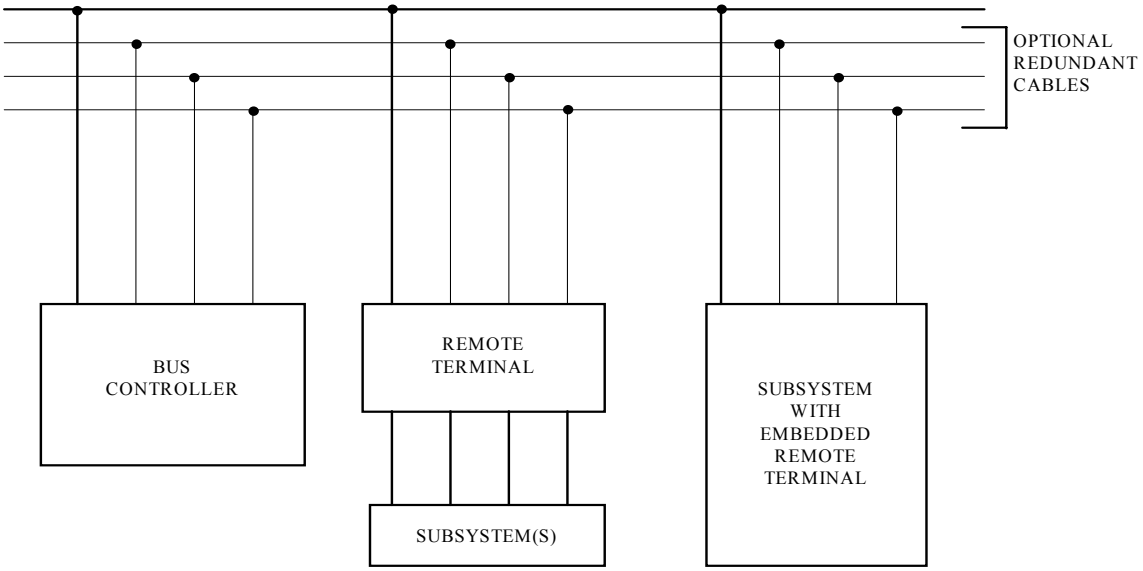


FIGURE 60. Sample multiplex data bus architecture.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-464 Electromagnetic Environmental Effects Requirements for Systems

Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

5.5.3 DEFINITIONS

5.5.3.1 Bit.

Contraction of binary digit: may be either zero or one. In information theory a binary digit is equal to one binary decision or the designation of one or two possible values or states of anything used to store or convey information.

5.5.3.2 Bit rate.

The number of bits transmitted per second.

5.5.3.3 Pulse code modulation (PCM).

The form of modulation in which the modulation signal is sampled, quantized, and coded so that each element of information consists of different types or numbers of pulses and spaces.

5.5.3.4 Time division multiplexing (TDM).

The transmission of information from several signal sources through one communication system with different signal samples staggered in time to form a composite pulse train.

5.5.3.5 Half duplex.

Operation of a data transfer system in either direction over a single line, but not in both directions on that line simultaneously.

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5.5.3.6 Word.

In this document a word is a sequence of 16 bits plus sync and parity. There are three types of words: command, status, and data.

5.5.3.7 Message.

A single message is the transmission of a command word, status word, and data words if they are specified. For the case of a remote terminal to remote terminal (RT to RT) transmission, the message shall include the two command words, the two status words, and data words.

5.5.3.8 Subsystem.

The device or functional unit receiving data transfer service from the data bus.

5.5.3.9 Data bus.

Whenever data bus or bus is referred to in this document it shall imply all the hardware including twisted shielded pair cables, isolation resistors, transformers, etc., required to provide a single data path between the bus controller and all the associated remote terminals.

5.5.3.10 Terminal.

The electronic module necessary to interface the data bus with the subsystem and the subsystem with the data bus. Terminals may exist as separate line replaceable units (LRU)s or be contained within the elements of the subsystem.

5.5.3.11 Bus controller (BC).

The terminal assigned the task of initiating information transfers on the data bus.

5.5.3.12 Bus monitor (BM).

The terminal assigned the task of receiving bus traffic and extracting selected information to be used at a later time.

5.5.3.13 Remote terminal (RT).

All terminals not operating as the bus controller or as a bus monitor.

5.5.3.14 Asynchronous operation.

For the purpose of this standard, asynchronous operation is the use of an independent clock source in each terminal for message transmission. Decoding is achieved in receiving terminals using clock information derived from the message.

5.5.3.15 Dynamic bus control.

The operation of a data bus system in which designated terminals are offered control of the data bus.

5.5.3.16 Command/Response.

Operation of a data bus system such that remote terminals receive and transmit data only when commanded to do so by the bus controller.

5.5.3.17 Redundant data bus.

The use of more than one data bus to provide more than one data path between the subsystems, i.e., dual redundant data bus, tri-redundant data bus, etc.

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5.5.3.18 Broadcast.

Operation of a data bus system such that information transmitted by the bus controller or a remote terminal is addressed to more than one of the remote terminals connected to the data bus.

5.5.3.19 Mode code.

A means by which the bus controller can communicate with the multiplex bus related hardware, in order to assist in the management of information flow.

5.5.4 GENERAL REQUIREMENTS

5.5.4.1 Test and operating requirements.

All requirements as specified herein shall be valid over the environmental conditions which the multiplex data bus system shall be required to operate.

5.5.4.2 Data bus operation.

The multiplex data bus system in its most elemental configuration shall be as shown on FIGURE 60. The multiplex data bus system shall function asynchronously in a command /response mode, and transmission shall occur in a half-duplex manner. Sole control of information transmission on the bus shall reside with the bus controller, which shall initiate all transmissions. The information flow on the data bus shall be comprised of messages which are, in turn, formed by three types of words (command, data, and status) as defined in 5.5.4.3.3.5.

5.5.4.3 Characteristics.

5.5.4.3.1 Data form.

Digital data may be transmitted in any desired form, provided that the chosen form shall be compatible with the message and word formats defined in this standard. Any unused bit positions in a word shall be transmitted as logic zeros.

5.5.4.3.2 Bit priority.

The most significant bit shall be transmitted first with the least significant bits following in descending order of value in the data word. The number of bits required to define a quantity shall be consistent with the resolution or accuracy required. In the event that multiple precision quantities (information accuracy or resolution requiring more than 16 bits) are transmitted, the most significant bits shall be transmitted first, followed by the word(s) containing the lesser significant bits in numerical descending order. Bit packing of multiple quantities in a single data word is permitted.

5.5.4.3.3 Transmission method.

5.5.4.3.3.1 Modulation.

The signal shall be transferred over the data bus in serial digital pulse code modulation form.

5.5.4.3.3.2 Data code.

The data code shall be Manchester II bi-phase level. A logic one shall be transmitted as a bipolar coded signal 1/0 (i.e., a positive pulse followed by a negative pulse). A logic zero shall be a bipolar coded signal 0/1 (i.e., a negative pulse followed by a positive pulse). A transition through zero occurs at the midpoint of each bit time (see FIGURE 61).

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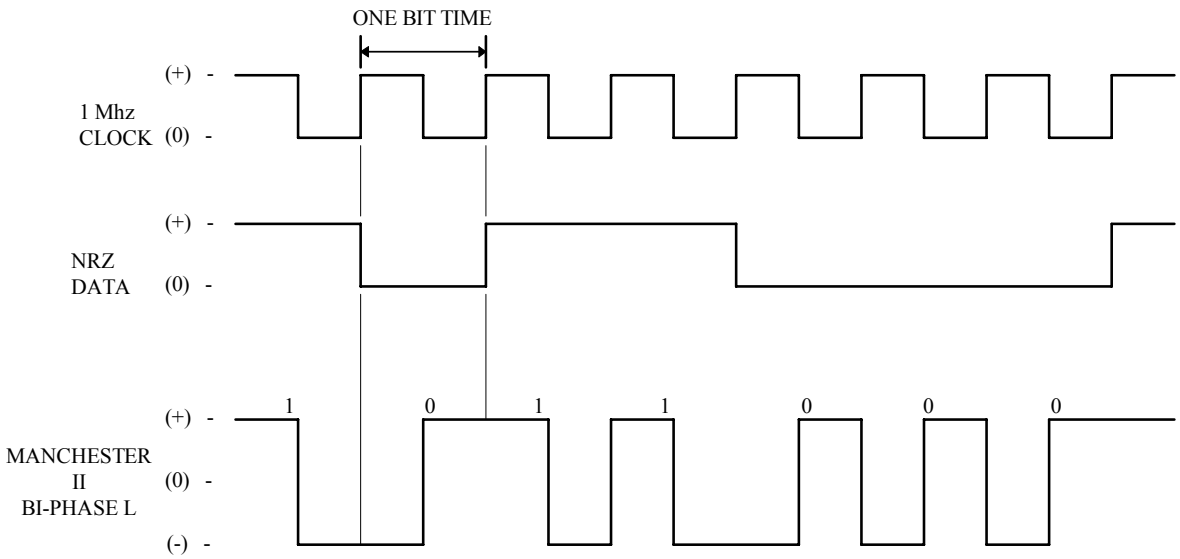


FIGURE 61. Data encoding.

5.5.4.3.3.3 Transmission bit rate.

The transmission bit rate on the bus shall be 1.0 megabit per second with a combined accuracy and long-term stability of ± 0.1 percent (i.e., ±1000 hertz (Hz)). The short-term stability (i.e., stability over 1.0 second interval) shall be at least 0.01 percent (i.e., ±100 Hz).

5.5.4.3.3.4 Word size.

The word size shall be 16 bits plus the sync waveform and the parity bit for a total of 20 bits times as shown on FIGURE 62.

5.5.4.3.3.5 Word formats.

The word formats shall be as shown on FIGURE 62 for the command, data, and status words.

5.5.4.3.3.5.1 Command word.

A command word shall be comprised of a sync waveform, remote terminal address field, transmit/receive (T/R) bit, subaddress/mode field, word count/mode code field, and a parity (P) bit (see FIGURE 62).

5.5.4.3.3.5.1.1 Sync.

The command sync waveform shall be an invalid Manchester waveform as shown in FIGURE 63. The width shall be three bit times, with the sync waveform being positive for the first one and one-half bit times, and then negative for the following one and one-half bit times. If the next bit following the sync waveform is a logic zero, then the last half of the sync waveform will have an apparent width of two clock periods due to the Manchester encoding.

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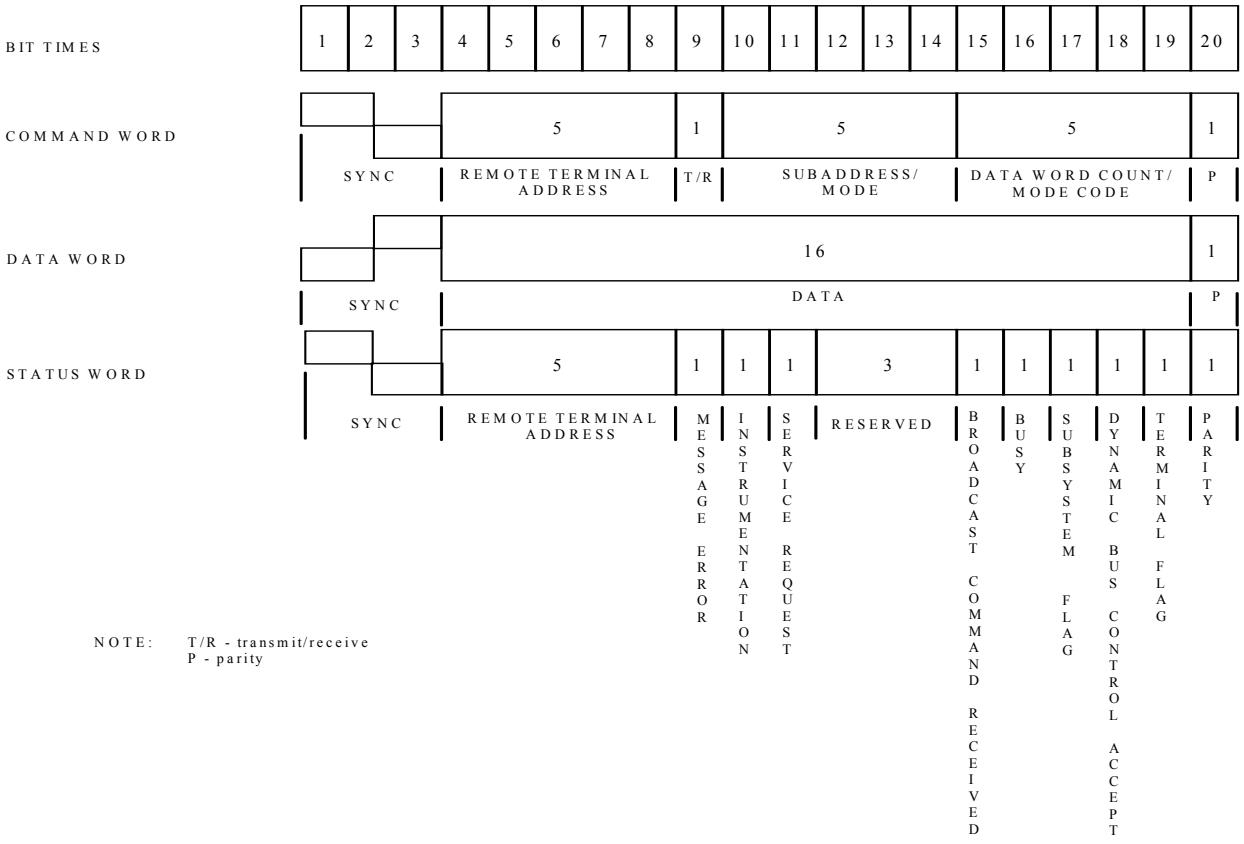


FIGURE 62. Word formats.

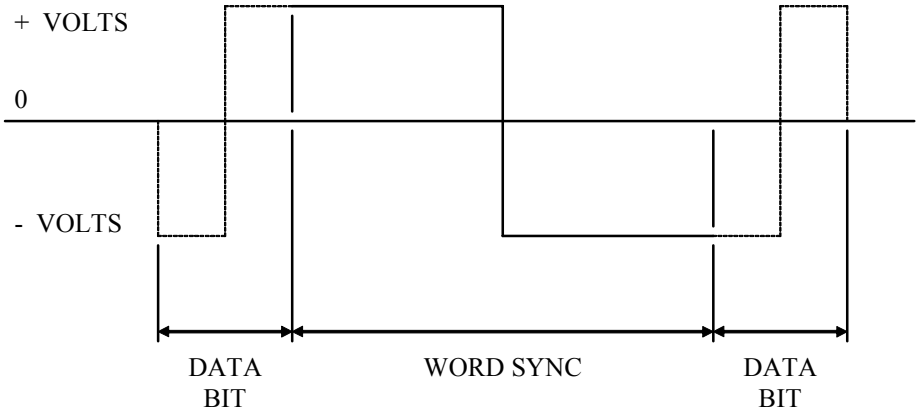


FIGURE 63. Command and status sync.

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5.5.4.3.3.5.1.2 Remote terminal address.

The next five bits following the sync shall be the RT address. Each RT shall be assigned a unique address. Decimal address 31 (11111) shall not be assigned as a unique address. In addition to its unique address, a RT shall be assigned decimal address 31 (11111) as the common address, if the broadcast option is used.

5.5.4.3.3.5.1.3 Transmit/receive.

The next bit following the remote terminal address shall be the T/R bit, which shall indicate the action required of the RT. A logic zero shall indicate the RT is to receive, and a logic one shall indicate the RT is to transmit.

5.5.4.3.3.5.1.4 Subaddress/mode.

The next five bits following the R/T bit shall be utilized to indicate an RT subaddress or use of mode control, as is dictated by the individual terminal requirements. The subaddress/mode values of 00000 and 11111 are reserved for special purposes, as specified in 5.5.4.3.3.5.1.7, and shall not be utilized for any other function.

5.5.4.3.3.5.1.5 Data word count/mode code.

The next five bits following the subaddress/mode field shall be the quantity of data words to be either sent out or received by the RT or the optional mode code as specified in 5.5.4.3.3.5.1.7. A maximum of 32 data words may be transmitted or received in any one message block. All 1's shall indicate a decimal count of 31, and all 0's shall indicate a decimal count of 32.

5.5.4.3.3.5.1.6 Parity.

The last bit in the word shall be used for parity over the preceding 16 bits. Odd parity shall be utilized.

5.5.4.3.3.5.1.7 Optional mode control.

For RT's exercising this option a subaddress/mode code of 00000 or 11111 shall imply that the contents of the data word count/mode code field are to be decoded as a five bit mode command. The mode code shall only be used to communicate with the multiplex bus related hardware, and to assist in the management of information flow, and not to extract data from or feed data to a functional subsystem. Codes 00000 through 01111 shall only be used for mode codes which do not require transfer of a data word. For these codes, the T/R bit shall be set to 1. Codes 10000 through 11111 shall only be used for mode codes which require transfer of a single data word. For these mode codes, the T/R bit shall indicate the direction of data word flow as specified in 5.5.4.3.3.5.1.3. No multiple data word transfer shall be implemented with any mode code. The mode codes are reserved for the specific functions as specified in TABLE XL and shall not be used for any other purpose. If the designer chooses to implement any of these functions, the specific codes, T/R bit assignments, and use of a data word, shall be used as indicated. The use of the broadcast command option shall only be applied to particular mode codes as specified in TABLE XL.

5.5.4.3.3.5.1.7.1 Dynamic bus control.

The controller shall issue a transmit command to an RT capable of performing the bus control function. This RT shall respond with a status word as specified in 5.5.4.3.3.5.3. Control of the data bus passes from the offering bus controller to the accepting RT upon completion of the transmission of the status word by the RT. If the RT rejects control of the data bus, the offering bus controller retains control of the data bus.

5.5.4.3.3.5.1.7.2 Synchronize (without data word).

This command shall cause the RT to synchronize (e.g., to reset the internal timer, to start a sequence, etc.). The RT shall transmit the status word as specified in 5.5.4.3.3.5.3.

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TABLE XL. Assigned mode codes.

T/R Bit	Mode Code	Function	Associated Data Word	Broadcast Command Allowed
1	00000	Dynamic bus Control	NO	NO
1	00001	Synchronize	NO	YES
1	00010	Transmit Status Word	NO	NO
1	00011	Initiate Self Test	NO	YES
1	00100	Transmitter Shutdown	NO	YES
1	00101	Override Transmitter Shutdown	NO	YES
1	00110	Inhibit Terminal Flag Bit	NO	YES
1	00111	Override Inhibit Terminal Flag Bit	NO	YES
1	01000	Reset Remote Terminal	NO	YES
1 ⋮ 1	01001 ⋮ 01111	Reserved ⋮ Reserved	NO ⋮ NO	TBD ⋮ TBD
1	10000	Transmit Vector Word	YES	NO
0	10001	Synchronize	YES	YES
1	10010	Transmit Last Command	YES	NO
1	10011	Transmit BIT Word	YES	NO
0	10100	Selected Transmitter Shutdown	YES	YES
0	10101	Override Selected Transmitter Shutdown	YES	YES
1 or 0 ⋮ 1 or 0	10110 ⋮ 11111	Reserved ⋮ Reserved	YES ⋮ YES	TBD ⋮ TBD
NOTE: To be determined (TBD)				

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5.5.4.3.3.5.1.7.3 Transmit status word.

This command shall cause the RT to transmit the status word associated with the last valid command word preceding this command. This mode command shall not alter the state of the status word.

5.5.4.3.3.5.1.7.4 Initiate self test.

This command shall be used to initiate self test within the RT. The RT shall transmit the status word as specified in 5.5.4.3.3.5.3.

5.5.4.3.3.5.1.7.5 Transmitter shutdown.

This command (to only be used with dual redundant bus systems) shall cause the RT to disable the transmitter associated with the redundant bus. The RT shall not comply with a command to shut down a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 5.5.4.3.3.5.3 after this command.

5.5.4.3.3.5.1.7.6 Override transmitter shutdown.

This command (to only be used with dual redundant bus systems) shall cause the RT to enable a transmitter which was previously disabled. The RT shall not comply with a command to enable a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 5.5.4.3.3.5.3 after this command.

5.5.4.3.3.5.1.7.7 Inhibit terminal flag (T/F) bit.

This command shall cause the RT to set T/F bit in the status word specified in 5.5.4.3.3.5.3 to logic zero until otherwise commanded. The RT shall transmit the status word as specified in 5.5.4.3.3.5.3.

5.5.4.3.3.5.1.7.8 Override inhibit T/F bit.

This command shall cause the RT to override the inhibit T/F bit specified in 5.5.4.3.3.5.1.7.7. The RT shall transmit the status word as specified in 5.5.4.3.3.5.3.

5.5.4.3.3.5.1.7.9 Reset remote terminal.

This command shall be used to reset the RT to power up initialized state. The RT shall first transmit its status word, and then reset.

5.5.4.3.3.5.1.7.10 Reserved mode codes (01001 to 01111).

These mode codes are reserved for future use and shall not be used.

5.5.4.3.3.5.1.7.11 Transmit vector word.

This command shall cause the RT to transmit a status word as specified in 5.5.4.3.3.5.3 and a data word containing service request information.

5.5.4.3.3.5.1.7.12 Synchronize (with data word).

The RT shall receive a command word followed by a data word as specified in 5.5.4.3.3.5.2. The data word shall contain synchronization information for the RT. After receiving the command and data word, the RT shall transmit the status word as specified in 5.5.4.3.3.5.3.

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5.5.4.3.3.5.1.7.13 Transmit last command word.

This command shall cause the RT to transmit its status word as specified in 5.5.4.3.3.5.3 followed by a single data word which contains bits 4-19 of the last command word, excluding a transmit last command word mode code received by the RT. This mode command shall not alter the state of the RT's status word.

5.5.4.3.3.5.1.7.14 Transmit built-in-test (BIT) word.

This command shall cause the RT to transmit its status word as specified in 5.5.4.3.3.5.3 followed by a single data word containing the RT BIT data. This function is intended to supplement the available bits in the status word when the RT hardware is sufficiently complex to warrant its use. The data word, containing the RT BIT data, shall not be altered by the reception of a transmit last command or a transmit status word mode code. This function shall not be used to convey BIT data from the associated subsystem(s).

5.5.4.3.3.5.1.7.15 Selected transmitter shutdown.

This command shall cause the RT to disable the transmitter associated with a specified redundant data bus. The command is designed for use with systems employing more than two redundant buses. The transmitter that is to be disabled shall be identified in the data word following the command word in the format as specified in 5.5.4.3.3.5.2. The RT shall not comply with a command to shut down a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 5.5.4.3.3.5.3.

5.5.4.3.3.5.1.7.16 Override selected transmitter shutdown.

This command shall cause the RT to enable a transmitter which was previously disabled. The command is designed for use with systems employing more than two redundant buses. The transmitter that is to be enabled shall be identified in the data word following the command word in the format as specified in 5.5.4.3.3.5.2. The RT shall not comply with a command to enable a transmitter on the bus from which this command is received. In all cases, the RT shall respond with a status word as specified in 5.5.4.3.3.5.3.

5.5.4.3.3.5.1.7.17 Reserved mode codes (10110 to 11111).

These mode codes are reserved for future use and shall not be used.

5.5.4.3.3.5.2 Data word.

A data word shall be comprised of a sync waveform, data bits, and a parity bit (see FIGURE 62).

5.5.4.3.3.5.2.1 Sync.

The data sync waveform shall be an invalid Manchester waveform as shown on FIGURE 64. The width shall be three bit times, with the waveform being negative for the first one and one-half bit times, and then positive for the following one and one-half bit times. Note that if the bits preceding and following the sync are logic ones, then the apparent width of the sync waveform will be increased to four bit times.

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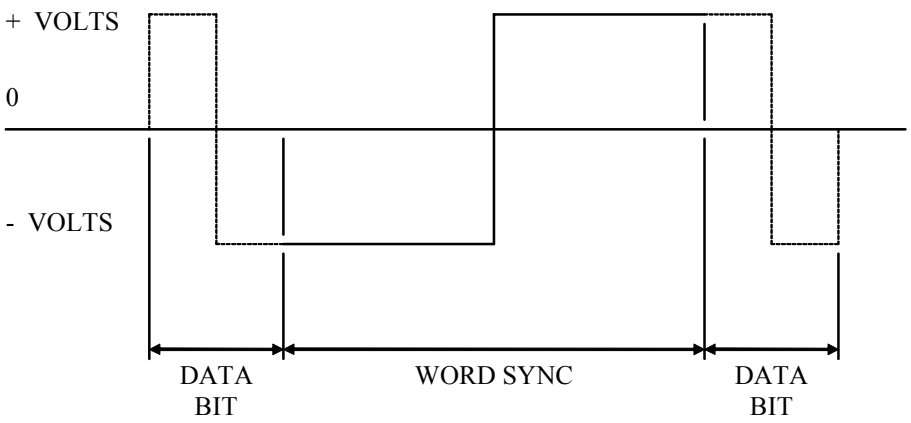


FIGURE 64. Data sync.

5.5.4.3.3.5.2.2 Data.

The sixteen bits following the sync shall be utilized for data transmission as specified in 5.5.4.3.2.

5.5.4.3.3.5.2.3 Parity.

The last bit shall be utilized for parity as specified in 5.5.4.3.3.5.1.6.

5.5.4.3.3.5.3 Status word.

A status word shall be comprised of a sync waveform, RT address, message error bit, instrumentation bit, service request bit, three reserved bits, broadcast command received bit, busy bit, subsystem flag bit, dynamic bus control acceptance bit, terminal flag bit, and a parity bit. For optional broadcast operation, transmission of the status word shall be suppressed as specified in 5.5.4.3.3.6.7.

5.5.4.3.3.5.3.1 Sync.

The status sync waveform shall be as specified in 5.5.4.3.3.5.1.1.

5.5.4.3.3.5.3.2 RT address.

The next five bits following the sync shall contain the address of the RT which is transmitting the status word as defined in 5.5.4.3.3.5.1.2.

5.5.4.3.3.5.3.3 Message error bit.

The status word bit at bit time nine (see FIGURE 62) shall be utilized to indicate that one or more of the data words associated with the preceding receive command word from the bus controller has failed to pass the RT's validity tests as specified in 5.5.4.4.1.1. The bit shall also be set under the conditions specified in 5.5.4.4.1.2, 5.5.4.4.3.4 and 5.5.4.4.3.6. A logic one shall indicate the presence of a message error, and a logic zero shall show its absence. All RT's shall implement the message error bit.

5.5.4.3.3.5.3.4 Instrumentation bit.

The status word at bit time ten (see FIGURE 62) shall be reserved for the instrumentation bit and shall always be a logic zero. This bit is intended to be used in conjunction with a logic one in bit time ten of the command word to distinguish between a command word and a status word. The use of the instrumentation bit is optional.

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5.5.4.3.3.5.3.5 Service request bit.

The status word bit at bit time eleven (see FIGURE 62) shall be reserved for the service request bit. The use of the bit is optional. This bit when used, shall indicate the need for the bus controller to take specific predefined actions relative to either the RT or associated subsystem. Multiple subsystems, interfaced to a single RT, which individually require a service request signal shall logically OR their individual signals into the single status word bit. In the event this logical OR is performed, then the designer must make provisions in a separate data word to identify the specific requesting subsystem. The service request bit is intended to be used only to trigger data transfer operations which take place on an exception rather than periodic basis. A logic one shall indicate the presence of a service request, and a logic zero its absence. If this function is not implemented, the bit shall be set to zero.

5.5.4.3.3.5.3.6 Reserved status bits.

The status word bits at bit times twelve through fourteen are reserved for future use and shall not be used. These bits shall be set to logic zero.

5.5.4.3.3.5.3.7 Broadcast command received bit.

The status word at bit time fifteen shall be set to a logic one to indicate that the preceding valid command word was a broadcast command and a logic zero shall show it was not a broadcast command. If the broadcast command option is not used, this bit shall be set to a logic zero.

5.5.4.3.3.5.3.8 Busy bit.

The status word bit at bit time sixteen (see FIGURE 62) shall be reserved for the busy bit. The use of this bit is optional. This bit, when used, shall indicate that the RT or subsystem is unable to move data to or from the subsystem in compliance with the bus controller's command. A logic one shall indicate the presence of a busy condition, and a logic zero its absence. In the event the busy bit is set in response to a transmit command, then the RT shall transmit its status word only. If this function is not implemented, the bit shall be set to logic zero.

5.5.4.3.3.5.3.9 Subsystem flag bit.

The status word bit at bit time seventeen (see FIGURE 62) shall be reserved for the subsystem flag bit. The use of this bit is optional. This bit, when used, shall flag a subsystem fault condition, and alert the bus controller to potentially invalid data. Multiple subsystems, interfaced to a single RT, which individually require a subsystem flag bit signal shall logically OR their individual signals into the single status word bit. In the event the logical OR is performed, then the designer must take provisions in a separate data word to identify the specific reporting subsystem. A logic one shall indicate the presence of the flag, and a logic zero its absence. If not used, this bit shall be set to logic zero.

5.5.4.3.3.5.3.10 Dynamic bus control acceptance bit.

The status word bit at bit time eighteen (see FIGURE 62) shall be reserved for the acceptance of dynamic bus control. This bit shall be used if the RT implements the optional dynamic bus control function. This bit, when used, shall indicate acceptance or rejection of a dynamic bus control offer as specified in 5.5.4.3.3.5.1.7.1. A logic one shall indicate acceptance of control, and a logic zero shall indicate rejection of control. If this function is not used, this bit shall be set to a logic zero.

5.5.4.3.3.5.3.11 Terminal flag bit.

The status word bit at bit time nineteen (see FIGURE 62) shall be reserved for the terminal flag function. The use of this bit is optional. This bit, when used, shall flag a RT fault condition. A logic one shall indicate the presence of the flag, and logic zero, its absence. If not used, this bit shall be set to logic zero.

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5.5.4.3.3.5.3.12 Parity bit.

The least significant bit in the status word shall be utilized for parity as specified in 5.5.4.3.3.5.1.6.

5.5.4.3.3.5.4 Status word reset.

The status word bit, with the exception of the address, shall be set to logic zero after a valid command word is received by the RT with the exception as specified in 5.5.4.3.3.5.1.7. If the conditions which caused bits in the status word to be set (e.g., terminal flag) continue after the bits are reset to logic zero, then the affected status word bit shall be again set, and then transmitted on the bus as required.

5.5.4.3.3.6 Message formats.

The messages transmitted on the data bus shall be in accordance with the formats on FIGURE 65 and FIGURE 66. The maximum and minimum response times shall be as stated in 5.5.4.3.3.7 and 5.5.4.3.3.8. No message formats, other than those defined herein, shall be used on the bus.

5.5.4.3.3.6.1 Bus controller to remote terminal transfers.

The bus controller shall issue a receive command followed by the specified number of data words. The RT shall, after message validation, transmit a status word back to the controller. The command and data words shall be transmitted in a contiguous fashion with no interword gaps.

5.5.4.3.3.6.2 Remote terminal to bus controller transfers.

The bus controller shall issue a transmit command to the RT. The RT shall, after command word validation, transmit a status word back to the bus controller, followed by the specified number of data words. The status and data words shall be transmitted in a contiguous fashion with no interword gaps.

5.5.4.3.3.6.3 Remote terminal to remote terminal transfers.

The bus controller shall issue a receive command to RT A followed contiguously by a transmit command to RT B. RT B shall, after command validation, transmit a status word followed by the specified number of data words. The status and data words shall be transmitted in a contiguous fashion with no gap. At the conclusion of the data transmission by RT B, RT A shall transmit a status word within the specified time period.

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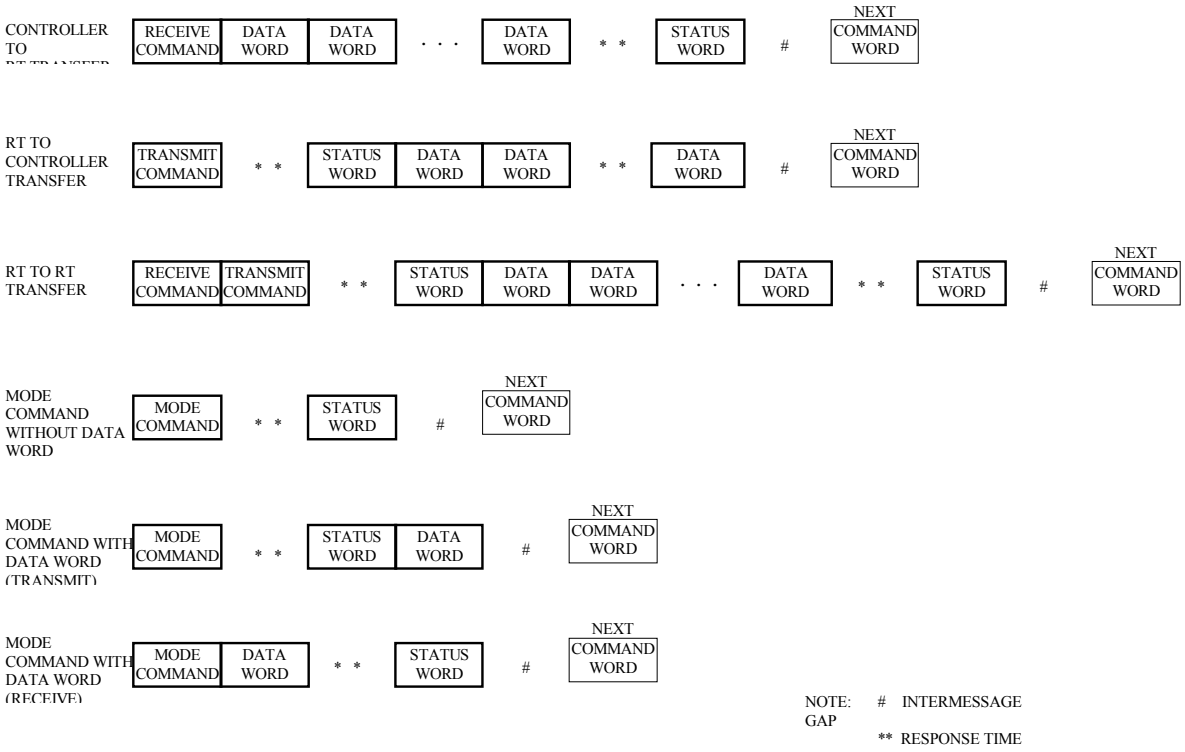


FIGURE 65. Information transfer formats.

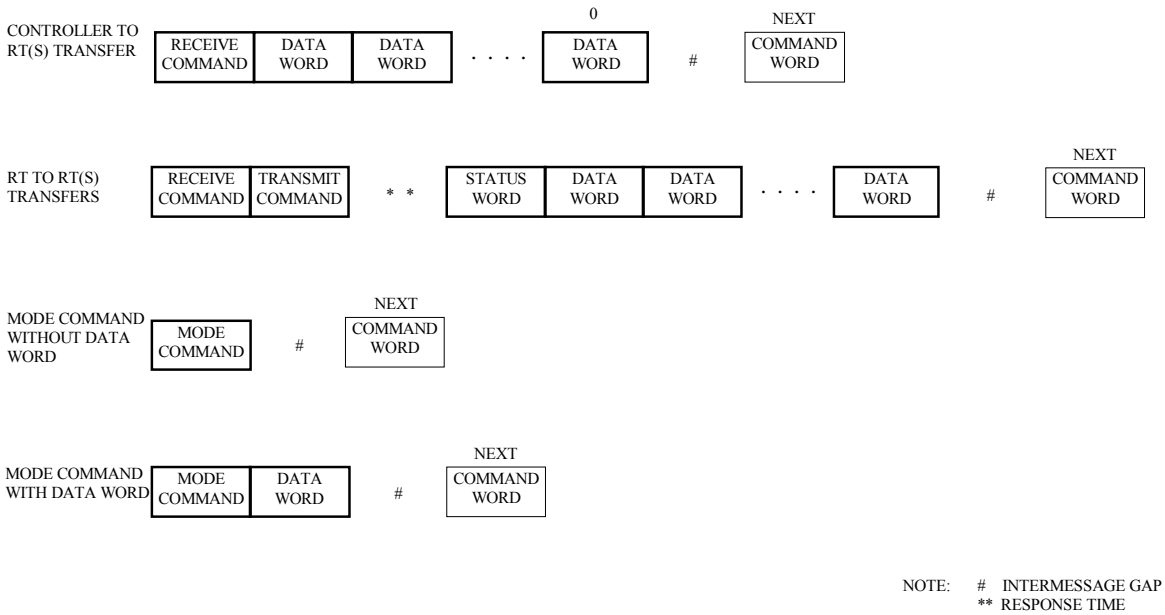


FIGURE 66. Broadcast information transfer formats.

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5.5.4.3.3.6.4 Mode command without data word.

The bus controller shall issue a transmit command to the RT using a mode code specified in TABLE XL. The RT shall, after command word validation, transmit a status word.

5.5.4.3.3.6.5 Mode command with data word (transmit).

The bus controller shall issue a transmit command to the RT using a mode code specified in TABLE XL. The RT shall, after command word validation, transmit a status word followed by one data word. The status word and data word shall be transmitted in a contiguous fashion with no gap.

5.5.4.3.3.6.6 Mode command with data word (receive).

The bus controller shall issue a receive command to the RT using a mode code specified in TABLE XL, followed by one data word. The command word and data word shall be transmitted in a contiguous fashion with no gap. The RT shall, after command and data word validation, transmit a status word back to the controller.

5.5.4.3.3.6.7 Optional broadcast command.

See 5.5.10.6 for additional information on the use of the broadcast command.

5.5.4.3.3.6.7.1 Bus controller to remoter terminal(s) transfer (broadcast).

The bus controller shall issue a receive command word with 11111 in the RT address field followed by the specified number of data words. The command word and data words shall be transmitted in a contiguous fashion with no gap. The RT(s) with the broadcast option shall after message validation, set the broadcast command received bit in the status word as specified in 5.5.4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.6.7.2 Remote terminal to remote terminal(s) transfers (broadcast).

The bus controller shall issue a receive command word with 11111 in the RT address field followed by a transmit command to RT A using the RT's address. RT A shall, after command word validation, transmit a status word followed by the specified number of data words. The status and data words shall be transmitted in a contiguous fashion with no gap. The RT(s) with the broadcast option, excluding RT A, shall after message validation, set the broadcast received bit in the status word as specified in 5.5.4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.6.7.3 Mode command without data word (broadcast).

The bus controller shall issue a transmit command word with 11111 in the RT address field, and a mode code specified in TABLE XL. The RT(s) with the broadcast option shall after command word validation, set the broadcast received bit in the status word as specified in 5.5.4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.6.7.4 Mode command with data word (broadcast).

The bus controller shall issue a receive command word with 11111 in the RT address field and a mode code specified in TABLE XL, followed by one data word. The command word and data word shall be transmitted in a contiguous fashion with no gap. The RT(s) with the broadcast option shall after message validation, set the broadcast received bit in the status word as specified in 5.5.4.3.3.5.3.7 and shall not transmit the status word.

5.5.4.3.3.7 Intermessage gap.

The bus controller shall provide a minimum gap time of 4.0 microseconds (μs) between messages as shown on FIGURE 65 and FIGURE 66. This time period, shown as T on FIGURE 67, is measured at point A of the bus controller as shown on FIGURE 68 or FIGURE 69. The time is measured from the mid-bit zero crossing of the last bit of the preceding message to mid-zero crossing of the next command word sync.

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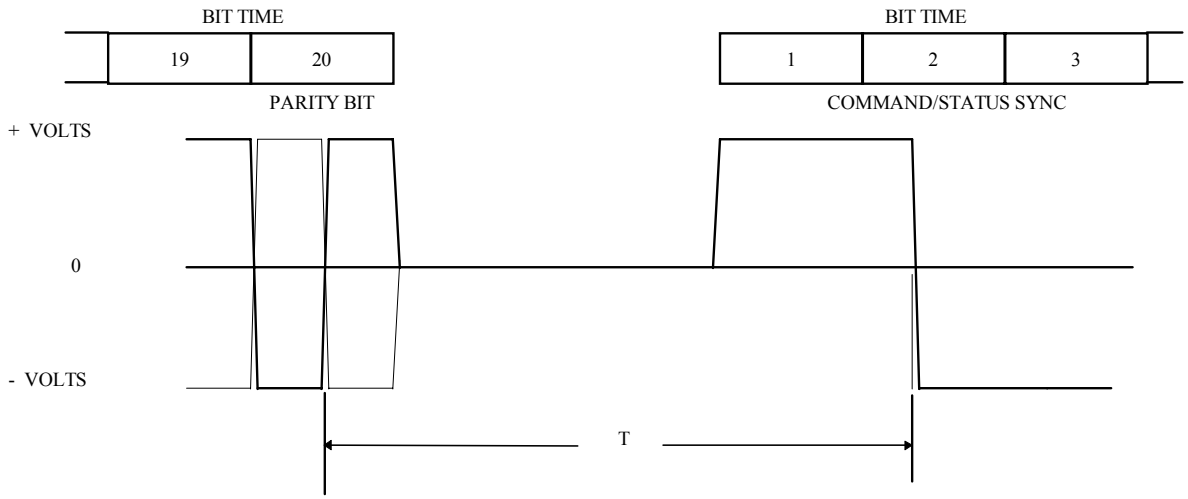


FIGURE 67. Intermessage gap and response time.

5.5.4.3.3.8 Response time.

The RT shall respond, in accordance with 5.5.4.3.3.6, to a valid command word within the time period of 4.0 to 12.0 μs. This time period, shown as T on FIGURE 67, is measured at point A of the RT as shown on FIGURE 68 or FIGURE 69. The time is measured from the mid bit zero crossing of the last word specified in 5.5.4.3.3.6 and as shown on FIGURE 65 and FIGURE 66 to the mid-zero crossing of the status word sync.

5.5.4.3.3.9 Minimum no-response time-out.

The minimum time that a terminal shall wait before considering that a response as specified in 5.5.4.3.3.8 has not occurred shall be 14.0 μs. The time is measured from the mid-bit zero crossing of the last bit of the last word to the mid-zero crossing of the expected status word sync at point A of the terminal as shown in FIGURE 68 or FIGURE 69.

5.5.4.4 Terminal operation.

5.5.4.4.1 Common operation.

Terminals shall have common operating capabilities as specified in the following paragraphs.

5.5.4.4.1.1 Word validation.

The terminal shall insure that each word conforms to the following minimum criteria:

a. The word begins with valid sync field.
b. The bits are in a valid Manchester II code.
c. The information field has 16 bits plus parity.
d. The word parity is odd.

When a word fails to conform to the preceding criteria, the word shall be considered invalid.

5.5.4.4.1.2 Transmission continuity.

The terminal shall verify that the message is contiguous as defined in 5.5.4.3.3.6. Improperly timed data syncs shall be considered a message error.

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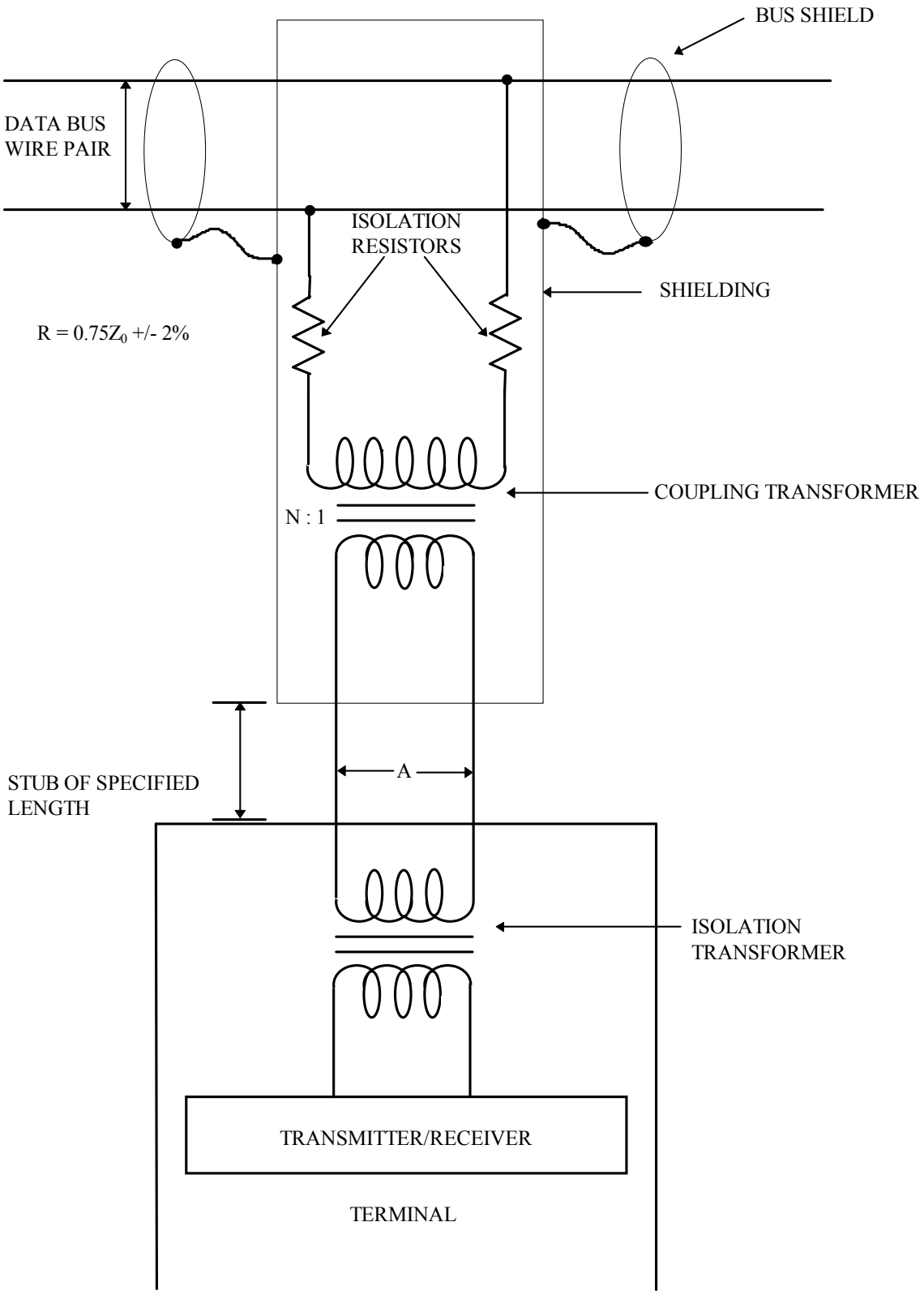


FIGURE 68. Data bus interface using transformer coupling.

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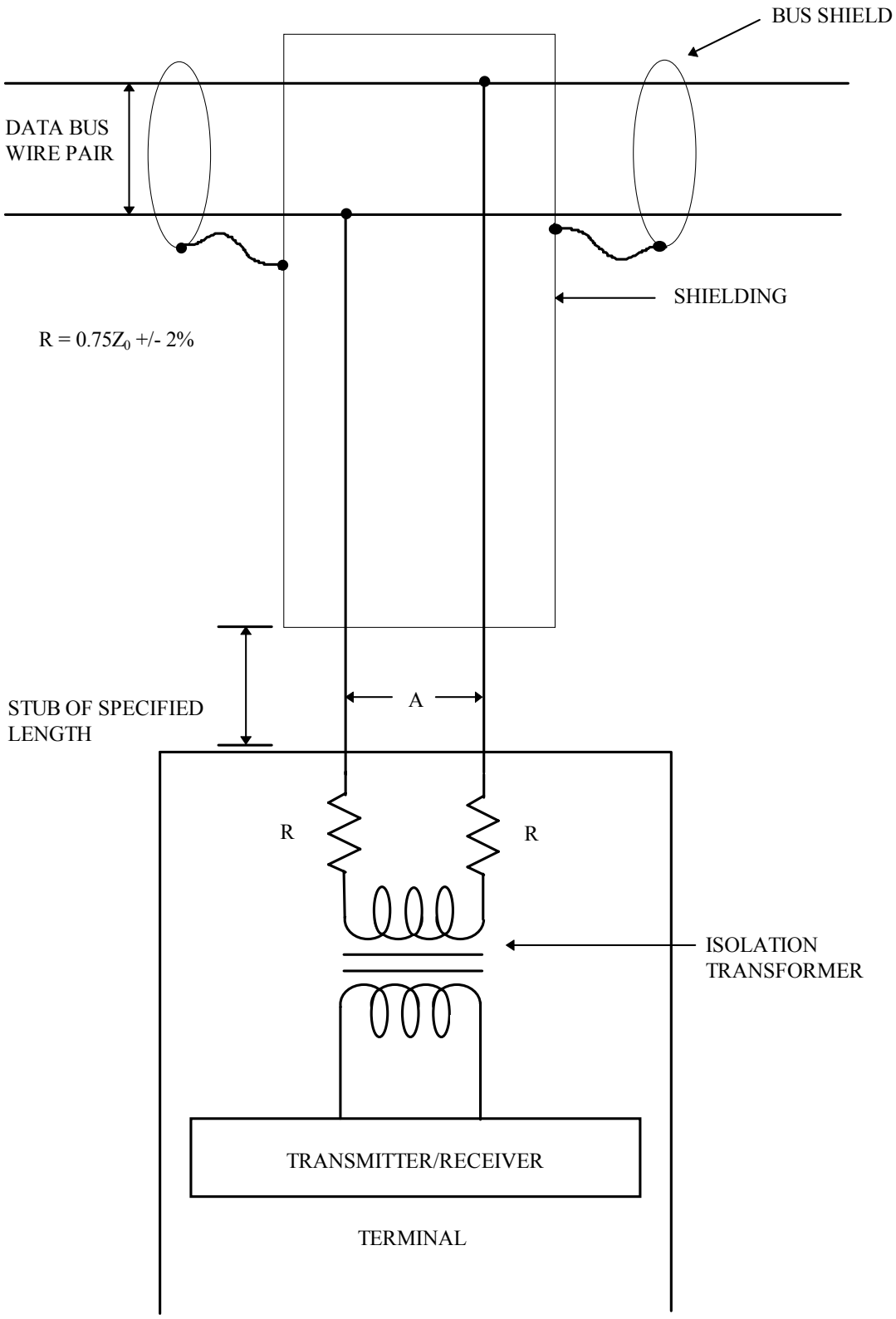


FIGURE 69. Data interface using direct coupling.

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5.5.4.4.1.3 Terminal fail-safe.

The terminal shall contain a hardware implemented time-out to preclude a signal transmission of greater than 800.0 µs. This hardware shall not preclude a correct transmission in response to a command. Reset of this time-out function shall be performed by the reception of a valid command on the bus on which the time-out has occurred.

5.5.4.4.2 Bus controller operation.

A terminal operating as a bus controller shall be responsible for sending data bus commands, participating in data transfers, receiving status responses, and monitoring system status as defined in this standard. The bus controller function may be embodied as either a stand-alone terminal, whose sole function is to control the data bus(s), or contained within a subsystem. Only one terminal shall be in active control of a data bus at any one time.

5.5.4.4.3 Remote terminal.

5.5.4.4.3.1 Operation.

A remote terminal (RT) shall operate in response to valid commands received from the bus controller. The RT shall accept a command word as valid when the command word meets the criteria of 5.5.4.4.1.1, and the command word contains a terminal address which matches the RT address or an address of 11111, if the RT has the broadcast option. No combination of RT address bits, T/R bit, subaddress/mode bits, and data word count/mode code bits of a command word shall result in invalid transmissions by the RT. Subsequent valid commands shall be properly responded to by the RT.

5.5.4.4.3.2 Superseding valid commands.

The RT shall be capable of receiving a command word on the data bus after the minimum intermessage gap time as specified in 5.5.4.3.3.7 has been exceeded, when the RT is not in the time period T as specified in 5.5.4.3.3.8 prior to the transmission of a status word, and when it is not transmitting on that data bus. A second valid command word sent to an RT shall take precedence over the previous command. The RT shall respond to the second valid command as specified in 5.5.4.3.3.8.

5.5.4.4.3.3 Invalid commands.

A remote terminal shall not respond to a command word which fails to meet the criteria specified in 5.5.4.4.3.1.

5.5.4.4.3.4 Illegal command.

An illegal command is a valid command as specified in 5.5.4.4.3.1, where the bits in the subaddress/mode field, data word count/mode code field, and the T/R bit indicate a mode command, subaddress, or word count that has not been implemented in the RT. It is the responsibility of the bus controller to assure that no illegal commands are sent out. The RT designer has the option of monitoring for illegal commands. If an RT that is designed with this option detects an illegal command and the proper number of contiguous valid data words as specified by the illegal command word, it shall respond with a status word only, setting the message error bit, and not use the information received.

5.5.4.4.3.5 Valid data reception.

The remote terminal shall respond with a status word when a valid command word and the proper number of contiguous valid data words are received, or a single valid data word associated with a mode code is received. Each data word shall meet the criteria specified in 5.5.4.4.1.1.

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5.5.4.4.3.6 Invalid data reception.

Any data word(s) associated with a valid receive command that does not meet the criteria specified in 5.5.4.4.1.1 and 5.5.4.4.1.2 or an error in the data word count shall cause the remote terminal to set the message error bit in the status word to a logic one and suppress the transmission of the status word. If a message error has occurred, then the entire message shall be considered invalid.

5.5.4.4.4 Bus monitor operation.

A terminal operating as a bus monitor shall receive bus traffic and extract selected information. While operating as a bus monitor, the terminal shall not respond to any messages except one containing its own unique address if one is assigned. All information obtained while acting as a bus monitor shall be strictly used for off-line applications (e.g., flight test recording, maintenance recording or mission analysis) or to provide the back-up bus controller sufficient information to take over as the bus controller.

5.5.4.5 Hardware characteristics.

5.5.4.5.1 Data bus characteristics.

5.5.4.5.1.1 Cable.

The cable used for the main bus and all stubs shall be a two conductor, twisted, shielded, jacketed cable. The wire-to-wire distributed capacitance shall not exceed 30.0 picofarads per foot. The cables shall be formed with not less than four twists per foot where a twist is defined as a 360 degree rotation of the wire pairs; and, the cable shield shall provide a minimum of 75 percent coverage.

5.5.4.5.1.2 Characteristic impedance.

The nominal characteristic impedance of the cable (Z_0) shall be within the range of 70.0 ohms to 85.0 ohms at a sinusoidal frequency of 1.0 megahertz (MHz).

5.5.4.5.1.3 Cable attenuation.

At the frequency of 5.5.4.5.1.2, the cable power loss shall not exceed 1.5 decibels (dB) / 100 feet (ft).

5.5.4.5.1.4 Cable termination.

The two ends of the cable shall be terminated with a resistance, equal to the selected cable nominal characteristic impedance (Z_0) ± 2.0 percent.

5.5.4.5.1.5 Cable stub requirements.

The cable shall be coupled to the terminal as shown on FIGURE 68 or FIGURE 69. The use of long stubs is discouraged, and the length of a stub should be minimized. However, if installation requirements dictate, stub lengths exceeding those lengths specified in 5.5.4.5.1.5.1 and 5.5.4.5.1.5.2 are permissible.

5.5.4.5.1.5.1 Transformer Coupled stubs.

The length of a Transformer Coupled stub should not exceed 20 feet. If a Transformer Coupled stub is used, then the following shall apply.

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5.5.4.5.1.5.1.1 Coupling transformer.

A coupling transformer, as shown on FIGURE 68, shall be required. This transformer shall have a turns ratio of 1:1.41 ± 3.0 percent, with the higher turns on the isolation resistor side of the stub.

5.5.4.5.1.5.1.1.1 Transformer input impedance.

The open circuit impedance as seen at point B on FIGURE 70 shall be greater than 3000 ohms over the frequency range of 75.0 kilohertz (kHz) to 1.0 megahertz (MHz), when measured with a 1.0 V root-mean-square (RMS) sin wave.

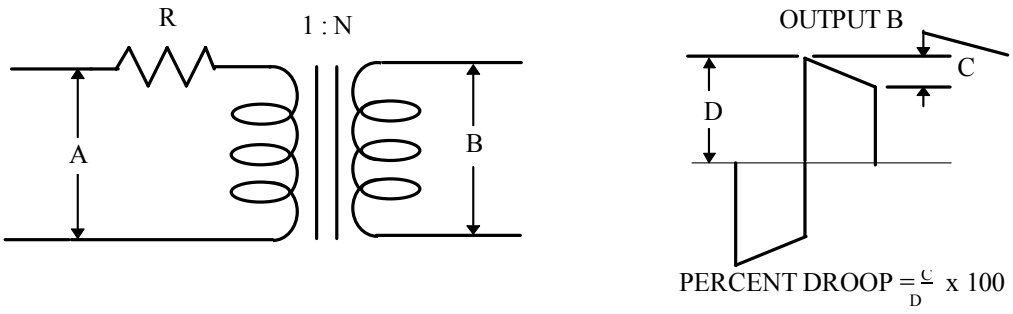


FIGURE 70. Coupling transformer.

5.5.4.5.1.5.1.1.2 Transformer waveform integrity.

The droop of the transformer using the test configuration shown of FIGURE 70 at point B, shall not exceed 20.0 percent. Overshoot and ringing as measured at point B shall be less than ± 1.0 V peak. For this test, R shall equal 360.0 ohms ± 5.0 percent and the input A of FIGURE 70 shall be a 250.0 kHz square wave, 27.0 V peak-to-peak, with a rise and fall time no greater than 100 nanoseconds (ns).

5.5.4.5.1.5.1.1.3 Transformer common mode rejection.

The coupling transformer shall have a common mode rejection ratio greater than 45.0 dB at 1.0 MHz.

5.5.4.5.1.5.1.2 Fault isolation.

An isolation resistor shall be placed in series with each connection to the data bus cable. This resistor shall have a value of 0.75 Z_0 ohms plus or minus 2.0 percent, where Z_0 is the selected cable nominal characteristic impedance. The impedance placed across the data bus cable shall be no less than 1.5 Z_0 ohms for any failure of the coupling transformer, cable stub, or terminal transmitter/receiver.

5.5.4.5.1.5.1.3 Cable coupling.

All coupling transformers and isolation resistors, as specified in 5.5.4.5.1.5.1.1 and 5.5.4.5.1.5.1.2, shall have continuous shielding which will provide a minimum of 75 percent coverage. The isolation resistors and coupling transformers shall be placed at minimum possible distance from the junction of the stub to the main bus.

5.5.4.5.1.5.1.4 Stub voltage requirements.

Every data bus shall be designed such that all stubs at point A of FIGURE 68 shall have a peak-to-peak amplitude, line-to-line within the range of 1.0 and 14.0 V for a transmission by any terminal on the bus. This shall include the maximum reduction of data bus signal amplitude in the event that one of the terminals has a fault which causes it to reflect a fault impedance specified in 5.5.4.5.1.5.1.2 on the data bus. This shall also include the worse case output voltage of the terminals as specified in 5.5.4.5.2.1.1.1 and 5.5.4.5.2.2.1.1.

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5.5.4.5.1.5.2 Direct coupled stubs.

The length of a Direct Coupled stub should not exceed 1 foot. Refer to 10.5 for comments concerning Direct Coupled stubs. If a Direct Coupled stub is used, then the following shall apply.

5.5.4.5.1.5.2.1 Fault isolation.

An isolation resistor shall be placed in series with each connection to the data bus cable. This resistor shall have a value of 55.0 ohms plus or minus 2.0 percent. The isolation resistors shall be placed within the RT as shown on FIGURE 69.

5.5.4.5.1.5.2.2 Cable coupling.

All bus-stub junctions shall have continuous shielding which will provide a minimum of 75 percent coverage.

5.5.4.5.1.5.2.3 Stub voltage requirements.

Every data bus shall be designed such that all stubs at point A of FIGURE 69 shall have a peak-to-peak amplitude, line-to-line within the range of 1.4 and 20.0 V for a transmission by any terminal on the bus. This shall include the maximum reduction of data bus signal amplitude in the event that one of the terminals has a fault which causes it to reflect a fault impedance of 110 ohms on the data bus. This shall also include the severe output voltage of the terminals as specified in 5.5.4.5.2.1.1.1 and 5.5.4.5.2.2.1.1.

5.5.4.5.1.5.3 Wiring and cabling for EMC.

For purposes of electromagnetic capability (EMC), the wiring and cabling provisions of MIL-STD-464 shall apply.

5.5.4.5.2 Terminal characteristics.

5.5.4.5.2.1 Terminals with Transformer Coupled stubs.

5.5.4.5.2.1.1 Terminal output characteristics.

The following characteristics shall be measured with R_L, as shown on FIGURE 71, equal to 70.0 ohms ± 2.0 percent.

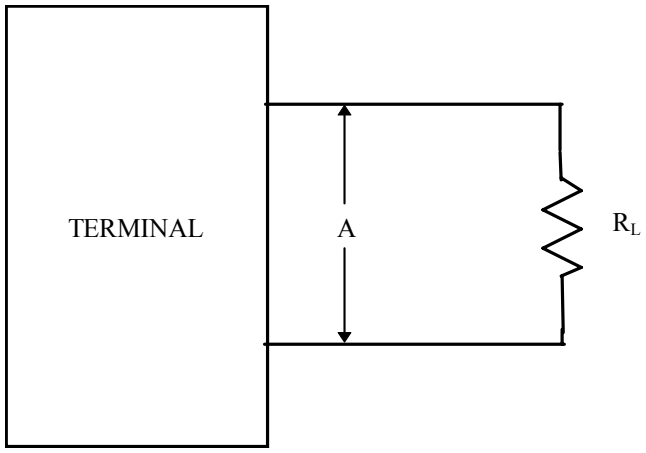


FIGURE 71. Terminal I/O characteristics for transformer coupled and direct coupled stubs.

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5.5.4.5.2.1.1.1 Output voltage levels.

The terminal output voltage levels shall be measured using the test configuration shown on FIGURE 71. The terminal output voltage shall be within the range of 18.0 to 27.0 V, peak-to-peak, line-to-line, when measured at point A on FIGURE 71.

5.5.4.5.2.1.1.2 Output waveform.

The waveform, when measure at point A on FIGURE 71 shall have zero crossing deviations which are equal to, or less than, 25.0 ns from the ideal crossing point, measured with respect to the previous zero crossing (i.e., .5 ± .025 µs, 1.0 ± .025 µs, 1.5 ± .025 µs, and 2.0 ± .025 µs). The rise and fall time of this waveform shall be from 100.0 to 300.0 ns when measured from levels of 10 to 90 percent of full waveform peak-to-peak, line-to-line, voltage as shown on FIGURE 72. Any distortion of the waveform including overshoot and ringing shall not exceed ± 900.0 millivolts (mV) peak, line-to-line, as measured at point A, FIGURE 71.

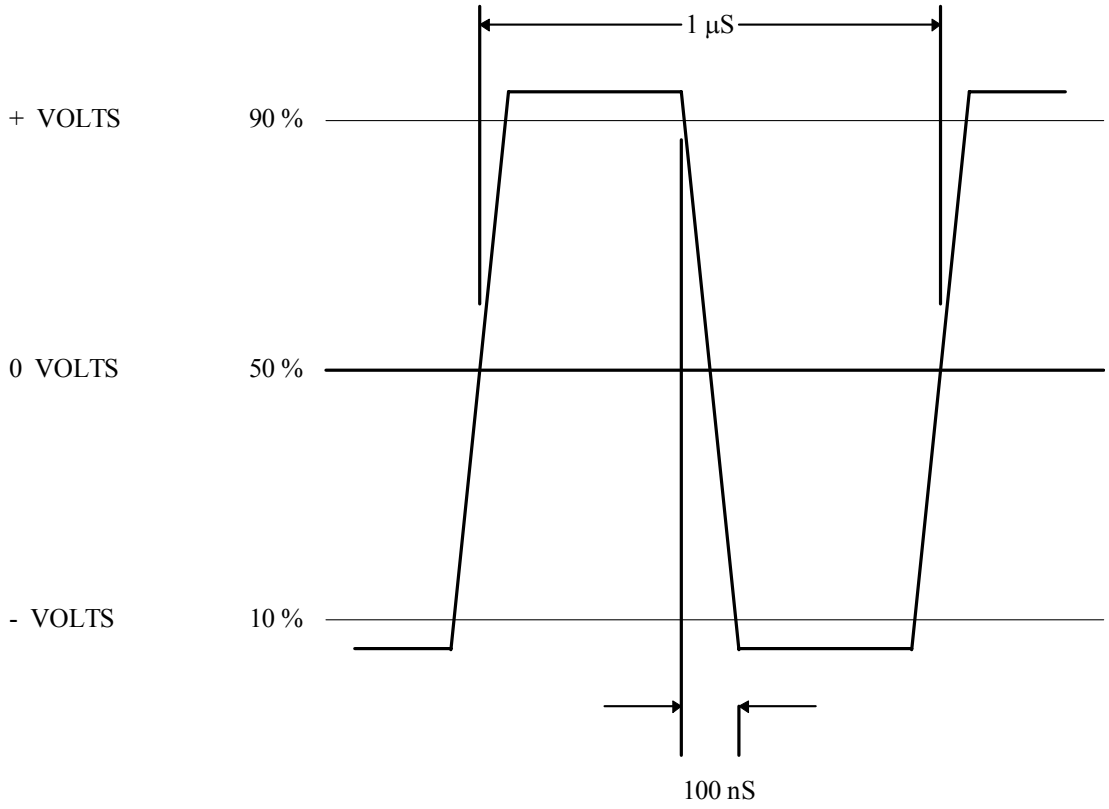


FIGURE 72. Output waveform.

5.5.4.5.2.1.1.3 Output noise.

Any noise transmitted when the terminal is receiving or has power removed, shall not exceed a value of 14.0 mV, RMS, line-to-line, as measured at point A, FIGURE 71.

5.5.4.5.2.1.1.4 Output symmetry.

From the time beginning 2.5 µs after the mid-bit crossing of the parity bit of the last word transmitted by a terminal, the maximum voltage at point A of FIGURE 71 shall be no greater than ± 250.0 mV peak, line-to-line. This shall be tested with the terminal transmitting the maximum number of words it is designed to transmit, up to 33. This test shall be run six times with each word in a contiguous block of words having the same bit pattern. The six word

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contents that shall be used are 8000_{16}, $7FFF_{16}$, 0000_{16}, $FFFF_{16}$, 5555_{16}, $AAAA_{16}$. The output of the terminal shall be as specified in 5.5.4.5.2.1.1.1 and 5.5.4.5.2.1.1.2.

5.5.4.5.2.1.2 Terminal input characteristics.

The following characteristics shall be measured independently.

5.5.4.5.2.1.2.1 Input waveform compatibility.

The terminal shall be capable of receiving and operating with the incoming signals specified herein, and shall accept waveform varying from square wave to sine wave with a maximum zero crossing deviation from the ideal with respect to the previous zero crossing of ± 150 ns, (i.e., 2.0 ± .15 µs, 1.5 ± .15 µs, 1.0 ± .15 µs, .5 ± .15 µs). The terminal shall respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of .86 to 14.0 V. The terminal shall not respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of 0.0 to .20 V. The voltages are measured at point A on FIGURE 68.

5.5.4.5.2.1.2.2 Common mode rejections.

Any signals from direct current (DC) to 2.0 MHz, with amplitudes equal to or less than ± 10.0 V peak, line-to-ground, measured at point A on FIGURE 68, shall not degrade the performance of the receiver.

5.5.4.5.2.1.2.3 Input impedance.

The magnitude of the terminal input impedance, when the RT is not transmitting, or has power removed, shall be a minimum of 1000.0 ohms within the frequency range of 75.0 kHz to 1.0 MHz. This impedance is that measured line-to-line at point A on FIGURE 68.

5.5.4.5.2.1.2.4 Noise rejection.

The terminal shall exhibit a maximum word error rate of one part in 10^7, on all words received by the terminal, after validation checks specified in 5.5.4.4, when operating in the presence of AWGN distributed over a bandwidth of 1.0 kHz to 4.0 MHz at an RMS amplitude of 140 mV. A word error shall include any fault which causes the message error bit to be set in the terminal's status word, or one which causes a terminal to not respond to a valid command. The word error rate shall be measured with a 2.1 V peak-to-peak, line-to-line, input to the terminal as measured at point A on FIGURE 68. The noise tests shall be run continuously until, for a particular number of failures, the number of words received by the terminal, including both command and data words, exceeds the required number for acceptance of the terminal, or is less than the required number for rejection of the terminal, as specified in TABLE XLI. All data words used in the tests shall contain random bit patterns. These bit patterns shall be unique for each data word in a message, and shall change randomly from message to message.

5.5.4.5.2.2 Terminals with Direct Coupled stubs.

5.5.4.5.2.2.1 Terminal output characteristics.

The following characteristics shall be measured with R_L, as shown on FIGURE 71, equal to 35.0 ohms ± 2.0 percent.

5.5.4.5.2.2.1.1 Output levels.

The terminal output voltage levels shall be measured using the test configuration shown on FIGURE 71. The terminal output voltage shall be within the range of 6.0 to 9.0 V, peak-to-peak, line-to-line, when measured at point A on FIGURE 71.

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TABLE XLI. Criteria for acceptance or rejection for the noise rejection test.

TOTAL WORDS RECEIVED BY THE TERMINAL

(in multiples of 10^7)

No. of Errors	Reject (Equal or less)	Accept (Equal or more)
0	N/A	4.40
1	N/A	5.21
2	N/A	6.02
3	N/A	6.83
4	N/A	7.64
5	N/A	8.45
6	.45	9.27
7	1.26	10.08
8	2.07	10.89
9	2.88	11.70
10	3.69	12.51
11	4.50	13.32
12	5.31	14.13
13	6.12	14.94
14	6.93	15.75
15	7.74	16.56
16	8.55	17.37
17	9.37	18.19
18	10.18	19.00
19	10.99	19.81
20	11.80	20.62
21	12.61	21.43
22	13.42	22.24
23	14.23	23.05
24	15.04	23.86
25	15.85	24.67
26	16.66	25.48
27	17.47	26.29
28	18.29	27.11
29	19.10	27.92
30	19.90	28.73
31	20.72	29.54
32	21.53	30.35
33	22.34	31.16
34	23.15	31.97

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TABLE XLI. Criteria for acceptance or rejection for the noise rejection test – Continued.

No. of Errors	Reject (Equal or less)	Accept (Equal or more)
35	23.96	32.78
36	24.77	33.00
37	25.58	33.00
38	26.39	33.00
39	27.21	33.00
40	28.02	33.00
41	33.00	N/A

5.5.4.5.2.2.1.2 Output waveform.

The waveform, when measured at point A on FIGURE 71, shall have zero crossing deviations which are equal to, or less than, 25.0 ns from the ideal crossing point, measured with respect to the previous zero crossing (i.e., .5 ± .025 µs, 1.0 ± .025 µs, 1.5 ± .025 µs, and 2.0 ± .025 µs). The rise and fall time of this waveform shall be from 100.0 to 300.0 ns when measured from levels of 10 to 90 percent of full waveform peak-to-peak, line-to-line, voltage as shown on FIGURE 72. Any distortion of the waveform including overshoot and ringing shall not exceed ± 300.0 mV peak, line-to-line, as measured at point A, FIGURE 71.

5.5.4.5.2.2.1.3 Output noise.

Any noise transmitted when the terminal is receiving or has power removed, shall not exceed a value of 5.0 mV, RMS, line-to-line, as measured at point A on FIGURE 71.

5.5.4.5.2.2.1.4 Output symmetry.

From the time beginning 2.5 µs after the mid-bit crossing of the parity bit of the last word transmitted by a terminal, the maximum voltage at point A of FIGURE 71 shall be no greater than ± 90.0 mV peak, line-to-line. This shall be tested with the terminal transmitting the maximum number of words it is designed to transmit, up to 33. This test shall be run six times with each word in a contiguous block of words having the same bit pattern. The six word contents that shall be used are 8000_{16}, $7FFF_{16}$, 0000_{16}, $FFFF_{16}$, 5555_{16}, $AAAA_{16}$. The output of the terminal shall be as specified in 5.5.4.5.2.2.1.1 and 5.5.4.5.2.2.1.2.

5.5.4.5.2.2.2 Terminal input characteristics.

The following characteristics shall be measured independently.

5.5.4.5.2.2.2.1 Input waveform compatibility.

The terminal shall be capable of receiving and operating with the incoming signals specified herein, and shall accept waveform varying from square wave to sine wave with a maximum zero crossing deviation from the ideal with respect to the previous zero crossing of plus or minus 150 ns, (i.e., 2.0 ± .15 µs, 1.5 ± .15 µs, 1.0 ± .15 µs, .5 ± .15 µs). The terminal shall respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of 1.2 to 20.0 V. The terminal shall not respond to an input signal whose peak-to-peak amplitude, line-to-line, is within the range of 0.0 to .28 V. The voltages are measured at point A on FIGURE 69.

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5.5.4.5.2.2.2.2 Common mode rejections.

Any signals from DC to 2.0 MHz, with amplitudes equal to or less than ± 10.0 V peak, line-to-ground, measured at point A on FIGURE 69, shall not degrade the performance of the receiver.

5.5.4.5.2.2.2.3 Input impedance.

The magnitude of the terminal input impedance, when the RT is not transmitting, or has power removed, shall be a minimum of 2000.0 ohms within the frequency range of 75.0 kHz to 1.0 MHz. This impedance is that measured line-to-line at point A on FIGURE 69.

5.5.4.5.2.2.2.4 Noise rejection.

The terminal shall exhibit a maximum word error rate of one part in 10^7, on all words received by the terminal, after validation checks specified in 4.4, when operating in the presence of AWGN distributed over a bandwidth of 1.0 kHz to 4.0 MHz at an RMS amplitude of 200 mV. A word error shall include any fault which causes the message error bit to be set in the terminal's status word, or one which causes a terminal to not respond to a valid command. The word error rate shall be measured with a 3.0 V peak-to-peak, line-to-line, input to the terminal as measured at point A on FIGURE 69. The noise tests shall be run continuously until, for a particular number of failures, the number of words received by the terminal, including both command and data words, exceeds the required number for acceptance of the terminal, or is less than the required number for rejection of the terminal, as specified in TABLE XLI. All data words used in the tests shall contain random bit patterns. These bit patterns shall be unique for each data word in a message, and shall change randomly from message to message.

5.5.4.6 Redundant data bus requirements.

If redundant data buses are used, the requirements as specified in the following paragraphs shall apply to those data buses.

5.5.4.6.1 Electrical isolation.

All terminals shall have a minimum of 45 dB isolation between data buses. Isolation here means the ratio in dB between the output voltage on the active data bus and the output voltage on the inactive data bus. This shall be measured using the test configuration specified in 5.5.4.5.2.1.1 or 5.5.4.5.2.2.1 for each data bus. Each data bus shall be alternately activated with all measurements being taken at point A on FIGURE 71 for each data bus.

5.5.4.6.2 Single event failures.

All data buses shall be routed to minimize the possibility that a single event failure to a data bus shall cause the loss of more than that particular data bus.

5.5.4.6.3 Dual standby redundant data bus.

If a dual redundant data bus is used, then it shall be a dual standby redundant data bus as specified in the following paragraphs.

5.5.4.6.3.1 Data bus activity.

Only one data bus can be active at any given time except as specified in 5.5.4.6.3.2.

5.5.4.6.3.2 Superseding valid commands.

If while operating on a command, a terminal receives another valid command, from the other data bus, it shall reset and respond to the new command on the data bus on which the new command is received. The terminal shall respond to the new command as specified in 5.5.4.3.3.8.

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5.5.5 DETAILED REQUIREMENTS

(Not Applicable)

5.5.6 Reserved

5.5.7 Reserved

5.5.8 Reserved

5.5.9 Reserved

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5.5.10 General.

The following paragraphs in this appendix are presented in order to discuss certain aspects of the standard in a general sense. They are intended to provide a user of the standard more insight into the aspects discussed.

5.5.10.1 Redundancy.

It is intended that this standard be used to support rather than to supplant the system design process. However, it has been found, through application experience in various aircraft, that the use of a dual standby redundancy technique is very desirable for use in integrating mission avionics. For this reason, this redundancy scheme is defined in 5.5.4.6 of this standard. None the less, the system designer should utilize this standard as the needs of a particular application dictate. The use of redundancy, the degree to which it is implemented, and the form which it takes must be determined on an individual application basis. FIGURE 73 and FIGURE 74 illustrate some possible approaches to dual redundancy. These illustrations are not intended to be inclusive, but rather representative. It should be noted that analogous approaches exist for the triple and quad redundant cases.

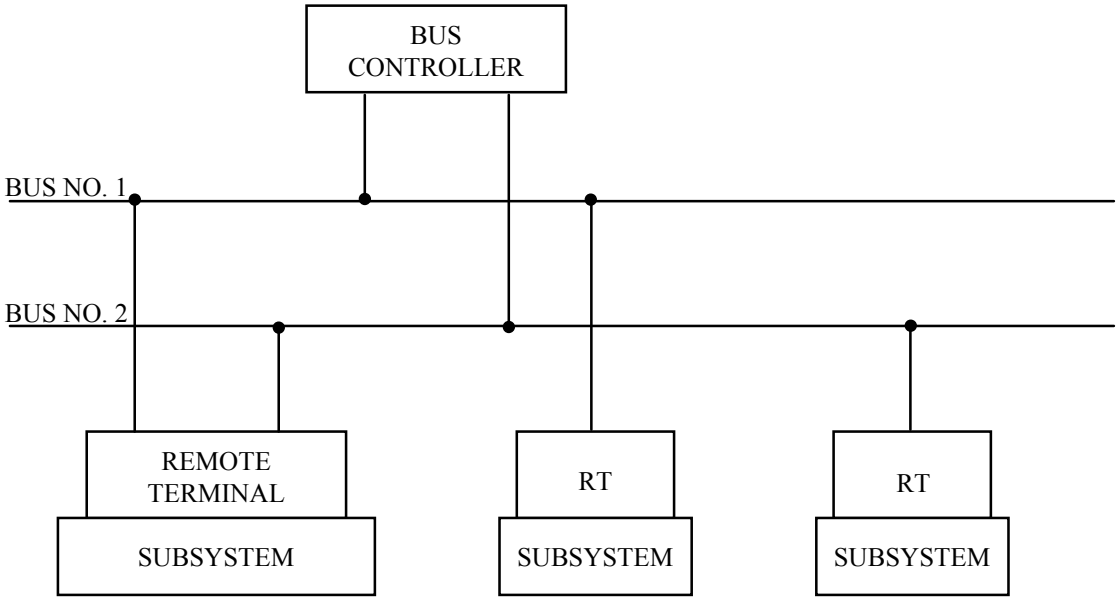
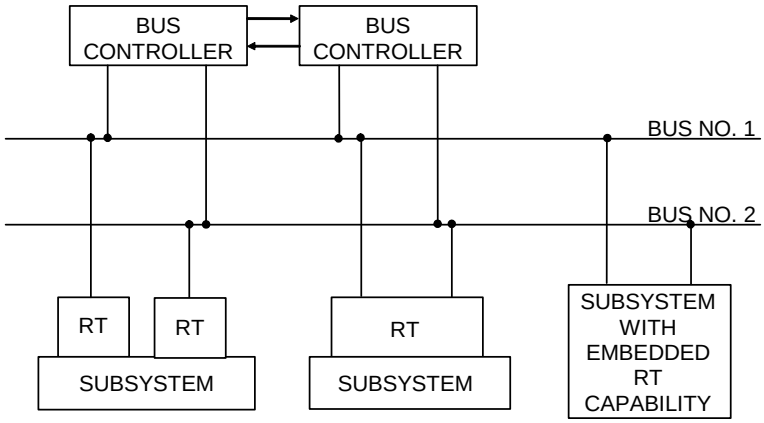


FIGURE 73. Illustration of possible redundancy.

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NOTE: RT - Remote Terminal

FIGURE 74. Illustration of possible redundancy.

5.5.10.2 Bus controller.

The bus controller is a key part of the data bus system. The functions of the bus controller, in addition to the issuance of commands, must include the constant monitoring of the data bus and the traffic on the bus. It is envisioned that most of the routine minute details of bus monitoring (e.g., parity checking, terminal non-response time-out, etc.) will be embodied in hardware, while the algorithms for bus control and decision making will reside in software. It is also envisioned that, in general, the bus controller will be a general purpose computer with a special input/output (I/O) to interface with the data bus. It is of extreme importance in bus controller design that the bus controller be readily able to accommodate terminals of differing protocol's and status word bits used. Equipment designed to MIL-STD-1553A will be in use for a considerable period of time; thus, bus controllers must be capable of adjusting to their differing needs. It is also important to remember that the bus controller will be the focal point for modification and growth within the multiplex system, and thus the software must be written in such a manner as to permit modification with relative ease.

5.5.10.3 Multiplex selection criteria.

The selection of candidate signals for multiplexing is a function of the particular application involved, and criteria will in general vary from system to system. Obviously, those signals which have bandwidths of 400 Hz or less are prime candidates for inclusion on the bus. It is also obvious that video, audio, and high speed parallel digital signals should be excluded. The area of questionable application is usually between 400 Hz and 3 KHz bandwidth. The transfer of these signals on the data bus will depend heavily upon the loading of the bus in a particular application. The decision must be based on projected future bus needs as well as the current loading. Another class of signals which in general are not suitable for multiplexing are those which can be typified by a low rate (over a mission) but possessing a high priority or urgency. Examples of such signals might be a nuclear event detector output or a missile launch alarm from a warning receiver. Such signals are usually better left hardwired, but they may be accommodated by the multiplex system if a direct connection to the bus controller's interrupt hardware is used to trigger a software action in response to the signal.

5.5.10.4 High reliability requirements.

The use of simple parity for error detection within the multiplex bus system was dictated by a compromise between the need for reliable data transmission, system overhead, and remote terminal simplicity. Theoretical and empirical evidence indicates that an undetected bit error rate of 10^{-12} can be expected from a practical multiplex system built to this standard. If a particular signal requires a bit error rate which is better than that provided by the parity checking, then it is incumbent upon the system designer to provide the reliability within the constraints of the standard or to not include this signal within the multiplex bus system. A possible approach in this case would be to have the signal source and sink provide appropriate error detection and correction encoding/decoding and employ extra data words

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to transfer the information. Another approach would be to partition the message, transmit a portion at a time, and then verify (by interrogation) the proper transfer of each segment.

5.5.10.5 Stubbing.

Stubbing is the method wherein a separate line is connected between the primary data bus line and a terminal. The direct connection of a stub line causes a mismatch which appears on the waveforms. This mismatch can be reduced by filtering at the receiver and by using bi-phase modulation. Stubs are often employed not only as a convenience in bus layout but as a means of coupling a unit to the line in such a manner that a fault on the stub or terminal will not greatly affect the transmission line operation. In this case, a network is employed in the stub line to provide isolation from the fault. These networks are also used for stubs that are of such length that the mismatch and reflection degrades bus operation. The preferred method of stubbing is to use Transformer Coupled stubs, as defined in 5.5.4.5.1.5.1. This method provides the benefits of DC isolation, increased common mode protection, a doubling of effective stub impedance, and fault isolation for the entire stub and terminal. Direct coupled stubs, as defined in 5.5.4.5.1.5.2 of this standard, should be avoided if at all possible. Direct coupled stubs provide no DC isolation or common mode rejection for the terminal external to its subsystem. Further, any shorting fault between the subsystems internal isolation resistors (usually on a circuit board) and the main bus junction will cause failure of that entire bus. It can be expected that when the Direct Coupled stub length exceeds 1.6 feet, that it will begin to distort the main bus waveforms. Note that this length includes the cable runs internal to a given subsystem.

5.5.10.6 Use of broadcast option.

The use of a broadcast message as defined in 5.5.4.3.3.6.7 of this standard represents a significant departure from the basic philosophy of this standard in that it is a message format which does not provide positive closed-loop control of bus traffic. The system designer is strongly encouraged to solve any design problems through the use of the three basic message formats without resorting to use of the broadcast option. If system designers do choose to use the broadcast command, they should carefully consider the potential effects of a missed broadcast message, and the subsequent implications for fault or error recovery design in the remote terminals and bus controllers.

5.5.10.7 Other related documents.

Several documents exist which are related to this standard. MIL-HDBK-1553 describes implementation practices for this standard and other related data. This standard is embodied in or referenced by the following international documents: NATO STANAG 3838, ASCC Air Standard 50/2, and UK DEF STAN 00-18 (PART 2)/Issue 1.

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5.5.11 Reserved

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5.5.14 Reserved

5.5.15 Reserved

5.5.16 Reserved

5.5.17 Reserved

5.5.18 Reserved

5.5.19 Reserved

5.5.20 Referenced documents.

Not applicable.

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5.5.21 Reserved

5.5.22 Reserved

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5.5.25 Reserved

5.5.26 Reserved

5.5.27 Reserved

5.5.28 Reserved

5.5.29 Reserved

5.5.30 General requirements.

5.5.30.1 Option selection.

This section of the appendix shall select those options required to further define portions of the standard to enhance tri-service interoperability. References in parentheses are to paragraphs in this standard which are affected.

5.5.30.2 Application.

Section 30 of this appendix shall apply to all dual standby redundant applications for the Army, Navy, and Air Force. All Air Force aircraft internal avionics applications shall be dual standby redundant, except where safety critical or flight critical requirements dictate a higher level of redundancy.

5.5.30.3 Unique address (5.5.4.3.3.5.1.2).

All remote terminals shall be capable of being assigned any unique address from decimal address 0 (00000) through decimal address 30 (11110). The address shall be established through an external connector, which is part of the system wiring and connects to the remote terminal. Changing the unique address of a remote terminal shall not require the physical modification or manipulation of any part of the remote terminal. The remote terminal shall, as a minimum, determine and validate its address during power-up conditions. No single point failure shall cause a terminal to validate a false address. The remote terminal shall not respond to any messages if it has determined its unique address is not valid.

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5.5.30.4 Mode codes (5.5.4.3.3.5.1.7).

5.5.30.4.1 Subaddress/mode (5.5.4.3.3.5.1.4).

An RT shall have the capability to respond to mode codes with both subaddress/mode of 00000 and 11111. Bus controllers shall have the capability to issue mode commands with both subaddress/mode of 00000 and 11111. The subaddress/mode of 00000 and 11111 shall not convey different information.

5.5.30.4.2 Required mode codes (5.5.4.3.3.5.1.7).

5.5.30.4.2.1 Remote terminal required mode codes.

An RT shall implement the following mode codes as a minimum:

Mode Code	Function
00010	Transmit status word
00100	Transmitter shutdown
00101	Override transmitter shutdown
01000	Reset remote terminal

5.5.30.4.2.2 Bus controller required mode codes.

The bus controller shall have the capability to implement all of the mode codes as defined in 5.5.4.3.3.5.1.7. For Air Force applications, the dynamic bus control mode command shall never be issued by the bus controller.

5.5.30.4.3 Reset remote terminal (5.5.4.3.3.5.1.7.9).

An RT receiving the reset remote terminal mode code shall respond with a status word as specified in 5.5.4.3.3.5.1.7.9 and then reset. While the RT is being reset, the RT shall respond to a valid command with any of the following: no response on either data bus, status word transmitted with the busy bit set, or normal response. If any data is transmitted from the RT while it is being reset, the information content of the data shall be valid. An RT receiving this mode code shall complete the reset function within 5.0 milliseconds following transmission of the status word specified in 5.5.4.3.3.5.1.7.9. The time shall be measured from the mid-bit zero crossing of the parity bit of the status word to the mid-sync zero crossing of the command word at point A on FIGURE 68 and FIGURE 69.

5.5.30.4.4 Initiate RT self test (5.5.4.3.3.5.1.7.4).

If the initiate self test mode command is implemented in the RT, then the RT receiving the initiate self test mode code shall respond with a status word as specified in 5.5.4.3.3.5.1.7.4 and then initiate the RT self test function. Subsequent valid commands may terminate the self-test function. While the RT self test is in progress, the RT shall respond to a valid command with any of the following: no response on either data bus, status word transmitted with the busy bit set, or normal response. If any data is transmitted from the RT while it is in self test, the information content of the data shall be valid. An RT receiving this mode code shall complete the self test function and have the results of the self test available within 100.0 milliseconds following transmission of the status word specified in 5.5.4.3.3.5.1.7.4. The time shall be measured from the mid-bit zero crossing of the parity bit of the status word to the mid-sync zero crossing of the command word at point A on FIGURE 68 and FIGURE 69.

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5.5.30.5 Status word bits (5.5.4.3.3.5.3).

5.5.30.5.1 Information content.

The status word transmitted by an RT shall contain valid information at all times, e.g., following RT power up, during initialization, and during normal operation.

5.5.30.5.2 Status bit requirements (5.5.4.3.3.5.3).

An RT shall implement the status bits as follows:

Message error bit (5.5.4.3.3.5.3.3) - Required

Instrumentation bit (5.5.4.3.3.5.3.4) - Always logic zero

Service request bit (5.5.4.3.3.5.3.5) - Optional

Reserved status bits (5.5.4.3.3.5.3.6) - Always logic zero

Broadcast command received bit (5.5.4.3.3.5.3.7) - If the RT implements the broadcast option, then this bit shall be required.

Busy bit (5.5.4.3.3.5.3.8) - As required by 5.5.30.5.3

Subsystem flag bit (5.5.4.3.3.5.3.9) - If an associated subsystem has the capability for self test, then this bit shall be required.

Dynamic bus control acceptance bit (5.5.4.3.3.5.3.10) - If the RT implements the dynamic bus control function, then this bit shall be required.

Terminal flag bit (5.5.4.3.3.5.3.11) - If an RT has the capability for self test, then this bit shall be required.

5.5.30.5.3 Busy bit (5.5.4.3.3.5.3.8).

The existence of busy conditions is discouraged. However, any busy condition, in the RT or the subsystem interface that would affect communication over the bus shall be conveyed via the busy bit. Busy conditions, and thus the setting of the busy bit, shall occur only as a result of particular commands/ messages sent to an RT. Thus for a non-failed RT, the bus controller can, with prior knowledge of the remote terminal characteristics, determine when the remote terminal can become busy and when it will not be busy. However, the RT may also set the busy bit (in addition to setting the terminal flag bit or subsystem flag bit) as a result of failure/fault conditions within the RT/subsystem.

5.5.30.6 Broadcast (5.5.4.3.3.6.7).

The only broadcast commands allowed to be transmitted on the data bus by the bus controller shall be the broadcast mode commands identified in TABLE XL. The broadcast option may be implemented in remote terminals.

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However, if implemented, the RT shall be capable of distinguishing between a broadcast and a non-broadcast message to the same subaddress for non-mode command messages. The RT address of 11111 is still reserved for broadcast and shall not be used for any other purpose.

5.5.30.7 Data wrap-around (5.5.4.3.3.5.1.4).

Remote terminals shall provide a receive subaddress to which one to N data words of any bit pattern can be received. Remote terminals shall provide a transmit subaddress from which a minimum of N data words can be transmitted. N is equal to the maximum word count from the set of all messages defined for the RT. A valid receive message to the data wrap-around receive subaddress followed by a valid transmit command to the data wrap-around transmit subaddress, with the same word count and without any intervening valid commands to that RT, shall cause the RT to respond with each data word having the same bit pattern as the corresponding received data word. A data wrap-around receive and transmit subaddress of 30 (11110) is desired.

5.5.30.8 Message formats (5.5.4.3.3.6).

Remote terminals shall, as a minimum, implement the following non-broadcast message formats as defined in 5.5.4.3.3.6: RT to BC transfers, BC to RT transfers, RT to RT transfers (receive and transmit), and mode command without data word transfers. For non-broadcast messages, the RT shall not distinguish between data received during a BC to RT transfer or data received during a RT to RT transfer (receive) to the same subaddress. The RT shall not distinguish between data to be transmitted during an RT to BC transfer or data to be transmitted during an RT to RT transfer (transmit) from the same subaddress. BCs shall have the capability to issue all message formats defined in 5.5.4.3.3.6.

5.5.30.9 RT to RT validation (5.5.4.3.3.9).

For RT to RT transfers, in addition to the validation criteria specified in 5.5.4.4.3.6, if a valid receive command is received by the RT and the first data word is received after 57.0 plus or minus 3.0 microseconds, the RT shall consider the message invalid and respond as specified in 5.5.4.4.3.6. The time shall be measured from the mid-bit zero crossing of the parity bit of the receive command to the mid-sync zero crossing of the first expected data word at point A as shown on FIGURE 68 and FIGURE 69. It is recommended that the receiving RT of an RT to RT transfer verify the proper occurrence of the transmit command word and status word as specified in 5.5.4.3.3.6.3.

5.5.30.10 Electrical characteristics (5.5.4.5).

5.5.30.10.1 Cable shielding (5.5.4.5.1.1).

The cable shield shall provide a minimum of 90.0 percent coverage.

5.5.30.10.2 Shielding (5.5.4.5.1).

All cable to connector junctions, cable terminations, and bus-stub junctions shall have continuous 360 degree shielding which shall provide a minimum of 75.0 percent coverage.

5.5.30.10.3 Connector polarity.

For applications that use concentric connectors or inserts for each bus, the center pin of the connector or insert shall be used for the high (positive) Manchester bi-phase signal. The inner ring shall be used for the low (negative) Manchester bi-phase signal.

5.5.30.10.4 Characteristic impedance (5.5.4.5.1.2).

The actual (not nominal) characteristic impedance of the data bus cable shall be within the range of 70.0 ohms to 85.0 ohms at a sinusoidal frequency of 1.0 megahertz.

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5.5.30.10.5 Stub coupling (5.5.4.5.1.5).

For Navy applications, each terminal shall have both transformer and Direct Coupled stub connections externally available. For Navy systems using these terminals, either transformer or Direct Coupled connections may be used. For Army and Air Force applications, each terminal shall have Transformer Coupled stub connections, but may also have Direct Coupled stub connections. For Army and Air Force systems, only Transformer Coupled stub connections shall be used. Unused terminal connections shall have a minimum of 75 percent shielding coverage.

5.5.30.10.6 Power on/off noise.

A terminal shall limit any spurious output during a power-up or power-down sequence. The maximum allowable output noise amplitude shall be ±250 mV peak, line-to-line for Transformer Coupled stubs and ±90 mV peak, line-to-line for Direct Coupled stubs, measured at point A of FIGURE 71.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

This standard is intended to be used for airborne military applications and applies to all U.S. Air Force aircraft internal avionics activities. This standard is intended to establish the performance, interface, and verification requirements for an extended waveform multiplexing capability (extended 1553) that retains backwards compatibility with the 21 September 1978 release of MIL-STD 1553B and notices 1 through 4.

6.2 Subject term (key word) listing.

Filtered waveform

Multi-carrier waveform

Open systems interconnection

Trapezoidal waveform

6.3 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:

Air Force – 11

Preparing activity:

Air Force - 11

(Project MCCR-2007-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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